

**CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD  
CENTRAL VALLEY REGION**

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**ORDER R5-2015-0123**

**NPDES NO. CA0081434**

**WASTE DISCHARGE REQUIREMENTS  
FOR THE CITY OF GALT  
WASTEWATER TREATMENT PLANT AND RECLAMATION FACILITY  
SACRAMENTO COUNTY**

The following Discharger is subject to waste discharge requirements (WDR's) set forth in this Order:

**Table 1. Discharger Information**

|                         |                                                                         |
|-------------------------|-------------------------------------------------------------------------|
| <b>Discharger</b>       | <b>City of Galt</b>                                                     |
| <b>Name of Facility</b> | <b>City of Galt Wastewater Treatment Plant and Reclamation Facility</b> |
| <b>Facility Address</b> | <b>10059 Twin Cities Road</b>                                           |
|                         | <b>Galt, CA 95632</b>                                                   |
|                         | <b>Sacramento County</b>                                                |

**Table 2. Discharge Location**

| <b>Discharge Point</b> | <b>Effluent Description</b> | <b>Discharge Point Latitude (No rth)</b> | <b>Discharge Point Longitude (West)</b> | <b>Receiving Water</b> |
|------------------------|-----------------------------|------------------------------------------|-----------------------------------------|------------------------|
| 001                    | Treated WWTP Effluent       | 38° 18' 14.88" N                         | 121° 19' 55.87" W                       | Laguna Creek           |

**Table 3. Administrative Information**

|                                                                                                                                                                                                                                                                                   |                         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| This Order was adopted on:                                                                                                                                                                                                                                                        | <b>11 December 2015</b> |
| This Order shall become effective on:                                                                                                                                                                                                                                             | <b>1 February 2016</b>  |
| This Order shall expire on:                                                                                                                                                                                                                                                       | <b>31 January 2021</b>  |
| The Discharger shall file a Report of Waste Discharge as an application for reissuance of WDR's in accordance with title 23, California Code of Regulations, and an application for reissuance of a National Pollutant Discharge Elimination System (NPDES) permit no later than: | <b>4 August 2020</b>    |
| The U.S. Environmental Protection Agency (U.S. EPA) and the California Regional Water Quality Control Board, Central Valley Region have classified this discharge as follows:                                                                                                     | <b>Major</b>            |

I, Pamela C. Creedon, Executive Officer, do hereby certify that this Order with all attachments is a full, true, and correct copy of the Order adopted by the California Regional Water Quality Control Board, Central Valley Region, on **11 December 2015**.

*ORIGINAL SIGNED BY*

\_\_\_\_\_  
**PAMELA C. CREEDON**, Executive Officer

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## I. FACILITY INFORMATION

Information describing the City of Galt Wastewater Treatment Plant and Reclamation Facility (Facility) is summarized in Table 1 and in sections I and II of the Fact Sheet (Attachment F). Section I of the Fact Sheet also includes information regarding the Facility's permit application.

## II. FINDINGS

The California Regional Water Quality Control Board, Central Valley Region (hereinafter Central Valley Water Board), finds:

- A. Legal Authorities.** This Order serves as WDR's pursuant to article 4, chapter 4, division 7 of the California Water Code (commencing with section 13260). This Order is also issued pursuant to section 402 of the federal Clean Water Act (CWA) and implementing regulations adopted by the U.S. EPA and chapter 5.5, division 7 of the Water Code (commencing with section 13370). It shall serve as an NPDES permit for point source discharges from this facility to surface waters. Separate permits have been issued for the surface water and land discharges. This Order regulates the wastewater treatment facility and discharges to Laguna Creek. WDR Order R5-2015-0125 regulates the reuse of undisinfected secondary wastewater and the land application of biosolids on the Discharger's land application area.
- B. Background and Rationale for Requirements.** The Central Valley Water Board developed the requirements in this Order based on information submitted as part of the application, through monitoring and reporting programs, and other available information. The Fact Sheet (Attachment F), which contains background information and rationale for the requirements in this Order, is hereby incorporated into and constitutes Findings for this Order. Attachments A through E and G through H are also incorporated into this Order.
- C. Provisions and Requirements Implementing State Law.** Not Applicable
- D. Monitoring and Reporting.** 40 C.F.R. section 122.48 requires that all NPDES permits specify requirements for recording and reporting monitoring results. Water Code sections 13267 and 13383 authorize the Central Valley Water Board to require technical and monitoring reports. The Monitoring and Reporting Program establishes monitoring and reporting requirements to implement federal and State requirements. The Monitoring and Reporting Program is provided in Attachment E.

The technical and monitoring reports in this Order are required in accordance with Water Code section 13267, which states the following in subsection (b)(1), *"In conducting an investigation specified in subdivision (a), the regional board may require that any person who has discharged, discharges, or is suspected of having discharged discharging, or who proposes to discharge waste within its region, or any citizen or domiciliary, or political agency or entity of this state who has discharged, discharges, or is suspected of having discharged or discharging, or who proposes to discharge, waste outside of its region could affect the quality of waters within its region shall furnish, under penalty of perjury, technical or monitoring program reports which the regional board requires. The burden, including costs, of these reports shall bear a reasonable relationship to the need for the report and the benefits to be obtained from the reports. In requiring those reports, the regional board shall provide the person with a written explanation with regard to the need for the reports, and shall identify the evidence that supports requiring that person to provide the reports."*

The Discharger owns and operates the Facility subject to this Order. The monitoring reports required by this Order are necessary to determine compliance with this Order. The need for the monitoring reports is discussed in the Fact Sheet.

- E. Notification of Interested Parties.** The Central Valley Water Board has notified the Discharger and interested agencies and persons of its intent to prescribe WDR's for the discharge and has provided them with an opportunity to submit their written comments and recommendations. Details of the notification are provided in the Fact Sheet.
- F. Consideration of Public Comment.** The Central Valley Water Board, in a public meeting, heard and considered all comments pertaining to the discharge. Details of the Public Hearing are provided in the Fact Sheet.

THEREFORE, IT IS HEREBY ORDERED that Order R5-2010-0099 is rescinded upon the effective date of this Order except for enforcement purposes, and, in order to meet the provisions contained in division 7 of the Water Code (commencing with section 13000) and regulations adopted thereunder, and the provisions of the CWA and regulations and guidelines adopted thereunder, the Discharger shall comply with the requirements in this Order. This action in no way prevents the Central Valley Water Board from taking enforcement action for past violations of the previous Order.

### **III. DISCHARGE PROHIBITIONS**

- A.** Discharge of wastewater from the Facility, as the Facility is specifically described in the Fact Sheet in sections II.A and II.E, in a manner different from that described in this Order is prohibited.
- B.** The by-pass or overflow of wastes to surface waters is prohibited, except as allowed by Federal Standard Provisions I.G. and I.H. (Attachment D).
- C.** Neither the discharge nor its treatment shall create a nuisance as defined in section 13050 of the Water Code.
- D.** The Discharger shall not allow pollutant-free wastewater to be discharged into the treatment or disposal system in amounts that significantly diminish the system's capability to comply with this Order. Pollutant-free wastewater means rainfall, groundwater, cooling waters, and condensates that are essentially free of pollutants.

### **IV. EFFLUENT LIMITATIONS AND DISCHARGE SPECIFICATIONS**

#### **A. Effluent Limitations – Discharge Point No. 001**

##### **1. Final Effluent Limitations –**

When discharging to surface water, the Discharger shall maintain compliance with the following effluent limitations at Discharge Point No. 001, with compliance measured at Monitoring Location Eff-001 (compliance with Total Coliform Organisms effluent limitation shall be measured at UVS-001) as described in the Monitoring and Reporting Program, Attachment E:

- a. The Discharger shall maintain compliance with the effluent limitations specified in Table 4:

**Table 4. Effluent Limitations**

| Parameter                                | Units                | Effluent Limitations |                |               |                       |                       |
|------------------------------------------|----------------------|----------------------|----------------|---------------|-----------------------|-----------------------|
|                                          |                      | Average Monthly      | Average Weekly | Maximum Daily | Instantaneous Minimum | Instantaneous Maximum |
| Biochemical Oxygen Demand (5-day @ 20°C) | mg/L                 | 10                   | 15             | 20            | --                    | --                    |
|                                          | lbs/day <sup>1</sup> | 250                  | 375            | 500           | --                    | --                    |
|                                          | lbs/day <sup>2</sup> | 375                  | 560            | 750           | --                    | --                    |
| Total Suspended Solids                   | mg/L                 | 10                   | 15             | 20            | --                    | --                    |
|                                          | lbs/day <sup>1</sup> | 250                  | 375            | 500           | --                    | --                    |
|                                          | lbs/day <sup>2</sup> | 375                  | 560            | 750           | --                    | --                    |
| Ammonia Nitrogen, Total (as N)           | mg/L                 | 1.3                  | 2.9            | --            | --                    | --                    |
|                                          | lbs/day <sup>1</sup> | 33                   | 73             | --            | --                    | --                    |
|                                          | lbs/day <sup>2</sup> | 49                   | 109            | --            | --                    | --                    |
| Arsenic                                  | µg/L                 | 10                   | --             | 19            | --                    | --                    |
| Nitrate plus Nitrite, Total (as N)       | mg/L                 | 10                   | 19             | --            | --                    | --                    |
| pH                                       | standard units       | --                   | --             | --            | 6.5                   | 8.2                   |
| Zinc, Total Recoverable                  | µg/L                 | 35                   | --             | 51            | --                    | --                    |

<sup>1</sup> Based on an average dry weather flow of 3.0 million gallons per day (MGD). Effective immediately and until Executive Officer's written approval of flow increase (Special Provisions VI.C.6.b).

<sup>2</sup> Based on an average dry weather flow of 4.5 MGD. Effective upon Executive Officer's written approval of flow increase (Special Provisions VI.C.6.b).

- b. **Percent Removal:** The average monthly percent removal of 5-day biochemical oxygen demand (BOD<sub>5</sub>) and total suspended solids (TSS) shall not be less than 85 percent.
- c. **Acute Whole Effluent Toxicity.** Survival of aquatic organisms in 96-hour bioassays of undiluted waste shall be no less than:
  - i. 70%, minimum for any one bioassay; and
  - ii. 90%, median for any three consecutive bioassays.
- d. **Total Coliform Organisms.** Effluent total coliform organisms shall not exceed:
  - i. 2.2 most probable number (MPN) per 100 mL, as a 7-day median;
  - ii. 23 MPN/100 mL, more than once in any 30-day period; and
  - iii. 240 MPN/100 mL, at any time.
- e. **Average Dry Weather Flow.** The average dry weather discharge flow shall not exceed:
  - i. 3.0 MGD, effective immediately until Executive Officer's written approval of flow increase (Special Provisions VI.C.6.b).
  - ii. 4.5 MGD, effective upon Executive Officer's written approval of flow increase (Special Provisions VI.C.6.b).
- f. **Mercury, Total.** for a calendar year, the total annual mass discharge of total mercury shall not exceed 0.05 pounds/year.

**2. Interim Effluent Limitations- Not Applicable**

**B. Land Discharge Specifications – Not Applicable**

**C. Recycling Specifications – Not Applicable**

**V. RECEIVING WATER LIMITATIONS**

**A. Surface Water Limitations**

The discharge shall not cause the following in Laguna Creek:

1. **Bacteria.** The fecal coliform concentration, based on a minimum of not less than five samples for any 30-day period, to exceed a geometric mean of 200 MPN/100 mL, nor more than 10 percent of the total number of fecal coliform samples taken during any 30-day period to exceed 400 MPN/100 mL.
2. **Biostimulatory Substances.** Water to contain biostimulatory substances which promote aquatic growths in concentrations that cause nuisance or adversely affect beneficial uses.
3. **Chemical Constituents.** Chemical constituents to be present in concentrations that adversely affect beneficial uses.
4. **Color.** Discoloration that causes nuisance or adversely affects beneficial uses.
5. **Dissolved Oxygen:**
  - a. The monthly median of the mean daily dissolved oxygen concentration to fall below 85 percent of saturation in the main water mass;
  - b. The 95 percentile dissolved oxygen concentration to fall below 75 percent of saturation; nor
  - c. The dissolved oxygen concentration to be reduced below 7.0 mg/L at any time.
6. **Floating Material.** Floating material to be present in amounts that cause nuisance or adversely affect beneficial uses.
7. **Oil and Grease.** Oils, greases, waxes, or other materials to be present in concentrations that cause nuisance, result in a visible film or coating on the surface of the water or on objects in the water, or otherwise adversely affect beneficial uses.
8. **pH.** The pH to be depressed below 6.5 nor raised above 8.5.
9. **Pesticides:**
  - a. Pesticides to be present, individually or in combination, in concentrations that adversely affect beneficial uses;
  - b. Pesticides to be present in bottom sediments or aquatic life in concentrations that adversely affect beneficial uses;
  - c. Total identifiable persistent chlorinated hydrocarbon pesticides to be present in the water column at concentrations detectable within the accuracy of analytical methods approved by USEPA or the Executive Officer;
  - d. Pesticide concentrations to exceed those allowable by applicable antidegradation policies (see State Water Board Resolution No. 68-16 and 40 CFR 131.12.);
  - e. Pesticide concentrations to exceed the lowest levels technically and economically achievable;

- f. Pesticides to be present in concentration in excess of the maximum contaminant levels (MCL's) set forth in CCR, Title 22, division 4, chapter 15; nor
  - g. Thiobencarb to be present in excess of 1.0 µg/L.
- 10. **Radioactivity:**
  - a. Radionuclides to be present in concentrations that are harmful to human, plant, animal, or aquatic life nor that result in the accumulation of radionuclides in the food web to an extent that presents a hazard to human, plant, animal, or aquatic life.
  - b. Radionuclides to be present in excess of the MCL's specified in Table 64442 of section 64442 and Table 64443 of section 64443 of Title 22 of the California Code of Regulations.
- 11. **Suspended Sediments.** The suspended sediment load and suspended sediment discharge rate of surface waters to be altered in such a manner as to cause nuisance or adversely affect beneficial uses.
- 12. **Settleable Substances.** Substances to be present in concentrations that result in the deposition of material that causes nuisance or adversely affects beneficial uses.
- 13. **Suspended Material.** Suspended material to be present in concentrations that cause nuisance or adversely affect beneficial uses.
- 14. **Taste and Odors.** Taste- or odor-producing substances to be present in concentrations that impart undesirable tastes or odors to fish flesh or other edible products of aquatic origin, or that cause nuisance, or otherwise adversely affect beneficial uses.
- 15. **Temperature.** The natural temperature to be increased by more than 5°F. Compliance to be determined based on the difference in temperature at Monitoring Locations RSW-001 and RSW-002.
- 16. **Toxicity.** Toxic substances to be present, individually or in combination, in concentrations that produce detrimental physiological responses in human, plant, animal, or aquatic life.
- 17. **Turbidity.**
  - a. Shall not exceed 2 Nephelometric Turbidity Units (NTU) where natural turbidity is less than 1 NTU;
  - b. Shall not increase more than 1 NTU where natural turbidity is between 1 and 5 NTUs;
  - c. Shall not increase more than 20 percent where natural turbidity is between 5 and 50 NTUs;
  - d. Shall not increase more than 10 NTU where natural turbidity is between 50 and 100 NTUs; nor
  - e. Shall not increase more than 10 percent where natural turbidity is greater than 100 NTUs.

**B. Groundwater Limitations – Not Applicable**

## VI. PROVISIONS

### A. Standard Provisions

1. The Discharger shall comply with all Standard Provisions included in Attachment D.
2. The Discharger shall comply with the following provisions. In the event that there is any conflict, duplication, or overlap between provisions specified by this Order, the more stringent provision shall apply:
  - a. If the Discharger's wastewater treatment plant is publicly owned or subject to regulation by California Public Utilities Commission, it shall be supervised and operated by persons possessing certificates of appropriate grade according to Title 23, CCR, division 3, chapter 26.
  - b. After notice and opportunity for a hearing, this Order may be terminated or modified for cause, including, but not limited to:
    - i. violation of any term or condition contained in this Order;
    - ii. obtaining this Order by misrepresentation or by failing to disclose fully all relevant facts;
    - iii. a change in any condition that requires either a temporary or permanent reduction or elimination of the authorized discharge; and
    - iv. a material change in the character, location, or volume of discharge.

The causes for modification include:

- i. *New regulations.* New regulations have been promulgated under section 405(d) of the CWA, or the standards or regulations on which the permit was based have been changed by promulgation of amended standards or regulations or by judicial decision after the permit was issued.
- ii. *Land application plans.* When required by a permit condition to incorporate a land application plan for beneficial reuse of sewage sludge, to revise an existing land application plan, or to add a land application plan.
- iii. *Change in sludge use or disposal practice.* Under 40 CFR section 122.62(a)(1), a change in the Discharger's sludge use or disposal practice is a cause for modification of the permit. It is cause for revocation and reissuance if the Discharger requests or agrees.

The Central Valley Water Board may review and revise this Order at any time upon application of any affected person or the Central Valley Water Board's own motion.

- c. If a toxic effluent standard or prohibition (including any scheduled compliance specified in such effluent standard or prohibition) is established under section 307(a) of the CWA, or amendments thereto, for a toxic pollutant that is present in the discharge authorized herein, and such standard or prohibition is more stringent than any limitation upon such pollutant in this Order, the Central Valley Water Board will revise or modify this Order in accordance with such toxic effluent standard or prohibition.

The Discharger shall comply with effluent standards and prohibitions within the time provided in the regulations that establish those standards or prohibitions, even if this Order has not yet been modified.



- d. This Order shall be modified, or alternately revoked and reissued, to comply with any applicable effluent standard or limitation issued or approved under sections 301(b)(2)(C) and (D), 304(b)(2), and 307(a)(2) of the CWA, if the effluent standard or limitation so issued or approved:
  - i. Contains different conditions or is otherwise more stringent than any effluent limitation in the Order; or
  - ii. Controls any pollutant limited in the Order.

The Order, as modified or reissued under this paragraph, shall also contain any other requirements of the CWA then applicable.

- e. The provisions of this Order are severable. If any provision of this Order is found invalid, the remainder of this Order shall not be affected.
- f. The Discharger shall take all reasonable steps to minimize any adverse effects to waters of the State or users of those waters resulting from any discharge or sludge use or disposal in violation of this Order. Reasonable steps shall include such accelerated or additional monitoring as necessary to determine the nature and impact of the non-complying discharge or sludge use or disposal.
- g. The Discharger shall ensure compliance with any existing or future pretreatment standard promulgated by USEPA under section 307 of the CWA, or amendment thereto, for any discharge to the municipal system.
- h. A copy of this Order shall be maintained at the discharge facility and be available at all times to operating personnel. Key operating personnel shall be familiar with its content.
- i. Safeguard to electric power failure:
  - i. The Discharger shall provide safeguards to assure that, should there be reduction, loss, or failure of electric power, the discharge shall comply with the terms and conditions of this Order.
  - ii. Upon written request by the Central Valley Water Board, the Discharger shall submit a written description of safeguards. Such safeguards may include alternate power sources, standby generators, retention capacity, operating procedures, or other means. A description of the safeguards provided shall include an analysis of the frequency, duration, and impact of power failures experienced over the past 5 years on effluent quality and on the capability of the Discharger to comply with the terms and conditions of the Order. The adequacy of the safeguards is subject to the approval of the Central Valley Water Board.
  - iii. Should the treatment works not include safeguards against reduction, loss, or failure of electric power, or should the Central Valley Water Board not approve the existing safeguards, the Discharger shall, within 90 days of having been advised in writing by the Central Valley Water Board that the existing safeguards are inadequate, provide to the Central Valley Water Board and USEPA a schedule of compliance for providing safeguards such that in the event of reduction, loss, or failure of electric power, the Discharger shall comply with the terms and conditions of this Order. The schedule of compliance shall, upon approval of the Central Valley Water Board, become a condition of this Order.

- j. The Discharger, upon written request of the Central Valley Water Board, shall file with the Board a technical report on its preventive (failsafe) and contingency (cleanup) plans for controlling accidental discharges, and for minimizing the effect of such events. This report may be combined with that required under the Central Valley Water Board Standard Provision contained in section VI.A.2.i of this Order.

The technical report shall:

- i. Identify the possible sources of spills, leaks, untreated waste by-pass, and contaminated drainage. Loading and storage areas, power outage, waste treatment unit outage, and failure of process equipment, tanks and pipes should be considered.
- ii. Evaluate the effectiveness of present facilities and procedures and state when they became operational.
- iii. Predict the effectiveness of the proposed facilities and procedures and provide an implementation schedule containing interim and final dates when they will be constructed, implemented, or operational.

The Central Valley Water Board, after review of the technical report, may establish conditions which it deems necessary to control accidental discharges and to minimize the effects of such events. Such conditions shall be incorporated as part of this Order, upon notice to the Discharger.

- k. A publicly owned treatment works whose waste flow has been increasing, or is projected to increase, shall estimate when flows will reach hydraulic and treatment capacities of its treatment and disposal facilities. The projections shall be made in January, based on the last 3 years' average dry weather flows, peak wet weather flows and total annual flows, as appropriate. When any projection shows that capacity of any part of the facilities may be exceeded in 4 years, the Discharger shall notify the Central Valley Water Board by 31 January. A copy of the notification shall be sent to appropriate local elected officials, local permitting agencies and the press. Within 120 days of the notification, the Discharger shall submit a technical report showing how it will prevent flow volumes from exceeding capacity or how it will increase capacity to handle the larger flows. The Central Valley Water Board may extend the time for submitting the report.
- l. The Discharger shall submit technical reports as directed by the Executive Officer. All technical reports required herein that involve planning, investigation, evaluation, or design, or other work requiring interpretation and proper application of engineering or geologic sciences, shall be prepared by or under the direction of persons registered to practice in California pursuant to California Business and Professions Code, sections 6735, 7835, and 7835.1. To demonstrate compliance with Title 16, CCR, sections 415 and 3065, all technical reports must contain a statement of the qualifications of the responsible registered professional(s). As required by these laws, completed technical reports must bear the signature(s) and seal(s) of the registered professional(s) in a manner such that all work can be clearly attributed to the professional responsible for the work.
- m. The Central Valley Water Board is authorized to enforce the terms of this permit under several provisions of the Water Code, including, but not limited to, sections 13385, 13386, and 13387.

- n. For publicly owned treatment works, prior to making any change in the point of discharge, place of use, or purpose of use of treated wastewater that results in a permanent decrease of flow in any portion of a watercourse, the Discharger must file a petition with the State Water Board, Division of Water Rights, and receive approval for such a change. (Water Code section 1211).
- o. In the event of any change in control or ownership of land or waste discharge facilities presently owned or controlled by the Discharger, the Discharger shall notify the succeeding owner or operator of the existence of this Order by letter, a copy of which shall be immediately forwarded to the Central Valley Water Board.

To assume operation under this Order, the succeeding owner or operator must apply in writing to the Executive Officer requesting transfer of the Order. The request must contain the requesting entity's full legal name, the state of incorporation if a corporation, address and telephone number of the persons responsible for contact with the Central Valley Water Board and a statement. The statement shall comply with the signatory and certification requirements in the federal Standard Provisions (Attachment D, section V.B) and state that the new owner or operator assumes full responsibility for compliance with this Order. Failure to submit the request shall be considered a discharge without requirements, a violation of the Water Code. Transfer shall be approved or disapproved in writing by the Executive Officer.

- p. Failure to comply with provisions or requirements of this Order, or violation of other applicable laws or regulations governing discharges from this facility, may subject the Discharger to administrative or civil liabilities, criminal penalties, and/or other enforcement remedies to ensure compliance. Additionally, certain violations may subject the Discharger to civil or criminal enforcement from appropriate local, state, or federal law enforcement entities.
- q. In the event the Discharger does not comply or will be unable to comply for any reason, with any prohibition, average monthly effluent limitations, average weekly effluent limitation, maximum daily effluent limitation, or receiving water limitation of this Order, the Discharger shall notify the Central Valley Water Board by telephone at (916) 464-3291 within 24 hours of having knowledge of such noncompliance, and shall confirm this notification in writing within five days, unless the Central Valley Water Board waives confirmation. The written notification shall state the nature, time, duration, and cause of noncompliance, and shall describe the measures being taken to remedy the current noncompliance and prevent recurrence including, where applicable, a schedule of implementation. Other noncompliance requires written notification as above at the time of the normal monitoring report.

## **B. Monitoring and Reporting Program (MRP) Requirements**

The Discharger shall comply with the MRP, and future revisions thereto, in Attachment E.

## **C. Special Provisions**

### **1. Reopener Provisions**

- a. Conditions that necessitate a major modification of a permit are described in 40 CFR section 122.62, including, but not limited to:
  - i. If new or amended applicable water quality standards are promulgated or approved pursuant to section 303 of the CWA, or amendments thereto, this permit may be reopened and modified in accordance with the new or amended standards.

- ii. When new information, that was not available at the time of permit issuance, would have justified different permit conditions at the time of issuance.
- b. This Order may be reopened for modification, or revocation and reissuance, as a result of the detection of a reportable priority pollutant generated by special conditions included in this Order. These special conditions may be, but are not limited to, fish tissue sampling, whole effluent toxicity, monitoring requirements on internal waste stream(s), and monitoring for surrogate parameters. Additional requirements may be included in this Order as a result of the special condition monitoring data.
- c. **Mercury.** If mercury is found to be causing toxicity based on acute or chronic toxicity test results, or if a TMDL program is adopted, this Order shall be reopened and the mass effluent limitation modified (higher or lower) or an effluent concentration limitation imposed. If the Central Valley Water Board determines that a mercury offset program is feasible for Dischargers subject to a NPDES permit, then this Order may be reopened to reevaluate the mercury mass loading limitation(s) and the need for a mercury offset program for the Discharger.
- d. **Whole Effluent Toxicity.** As a result of a Toxicity Reduction Evaluation (TRE), this Order may be reopened to include a numeric chronic toxicity limitation, a new acute toxicity limitation, and/or a limitation for a specific toxicant identified in the TRE. Additionally, if the State Water Board revises the SIP's toxicity control provisions that would require the establishment of numeric chronic toxicity effluent limitations, this Order may be reopened to include a numeric chronic toxicity effluent limitation based on the new provisions.
- e. **Water Effects Ratios (WER) and Metal Translators.** A default WER of 1.0 has been used in this Order for calculating criteria for applicable inorganic constituents. In addition, default dissolved-to-total metal translators have been used to convert water quality objectives from dissolved to total recoverable when developing effluent limitations. If the Discharger performs studies to determine site-specific WERs and/or site-specific dissolved-to-total metal translators, this Order may be reopened to modify the effluent limitations for the applicable inorganic constituents.
- f. **Drinking Water Policy.** On 26 July 2013 the Central Valley Water Board adopted Resolution No. R5-2013-0098 amending the Basin Plan and establishing a Drinking Water Policy. The State Water Board approved the Drinking Water Policy on 3 December 2013. This Order may be reopened to incorporate monitoring of drinking water constituents to implement the Drinking Water Policy.
- g. **Ultraviolet (UV) Disinfection Operating Specifications.** The UV operating specifications in this Order are based on the UV guidelines developed by the National Water Research Institute and American Water Works Association Research Foundation titled, "Ultraviolet Disinfection Guidelines for Drinking Water and Water Reuse." If the Discharger conducts a site-specific UV engineering study that identifies site-specific UV operating specifications that will achieve the virus inactivation equivalent to Title 22 disinfected tertiary recycled water, this Order may be reopened to modify the UV operating specifications.
- h. **Facility Re-Rating.** The Discharger is constructing facility improvements to improve nitrogen removal that are scheduled to be complete in 2016. The upgraded facility is expected to increase the design capacity of the Facility. If the Discharger conducts a Facility Re-Rating study that demonstrates the upgraded design capacity has increased, this Order may be reopened to modify the design average dry weather flow effluent limitations, as appropriate.

## 2. Special Studies, Technical Reports and Additional Monitoring Requirements

- a. **Toxicity Reduction Evaluation Requirements.** For compliance with the Basin Plan's narrative toxicity objective, this Order requires the Discharger to conduct chronic whole effluent toxicity (WET) testing, as specified in MRP section V. Furthermore, this Provision requires the Discharger to investigate the causes of, and identify corrective actions to reduce or eliminate effluent toxicity. If the discharge exceeds the numeric toxicity monitoring trigger during accelerated monitoring established in this Provision, the Discharger is required to initiate a Toxicity Reduction Evaluation (TRE) in accordance with an approved TRE Work Plan, and take actions to mitigate the impact of the discharge and prevent recurrence of toxicity. A TRE is a site-specific study conducted in a stepwise process to identify the source(s) of toxicity and the effective control measures for effluent toxicity. TREs are designed to identify the causative agents and sources of whole effluent toxicity, evaluate the effectiveness of the toxicity control options, and confirm the reduction in effluent toxicity. The Discharger currently has an approved TRE Work Plan in place. This Provision includes procedures for accelerated chronic toxicity monitoring and TRE initiation.
- i. **Accelerated Monitoring and TRE Initiation.** When the numeric toxicity monitoring trigger is exceeded during regular chronic toxicity monitoring, and the testing meets all test acceptability criteria, the Discharger shall initiate accelerated monitoring as required in the Accelerated Monitoring Specifications. The Discharger shall initiate a TRE to address effluent toxicity if any WET testing results exceed the numeric toxicity monitoring trigger during accelerated monitoring.
- ii. **Numeric Toxicity Monitoring Trigger.** The numeric toxicity monitoring trigger to initiate a TRE is  $> 1 \text{ TUc}$  (where  $\text{TUc} = 100/\text{NOEC}$ ). The monitoring trigger is not an effluent limitation; it is the toxicity threshold at which the Discharger is required to begin accelerated monitoring and initiate a TRE.
- iii. **Accelerated Monitoring Specifications.** If the numeric toxicity monitoring trigger is exceeded during regular chronic toxicity testing, the Discharger shall initiate accelerated monitoring within 14-days of notification by the laboratory of the exceedance. Accelerated monitoring shall consist of four chronic toxicity tests conducted once every two weeks using the species that exhibited toxicity. The following protocol shall be used for accelerated monitoring and TRE initiation:
- (a) If the results of four consecutive accelerated monitoring tests do not exceed the monitoring trigger, the Discharger may cease accelerated monitoring and resume regular chronic toxicity monitoring. However, notwithstanding the accelerated monitoring results, if there is adequate evidence of a pattern of effluent toxicity, the Executive Officer may require that the Discharger initiate a TRE.
- (b) If the source(s) of the toxicity is easily identified (e.g., temporary plant upset), the Discharger shall make necessary corrections to the facility and shall continue accelerated monitoring until four consecutive accelerated tests do not exceed the monitoring trigger. Upon confirmation that the effluent toxicity has been removed, the Discharger may cease accelerated monitoring and resume regular chronic toxicity monitoring.
- (c) If the result of any accelerated toxicity test exceeds the monitoring trigger, the Discharger shall cease accelerated monitoring and begin a TRE to

investigate the cause(s) of, and identify corrective actions to reduce or eliminate effluent toxicity. Within thirty (30) days of notification by the laboratory of any test result exceeding the monitoring trigger during accelerated monitoring, the Discharger shall submit a TRE Action Plan to the Central Valley Water Board including, at minimum:

- (1) Specific actions the Discharger will take to investigate and identify the cause(s) of toxicity, including a TRE WET monitoring schedule;
- (2) Specific actions the Discharger will take to mitigate the impact of the discharge and prevent the recurrence of toxicity; and
- (3) A schedule for these actions.

### **3. Best Management Practices and Pollution Prevention**

- a. **Salinity Evaluation and Minimization Plan.** The Discharger shall continue to implement a salinity evaluation and minimization plan to identify and address sources of salinity from the Facility. The Discharger shall evaluate the effectiveness of the salinity evaluation and minimization plan and provide a summary with the Report of Waste Discharge, due 180 days prior to the permit expiration date.

### **4. Construction, Operation and Maintenance Specifications**

- a. **Filtration System Operating Specifications.** To ensure the filtration system is operating properly to provide adequate disinfection of the wastewater, the turbidity of the filter effluent measured at Monitoring Location FIL-001 shall not exceed:
  - i. 2 NTU as a daily average;
  - ii. 5 NTU more than 5 percent of the time within a 24-hour period; and
  - iii. 10 NTU, at any time.
- b. **Ultraviolet (UV) Disinfection System Operating Specifications.** The UV disinfection system must be operated in accordance with an operations and maintenance program that assures adequate disinfection, and shall meet the following minimum specifications to provide virus inactivation equivalent to Title 22 Disinfected Tertiary Recycled Water:
  - i. **UV Dose.** The minimum hourly average UV dose in the UV reactor shall be 100 millijoules per square centimeter ( $\text{mJ}/\text{cm}^2$ ).
  - ii. **UV Transmittance.** The minimum hourly average UV transmittance (at 254 nanometers) in the wastewater measured at UVS-001 shall not fall below 55 percent.
  - iii. The lamp sleeves and cleaning system components must be visually inspected per the manufacturer's operations manual for physical wear (scoring, solarization, seal leaks, cleaning fluid levels, etc.) and to check the efficacy of the cleaning system.
  - iv. The lamp sleeves must be cleaned periodically as necessary to meet the UV dose requirements.
  - v. Lamps must be replaced per the manufacturer's operations manual, or sooner, if there are indications the lamps are failing to provide adequate disinfection. Lamp age and lamp replacement records must be maintained.

## 5. Special Provisions for Municipal Facilities (POTW's Only)

- a. **Collection System.** On 2 May 2006, the State Water Board adopted State Water Board Order No. 2006-0003-DWQ, Statewide General WDR's for Sanitary Sewer Systems. The Discharger shall be subject to the requirements of Order No. 2006-0003-DWQ and any future revisions thereto. Order No. 2006-0003-DWQ requires that all public agencies that currently own or operate sanitary sewer systems apply for coverage under the general WDR's. The Discharger has applied for and has been approved for coverage under Order 2006-0003-DWQ for operation of its wastewater collection system.

## 6. Other Special Provisions

- a. **Title 22, or Equivalent, Disinfection Requirements.** When discharging to surface water, wastewater shall be oxidized, coagulated, filtered, and adequately disinfected pursuant to the State Water Board, Division of Drinking Water (DDW) reclamation criteria, CCR, Title 22, division 4, chapter 3, (Title 22), or equivalent .
- b. **Permitted Discharge Increase (4.5 MGD).** The Discharger has requested to be permitted to discharge up to 4.5 MGD average dry weather flow year round to Laguna Creek upon completion of Facility upgrades. The permitted Average Dry Weather Flow at Discharge Point 001 may increase to 4.5 MGD upon compliance with the following conditions:
  - i. **Facility Improvements.** The Discharger shall have completed construction and startup of the Facility improvements, as identified in section II.A of the Fact Sheet in this Order.
  - ii. **Effluent Limitation and Receiving Water Compliance.** The discharge shall consistently comply with Final Effluent Limitations IV.A.1 and Receiving Water Limitations V.A. The Discharger shall provide evidence, certified by a licensed professional engineer, that the plant is operating properly.
  - iii. **Request for Increase.** The Discharger shall notify the Executive Officer of its compliance with item i and ii. above. The increase in permitted average dry weather flow to 4.5 MGD shall not be effective until the Executive Officer verifies compliance with Special Provisions VI.C.6.b.

## 7. Compliance Schedules – Not Applicable

## VII. COMPLIANCE DETERMINATION

- A. **BOD<sub>5</sub> and TSS Effluent Limitations (Sections IV.A.1.a-b).** Compliance with the final effluent limitations for BOD<sub>5</sub> and TSS required in Limitations and Discharge Requirements section IV.A.1.a shall be ascertained by 24-hour composite samples. Compliance with effluent limitations required in Limitations and Discharge Requirements section IV.A.1.b for percent removal shall be calculated using the arithmetic mean of BOD<sub>5</sub> and TSS in effluent samples collected over a monthly period as a percentage of the arithmetic mean of the values for influent samples collected at approximately the same times during the same period.
- B. **Total Mercury Mass Loading Effluent Limitations (Section IV.A.1.f).** The procedures for calculating mass loadings are as follows:
  1. The total pollutant mass load for each individual calendar month shall be determined using an average of all concentration data collected that month and the corresponding total monthly flow. All effluent monitoring data collected under the monitoring and reporting program, pretreatment program, and any special studies shall be used for

these calculations. The total annual mass loading shall be the sum of the individual calendar months.

2. In calculating compliance, the Discharger shall count all non-detect measures at one-half of the detection level. If compliance with the effluent limitation is not attained due to the non-detect contribution, the Discharger shall improve and implement available analytical capabilities and compliance shall be evaluated with consideration of the detection limits.
- C. Average Dry Weather Flow Effluent Limitations (Section IV.A.1.e).** The average dry weather discharge flow represents the daily average flow when groundwater is at or near normal and runoff is not occurring. Compliance with the average dry weather flow effluent limitations will be determined annually based on the average daily flow over three consecutive dry weather months (e.g., July, August, and September).
- D. Total Coliform Organisms Effluent Limitations (Section IV.A.1.d).** For each day that an effluent sample is collected and analyzed for total coliform organisms, the 7-day median shall be determined by calculating the median concentration of total coliform bacteria in the effluent utilizing the bacteriological results of the last 7 days. For example, if a sample is collected on a Wednesday, the result from that sampling event and all results from the previous 6 days (i.e., Tuesday, Monday, Sunday, Saturday, Friday, and Thursday) are used to calculate the 7-day median. If the 7-day median of total coliform organisms exceeds a most probable number (MPN) of 2.2 per 100 milliliters, the Discharger will be considered out of compliance.
- E. Mass Effluent Limitations.** The mass effluent limitations contained in the Final Effluent Limitations IV.A.1.a are based on the permitted average dry weather flow and calculated as follows:
- Mass (lbs/day) = Flow (MGD) x Concentration (mg/L) x 8.34 (conversion factor)
- If the effluent flow exceeds the permitted average dry weather flow during wet-weather seasons, the effluent mass limitations contained in Final Effluent Limitations IV.A.1.a shall not apply. If the effluent flow is below the permitted average dry weather flow during wet-weather seasons, the effluent mass limitations do apply.
- F. Priority Pollutant Effluent Limitations.** Compliance with effluent limitations for priority pollutants shall be determined in accordance with Section 2.4.5 of the SIP, as follows:
1. Dischargers shall be deemed out of compliance with an effluent limitation, if the concentration of the priority pollutant in the monitoring sample is greater than the effluent limitation and greater than or equal to the reporting level (RL).
  2. Dischargers shall be required to conduct a Pollutant Minimization Program (PMP) in accordance with section 2.4.5.1 of the SIP when there is evidence that the priority pollutant is present in the effluent above an effluent limitation and either:
    - a. A sample result is reported as detected, but not quantified (DNQ) and the effluent limitation is less than the RL; or
    - b. A sample result is reported as non-detect (ND) and the effluent limitation is less than the method detection limit (MDL).
  3. When determining compliance with an average monthly effluent limitation (AMEL) and more than one sample result is available in a month, the discharger shall compute the arithmetic mean unless the data set contains one or more reported determinations of DNQ or ND. In those cases, the discharger shall compute the median in place of the arithmetic mean in accordance with the following procedure:



- a. The data set shall be ranked from low to high, reported ND determinations lowest, DNQ determinations next, followed by quantified values (if any). The order of the individual ND or DNQ determinations is unimportant.
    - b. The median value of the data set shall be determined. If the data set has an odd number of data points, then the median is the middle value. If the data set has an even number of data points, then the median is the average of the two values around the middle unless one or both of the points are ND or DNQ, in which case the median value shall be the lower of the two data points where DNQ is lower than a value and ND is lower than DNQ.
  4. If a sample result, or the arithmetic mean or median of multiple sample results, is below the RL, and there is evidence that the priority pollutant is present in the effluent above an effluent limitation and the discharger conducts a PMP (as described in section 2.4.5.1), the discharger shall not be deemed out of compliance.
- G. Use of Regional Monitoring Program and other Receiving Water Data to determine compliance with Receiving Water Limitations.** Regional Monitoring Program data and other receiving water monitoring data that is not specifically required to be conducted by the Discharger under this permit, will not be used directly to determine that the discharge is in violation of this Permit. The Discharger may, however, conduct any site-specific receiving water monitoring deemed appropriate by the Discharger that is not conducted by the Delta RMP and submit that monitoring data. As described in Section VIII of Attachment E, such data may be used, if scientifically defensible, in conjunction with other receiving water data, effluent data, receiving water flow data, and other pertinent information to determine whether or not a discharge is in compliance with this Permit.
- H. Dissolved Oxygen Receiving Water Limitation (Section V.A.5.a-c).** The Facility provides a high level of treatment including tertiary filtration and nitrification/de-nitrification, which results in minimal dissolved oxygen impacts in the receiving water. Weekly dissolved oxygen receiving water monitoring is required in the Monitoring and Reporting Program (Attachment E) and is sufficient to evaluate the impacts of the discharge and compliance with this Order. Weekly receiving water monitoring data, measured at monitoring locations RSW-001 and RSW-002, will be used to determine compliance with part "c" of the dissolved oxygen receiving water limitation to ensure the discharge does not cause the dissolved oxygen concentrations in Laguna Creek to be reduced below 7.0 mg/L at any time. However, should more frequent dissolved oxygen and temperature receiving water monitoring be conducted to result in a minimum of six data points, Central Valley Water Board staff may evaluate compliance with parts "a" and "b". Furthermore, if dissolved oxygen concentrations measured upstream at RSW-001 AND downstream at RSW-001 are below the receiving water limit of 7.0 mg/L, the Facility will not be considered to have caused the receiving water to violate the receiving water limitation.

## ATTACHMENT A – DEFINITIONS

### **Arithmetic Mean ( $\mu$ )**

Also called the average, is the sum of measured values divided by the number of samples. For ambient water concentrations, the arithmetic mean is calculated as follows:

Arithmetic mean =  $\mu = \Sigma x / n$       where:  $\Sigma x$  is the sum of the measured ambient water concentrations, and  $n$  is the number of samples.

### **Average Monthly Effluent Limitation (AMEL)**

The highest allowable average of daily discharges over a calendar month, calculated as the sum of all daily discharges measured during a calendar month divided by the number of daily discharges measured during that month.

### **Average Weekly Effluent Limitation (AWEL)**

The highest allowable average of daily discharges over a calendar week (Sunday through Saturday), calculated as the sum of all daily discharges measured during a calendar week divided by the number of daily discharges measured during that week.

### **Bioaccumulative**

Those substances taken up by an organism from its surrounding medium through gill membranes, epithelial tissue, or from food and subsequently concentrated and retained in the body of the organism.

### **Carcinogenic**

Pollutants are substances that are known to cause cancer in living organisms.

### **Coefficient of Variation (CV)**

CV is a measure of the data variability and is calculated as the estimated standard deviation divided by the arithmetic mean of the observed values.

### **Daily Discharge**

Daily Discharge is defined as either: (1) the total mass of the constituent discharged over the calendar day (12:00 am through 11:59 pm) or any 24-hour period that reasonably represents a calendar day for purposes of sampling (as specified in the permit), for a constituent with limitations expressed in units of mass or; (2) the unweighted arithmetic mean measurement of the constituent over the day for a constituent with limitations expressed in other units of measurement (e.g., concentration).

The daily discharge may be determined by the analytical results of a composite sample taken over the course of one day (a calendar day or other 24-hour period defined as a day) or by the arithmetic mean of analytical results from one or more grab samples taken over the course of the day.

For composite sampling, if 1 day is defined as a 24-hour period other than a calendar day, the analytical result for the 24-hour period will be considered as the result for the calendar day in which the 24-hour period ends.

### **Detected, but Not Quantified (DNQ)**

DNQ are those sample results less than the RL, but greater than or equal to the laboratory's MDL. Sample results reported as DNQ are estimated concentrations.

### **Dilution Credit**

Dilution Credit is the amount of dilution granted to a discharge in the calculation of a water quality-based effluent limitation, based on the allowance of a specified mixing zone. It is calculated from the

dilution ratio or determined through conducting a mixing zone study or modeling of the discharge and receiving water.

#### **Effluent Concentration Allowance (ECA)**

ECA is a value derived from the water quality criterion/objective, dilution credit, and ambient background concentration that is used, in conjunction with the coefficient of variation for the effluent monitoring data, to calculate a long-term average (LTA) discharge concentration. The ECA has the same meaning as waste load allocation (WLA) as used in U.S. EPA guidance (Technical Support Document For Water Quality-based Toxics Control, March 1991, second printing, EPA/505/2-90-001).

#### **Enclosed Bays**

Enclosed Bays means indentations along the coast that enclose an area of oceanic water within distinct headlands or harbor works. Enclosed bays include all bays where the narrowest distance between the headlands or outermost harbor works is less than 75 percent of the greatest dimension of the enclosed portion of the bay. Enclosed bays include, but are not limited to, Humboldt Bay, Bodega Harbor, Tomales Bay, Drake's Estero, San Francisco Bay, Morro Bay, Los Angeles-Long Beach Harbor, Upper and Lower Newport Bay, Mission Bay, and San Diego Bay. Enclosed bays do not include inland surface waters or ocean waters.

#### **Estimated Chemical Concentration**

The estimated chemical concentration that results from the confirmed detection of the substance by the analytical method below the ML value.

#### **Estuaries**

Estuaries means waters, including coastal lagoons, located at the mouths of streams that serve as areas of mixing for fresh and ocean waters. Coastal lagoons and mouths of streams that are temporarily separated from the ocean by sandbars shall be considered estuaries. Estuarine waters shall be considered to extend from a bay or the open ocean to a point upstream where there is no significant mixing of fresh water and seawater. Estuarine waters included, but are not limited to, the Sacramento-San Joaquin Delta, as defined in Water Code section 12220, Suisun Bay, Carquinez Strait downstream to the Carquinez Bridge, and appropriate areas of the Smith, Mad, Eel, Noyo, Russian, Klamath, San Diego, and Otay rivers. Estuaries do not include inland surface waters or ocean waters.

#### **Inland Surface Waters**

All surface waters of the state that do not include the ocean, enclosed bays, or estuaries.

#### **Instantaneous Maximum Effluent Limitation**

The highest allowable value for any single grab sample or aliquot (i.e., each grab sample or aliquot is independently compared to the instantaneous maximum limitation).

#### **Instantaneous Minimum Effluent Limitation**

The lowest allowable value for any single grab sample or aliquot (i.e., each grab sample or aliquot is independently compared to the instantaneous minimum limitation).

#### **Maximum Daily Effluent Limitation (MDEL)**

The highest allowable daily discharge of a pollutant, over a calendar day (or 24-hour period). For pollutants with limitations expressed in units of mass, the daily discharge is calculated as the total mass of the pollutant discharged over the day. For pollutants with limitations expressed in other units of measurement, the daily discharge is calculated as the arithmetic mean measurement of the pollutant over the day.

### **Median**

The middle measurement in a set of data. The median of a set of data is found by first arranging the measurements in order of magnitude (either increasing or decreasing order). If the number of measurements ( $n$ ) is odd, then the median =  $X_{(n+1)/2}$ . If  $n$  is even, then the median =  $(X_{n/2} + X_{(n/2)+1})/2$  (i.e., the midpoint between the  $n/2$  and  $n/2+1$ ).

### **Method Detection Limit (MDL)**

MDL is the minimum concentration of a substance that can be measured and reported with 99 percent confidence that the analyte concentration is greater than zero, as defined in 40 C.F.R. part 136, Attachment B, revised as of July 3, 1999.

### **Minimum Level (ML)**

ML is the concentration at which the entire analytical system must give a recognizable signal and acceptable calibration point. The ML is the concentration in a sample that is equivalent to the concentration of the lowest calibration standard analyzed by a specific analytical procedure, assuming that all the method specified sample weights, volumes, and processing steps have been followed.

### **Mixing Zone**

Mixing Zone is a limited volume of receiving water that is allocated for mixing with a wastewater discharge where water quality criteria can be exceeded without causing adverse effects to the overall water body.

### **Not Detected (ND)**

Sample results which are less than the laboratory's MDL.

### **Ocean Waters**

The territorial marine waters of the State as defined by California law to the extent these waters are outside of enclosed bays, estuaries, and coastal lagoons. Discharges to ocean waters are regulated in accordance with the State Water Board's California Ocean Plan.

### **Persistent Pollutants**

Persistent pollutants are substances for which degradation or decomposition in the environment is nonexistent or very slow.

### **Pollutant Minimization Program (PMP)**

PMP means waste minimization and pollution prevention actions that include, but are not limited to, product substitution, waste stream recycling, alternative waste management methods, and education of the public and businesses. The goal of the PMP shall be to reduce all potential sources of a priority pollutant(s) through pollutant minimization (control) strategies, including pollution prevention measures as appropriate, to maintain the effluent concentration at or below the water quality-based effluent limitation. Pollution prevention measures may be particularly appropriate for persistent bioaccumulative priority pollutants where there is evidence that beneficial uses are being impacted. The Central Valley Water Board may consider cost effectiveness when establishing the requirements of a PMP. The completion and implementation of a Pollution Prevention Plan, if required pursuant to Water Code section 13263.3(d), shall be considered to fulfill the PMP requirements.

### **Pollution Prevention**

Pollution Prevention means any action that causes a net reduction in the use or generation of a hazardous substance or other pollutant that is discharged into water and includes, but is not limited to, input change, operational improvement, production process change, and product reformulation (as defined in Water Code section 13263.3). Pollution prevention does not include actions that merely shift a pollutant in wastewater from one environmental medium to another environmental medium, unless

clear environmental benefits of such an approach are identified to the satisfaction of the State Water Resources Control Board (State Water Board) or Central Valley Water Board.

### **Satellite Collection System**

The portion, if any, of a sanitary sewer system owned or operated by a different public agency than the agency that owns and operates the wastewater treatment facility that a sanitary sewer system is tributary to.

### **Source of Drinking Water**

Any water designated as municipal or domestic supply (MUN) in a Central Valley Water Board Basin Plan.

### **Standard Deviation ( $\sigma$ )**

Standard Deviation is a measure of variability that is calculated as follows:

$$\sigma = (\sum[(x - \mu)^2]/(n - 1))^{0.5}$$

where:

x is the observed value;

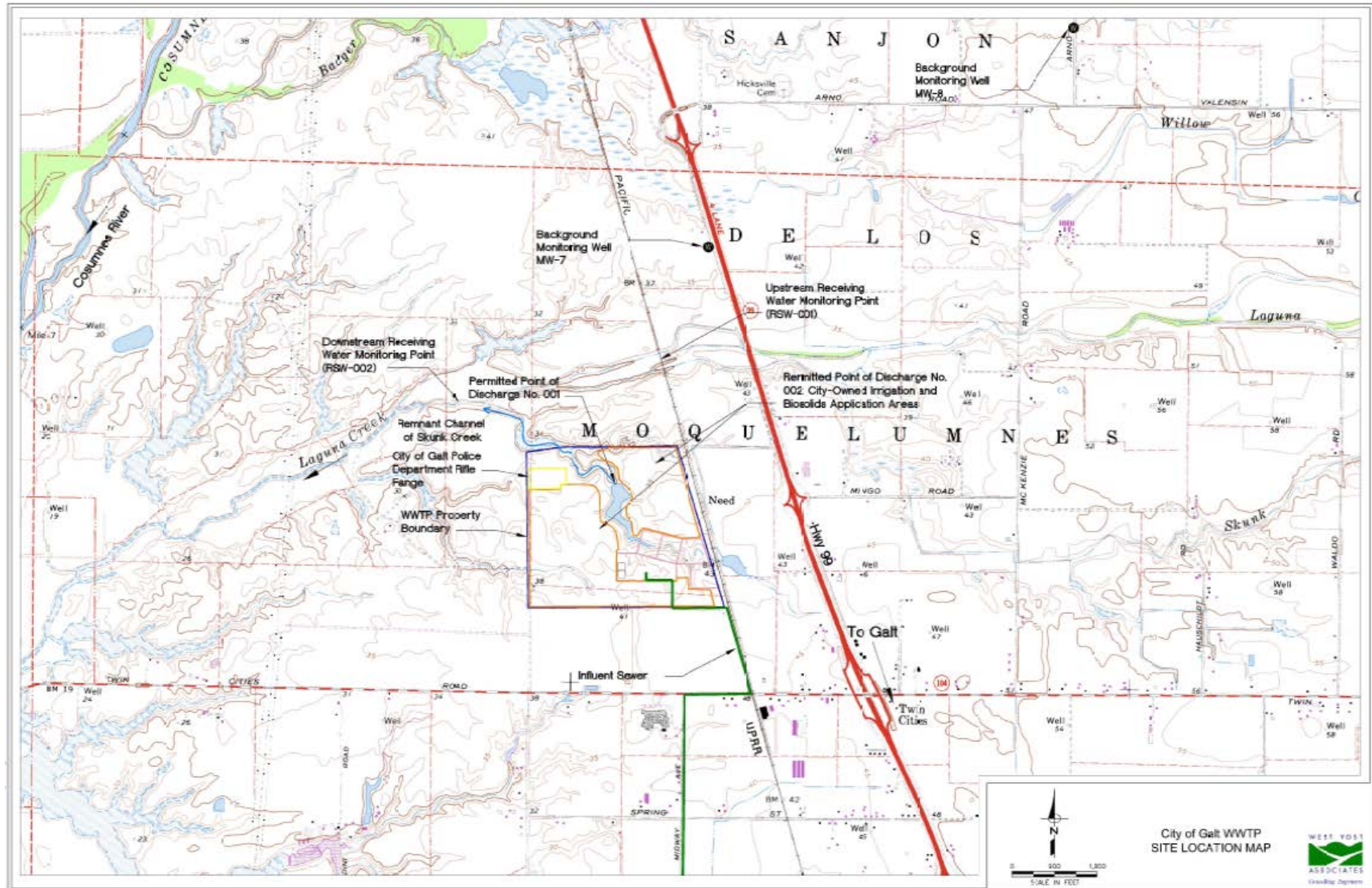
$\mu$  is the arithmetic mean of the observed values; and

n is the number of samples.

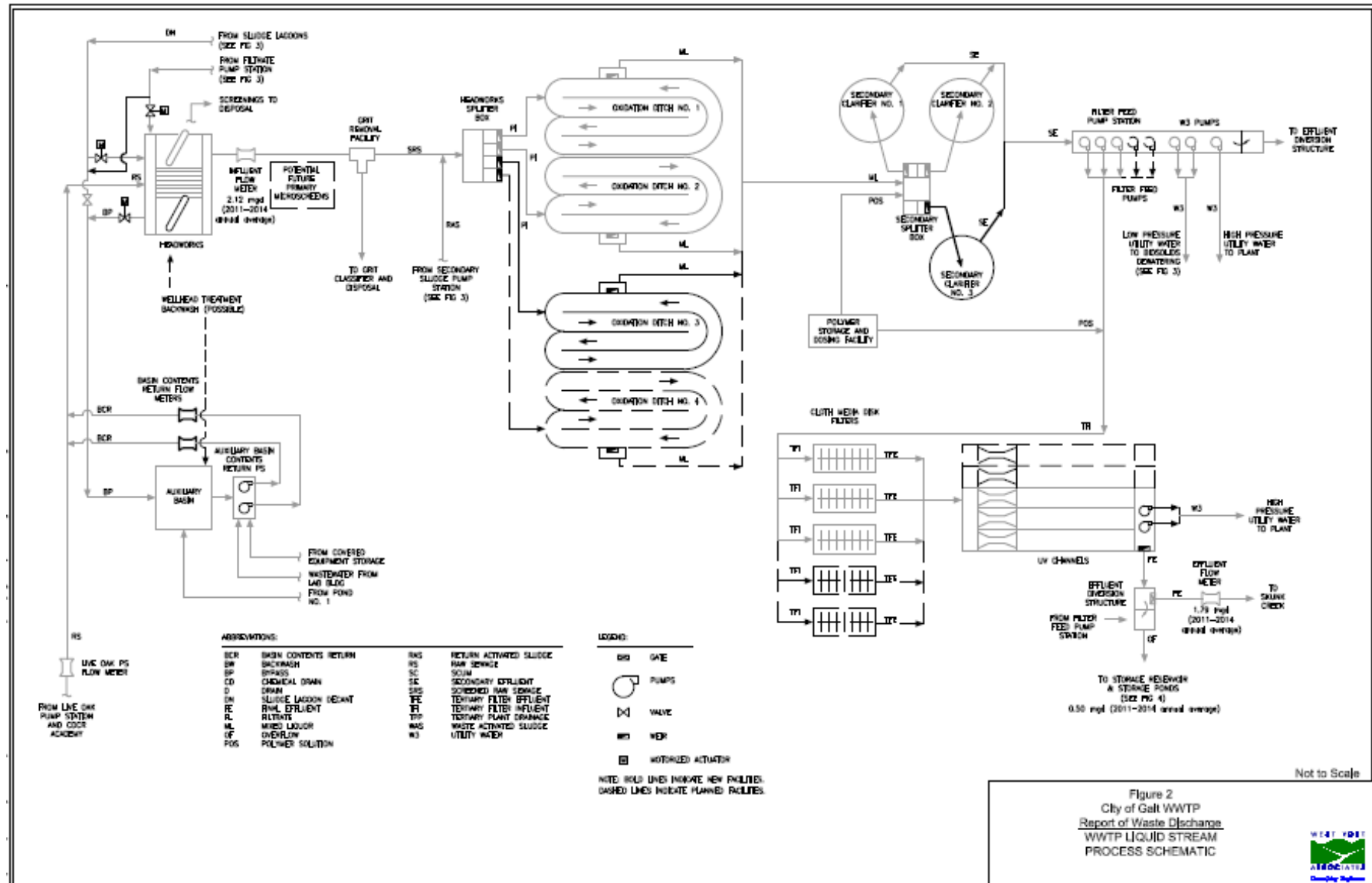
### **Toxicity Reduction Evaluation (TRE)**

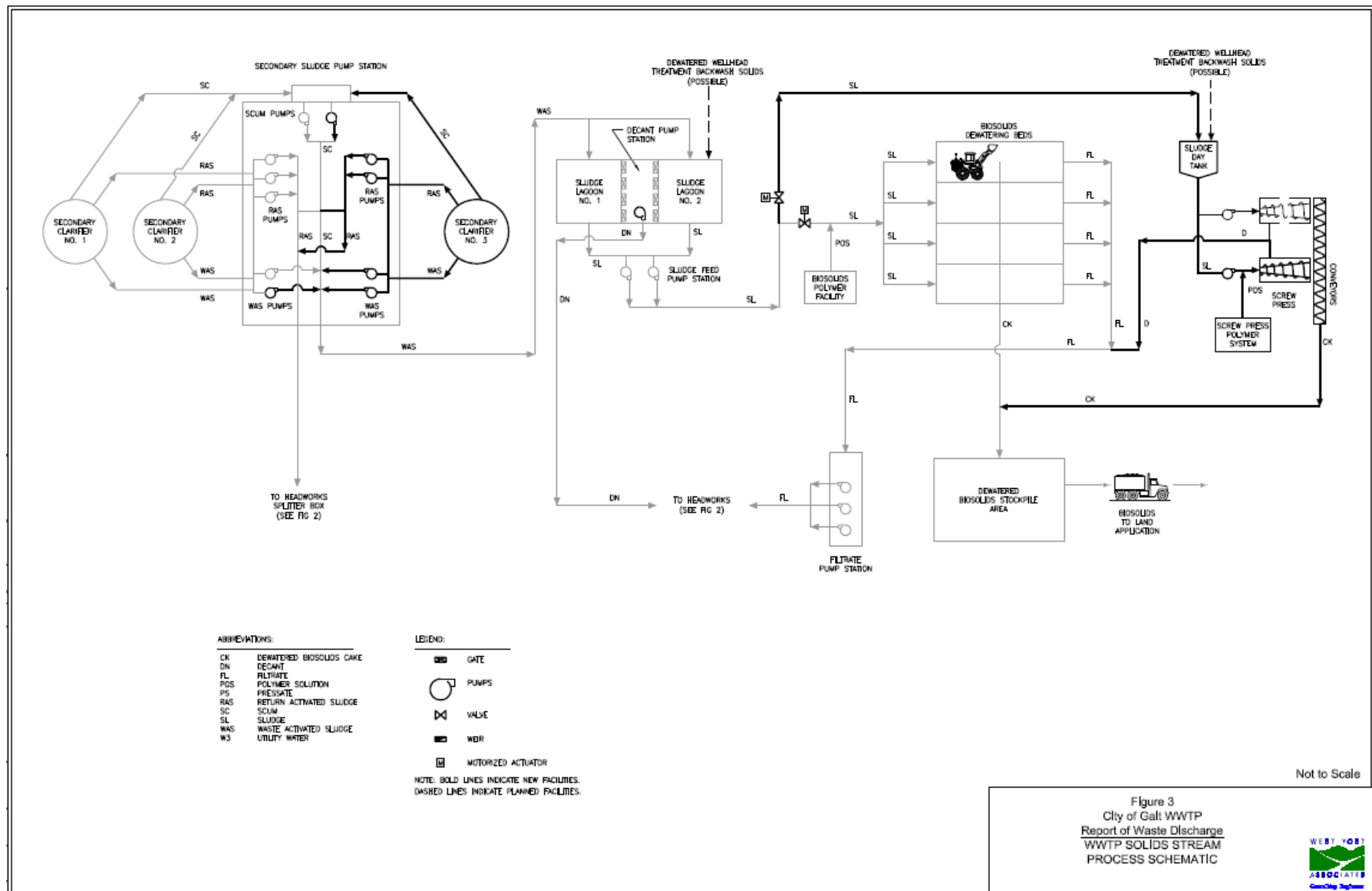
TRE is a study conducted in a step-wise process designed to identify the causative agents of effluent or ambient toxicity, isolate the sources of toxicity, evaluate the effectiveness of toxicity control options, and then confirm the reduction in toxicity. The first steps of the TRE consist of the collection of data relevant to the toxicity, including additional toxicity testing, and an evaluation of facility operations and maintenance practices, and best management practices. A Toxicity Identification Evaluation (TIE) may be required as part of the TRE, if appropriate. (A TIE is a set of procedures to identify the specific chemical(s) responsible for toxicity. These procedures are performed in three phases (characterization, identification, and confirmation) using aquatic organism toxicity tests.)

ATTACHMENT B – MAP



# ATTACHMENT C – FLOW SCHEMATIC







## **ATTACHMENT D – STANDARD PROVISIONS**

### **I. STANDARD PROVISIONS – PERMIT COMPLIANCE**

#### **A. Duty to Comply**

1. The Discharger must comply with all of the terms, requirements, and conditions of this Order. Any noncompliance constitutes a violation of the Clean Water Act (CWA) and the California Water Code and is grounds for enforcement action; permit termination, revocation and reissuance, or modification; denial of a permit renewal application; or a combination thereof. (40 C.F.R. § 122.41(a); Wat. Code, §§ 13261, 13263, 13265, 13268, 13000, 13001, 13304, 13350, 13385.)
2. The Discharger shall comply with effluent standards or prohibitions established under Section 307(a) of the CWA for toxic pollutants and with standards for sewage sludge use or disposal established under Section 405(d) of the CWA within the time provided in the regulations that establish these standards or prohibitions, even if this Order has not yet been modified to incorporate the requirement. (40 C.F.R. § 122.41(a)(1).)

#### **B. Need to Halt or Reduce Activity Not a Defense**

It shall not be a defense for a Discharger in an enforcement action that it would have been necessary to halt or reduce the permitted activity in order to maintain compliance with the conditions of this Order. (40 C.F.R. § 122.41(c).)

#### **C. Duty to Mitigate**

The Discharger shall take all reasonable steps to minimize or prevent any discharge or sludge use or disposal in violation of this Order that has a reasonable likelihood of adversely affecting human health or the environment. (40 C.F.R. § 122.41(d).)

#### **D. Proper Operation and Maintenance**

The Discharger shall at all times properly operate and maintain all facilities and systems of treatment and control (and related appurtenances) which are installed or used by the Discharger to achieve compliance with the conditions of this Order. Proper operation and maintenance also includes adequate laboratory controls and appropriate quality assurance procedures. This provision requires the operation of backup or auxiliary facilities or similar systems that are installed by a Discharger only when necessary to achieve compliance with the conditions of this Order. (40 C.F.R. § 122.41(e).)

#### **E. Property Rights**

1. This Order does not convey any property rights of any sort or any exclusive privileges. (40 C.F.R. § 122.41(g).)
2. The issuance of this Order does not authorize any injury to persons or property or invasion of other private rights, or any infringement of state or local law or regulations. (40 C.F.R. § 122.5(c).)

#### **F. Inspection and Entry**

The Discharger shall allow the Central Valley Water Board, State Water Board, U.S. EPA, and/or their authorized representatives (including an authorized contractor acting as their representative), upon the presentation of credentials and other documents, as may be required by law, to (33 U.S.C. § 1318(a)(4)(B); 40 C.F.R. § 122.41(i); Wat. Code, § 13267, 13383):

1. Enter upon the Discharger's premises where a regulated facility or activity is located or conducted, or where records are kept under the conditions of this Order (33 U.S.C § 1318(a)(4)(B)(ii); 40 C.F.R. § 122.41(i)(1); Wat. Code, §§ 13267, 13383);
2. Have access to and copy, at reasonable times, any records that must be kept under the conditions of this Order (33 U.S.C. § 1318(a)(4)(B)(ii); 40 C.F.R. § 122.41(i)(2); Wat. Code, §§ 13267, 13383);
3. Inspect and photograph, at reasonable times, any facilities, equipment (including monitoring and control equipment), practices, or operations regulated or required under this Order (33 U.S.C § 1318(a)(4)(B)(ii); 40 C.F.R. § 122.41(i)(3); Wat. Code, § 13267, 13383); and
4. Sample or monitor, at reasonable times, for the purposes of assuring Order compliance or as otherwise authorized by the CWA or the Water Code, any substances or parameters at any location. (33 U.S.C § 1318(a)(4)(B); 40 C.F.R. § 122.41(i)(4); Wat. Code, §§ 13267, 13383.)

#### **G. Bypass**

1. Definitions
  - a. "Bypass" means the intentional diversion of waste streams from any portion of a treatment facility. (40 C.F.R. § 122.41(m)(1)(i).)
  - b. "Severe property damage" means substantial physical damage to property, damage to the treatment facilities, which causes them to become inoperable, or substantial and permanent loss of natural resources that can reasonably be expected to occur in the absence of a bypass. Severe property damage does not mean economic loss caused by delays in production. (40 C.F.R. § 122.41(m)(1)(ii).)
2. Bypass not exceeding limitations. The Discharger may allow any bypass to occur which does not cause exceedances of effluent limitations, but only if it is for essential maintenance to assure efficient operation. These bypasses are not subject to the provisions listed in Standard Provisions – Permit Compliance I.G.3, I.G.4, and I.G.5 below. (40 C.F.R. § 122.41(m)(2).)
3. Prohibition of bypass. Bypass is prohibited, and the Central Valley Water Board may take enforcement action against a Discharger for bypass, unless (40 C.F.R. § 122.41(m)(4)(i)):
  - a. Bypass was unavoidable to prevent loss of life, personal injury, or severe property damage (40 C.F.R. § 122.41(m)(4)(i)(A));
  - b. There were no feasible alternatives to the bypass, such as the use of auxiliary treatment facilities, retention of untreated wastes, or maintenance during normal periods of equipment downtime. This condition is not satisfied if adequate back-up equipment should have been installed in the exercise of reasonable engineering judgment to prevent a bypass that occurred during normal periods of equipment downtime or preventive maintenance (40 C.F.R. § 122.41(m)(4)(i)(B)); and
  - c. The Discharger submitted notice to the Central Valley Water Board as required under Standard Provisions – Permit Compliance I.G.5 below. (40 C.F.R. § 122.41(m)(4)(i)(C).)
4. The Central Valley Water Board may approve an anticipated bypass, after considering its adverse effects, if the Central Valley Water Board determines that it will meet the three

conditions listed in Standard Provisions – Permit Compliance I.G.3 above. (40 C.F.R. § 122.41(m)(4)(ii).)

5. Notice

- a. Anticipated bypass. If the Discharger knows in advance of the need for a bypass, it shall submit a notice, if possible at least 10 days before the date of the bypass. (40 C.F.R. § 122.41(m)(3)(i).)
- b. Unanticipated bypass. The Discharger shall submit notice of an unanticipated bypass as required in Standard Provisions - Reporting V.E below (24-hour notice). (40 C.F.R. § 122.41(m)(3)(ii).)

**H. Upset**

Upset means an exceptional incident in which there is unintentional and temporary noncompliance with technology based permit effluent limitations because of factors beyond the reasonable control of the Discharger. An upset does not include noncompliance to the extent caused by operational error, improperly designed treatment facilities, inadequate treatment facilities, lack of preventive maintenance, or careless or improper operation. (40 C.F.R. § 122.41(n)(1).)

1. Effect of an upset. An upset constitutes an affirmative defense to an action brought for noncompliance with such technology based permit effluent limitations if the requirements of Standard Provisions – Permit Compliance I.H.2 below are met. No determination made during administrative review of claims that noncompliance was caused by upset, and before an action for noncompliance, is final administrative action subject to judicial review. (40 C.F.R. § 122.41(n)(2).)
2. Conditions necessary for a demonstration of upset. A Discharger who wishes to establish the affirmative defense of upset shall demonstrate, through properly signed, contemporaneous operating logs or other relevant evidence that (40 C.F.R. § 122.41(n)(3)):
  - a. An upset occurred and that the Discharger can identify the cause(s) of the upset (40 C.F.R. § 122.41(n)(3)(i));
  - b. The permitted facility was, at the time, being properly operated (40 C.F.R. § 122.41(n)(3)(ii));
  - c. The Discharger submitted notice of the upset as required in Standard Provisions – Reporting V.E.2.b below (24-hour notice) (40 C.F.R. § 122.41(n)(3)(iii)); and
  - d. The Discharger complied with any remedial measures required under Standard Provisions – Permit Compliance I.C above. (40 C.F.R. § 122.41(n)(3)(iv).)
3. Burden of proof. In any enforcement proceeding, the Discharger seeking to establish the occurrence of an upset has the burden of proof. (40 C.F.R. § 122.41(n)(4).)

**II. STANDARD PROVISIONS – PERMIT ACTION**

**A. General**

This Order may be modified, revoked and reissued, or terminated for cause. The filing of a request by the Discharger for modification, revocation and reissuance, or termination, or a notification of planned changes or anticipated noncompliance does not stay any Order condition. (40 C.F.R. § 122.41(f).)

**B. Duty to Reapply**

If the Discharger wishes to continue an activity regulated by this Order after the expiration date of this Order, the Discharger must apply for and obtain a new permit. (40 C.F.R. § 122.41(b).)

**C. Transfers**

This Order is not transferable to any person except after notice to the Central Valley Water Board. The Central Valley Water Board may require modification or revocation and reissuance of the Order to change the name of the Discharger and incorporate such other requirements as may be necessary under the CWA and the Water Code. (40 C.F.R. § 122.41(l)(3); 122.61.)

**III. STANDARD PROVISIONS – MONITORING**

- A.** Samples and measurements taken for the purpose of monitoring shall be representative of the monitored activity. (40 C.F.R. § 122.41(j)(1).)
- B.** Monitoring results must be conducted according to test procedures approved under 40 C.F.R. part 136 for the analyses of pollutants unless another method is required under 40 C.F.R. subchapters N or O. In the case of pollutants for which there are no approved methods under 40 C.F.R. part 136 or otherwise required under 40 C.F.R. subchapters N or O, monitoring must be conducted according to a test procedure specified in this Order for such pollutants. (40 C.F.R. § 122.41(j)(4); 122.44(i)(1)(iv).)

**IV. STANDARD PROVISIONS – RECORDS**

- A.** Except for records of monitoring information required by this Order related to the Discharger's sewage sludge use and disposal activities, which shall be retained for a period of at least five years (or longer as required by 40 C.F.R. part 503), the Discharger shall retain records of all monitoring information, including all calibration and maintenance records and all original strip chart recordings for continuous monitoring instrumentation, copies of all reports required by this Order, and records of all data used to complete the application for this Order, for a period of at least three (3) years from the date of the sample, measurement, report or application. This period may be extended by request of the Central Valley Water Board Executive Officer at any time. (40 C.F.R. § 122.41(j)(2).)
- B.** Records of monitoring information shall include:
  - 1. The date, exact place, and time of sampling or measurements (40 C.F.R. § 122.41(j)(3)(i));
  - 2. The individual(s) who performed the sampling or measurements (40 C.F.R. § 122.41(j)(3)(ii));
  - 3. The date(s) analyses were performed (40 C.F.R. § 122.41(j)(3)(iii));
  - 4. The individual(s) who performed the analyses (40 C.F.R. § 122.41(j)(3)(iv));
  - 5. The analytical techniques or methods used (40 C.F.R. § 122.41(j)(3)(v)); and
  - 6. The results of such analyses. (40 C.F.R. § 122.41(j)(3)(vi).)
- C.** Claims of confidentiality for the following information will be denied (40 C.F.R. § 122.7(b)):
  - 1. The name and address of any permit applicant or Discharger (40 C.F.R. § 122.7(b)(1)); and
  - 2. Permit applications and attachments, permits and effluent data. (40 C.F.R. § 122.7(b)(2).)

## **V. STANDARD PROVISIONS – REPORTING**

### **A. Duty to Provide Information**

The Discharger shall furnish to the Central Valley Water Board, State Water Board, or U.S. EPA within a reasonable time, any information which the Central Valley Water Board, State Water Board, or U.S. EPA may request to determine whether cause exists for modifying, revoking and reissuing, or terminating this Order or to determine compliance with this Order. Upon request, the Discharger shall also furnish to the Central Valley Water Board, State Water Board, or U.S. EPA copies of records required to be kept by this Order. (40 C.F.R. § 122.41(h); Wat. Code, §§ 13267, 13383.)

### **B. Signatory and Certification Requirements**

1. All applications, reports, or information submitted to the Central Valley Water Board, State Water Board, and/or U.S. EPA shall be signed and certified in accordance with Standard Provisions – Reporting V.B.2, V.B.3, V.B.4, and V.B.5 below. (40 C.F.R. § 122.41(k).)
2. All permit applications shall be signed by either a principal executive officer or ranking elected official. For purposes of this provision, a principal executive officer of a federal agency includes: (i) the chief executive officer of the agency, or (ii) a senior executive officer having responsibility for the overall operations of a principal geographic unit of the agency (e.g., Regional Administrators of U.S. EPA). (40 C.F.R. § 122.22(a)(3).)
3. All reports required by this Order and other information requested by the Central Valley Water Board, State Water Board, or U.S. EPA shall be signed by a person described in Standard Provisions – Reporting V.B.2 above, or by a duly authorized representative of that person. A person is a duly authorized representative only if:
  - a. The authorization is made in writing by a person described in Standard Provisions – Reporting V.B.2 above (40 C.F.R. § 122.22(b)(1));
  - b. The authorization specifies either an individual or a position having responsibility for the overall operation of the regulated facility or activity such as the position of plant manager, operator of a well or a well field, superintendent, position of equivalent responsibility, or an individual or position having overall responsibility for environmental matters for the company. (A duly authorized representative may thus be either a named individual or any individual occupying a named position.) (40 C.F.R. § 122.22(b)(2)); and
  - c. The written authorization is submitted to the Central Valley Water Board and State Water Board. (40 C.F.R. § 122.22(b)(3).)
4. If an authorization under Standard Provisions – Reporting V.B.3 above is no longer accurate because a different individual or position has responsibility for the overall operation of the facility, a new authorization satisfying the requirements of Standard Provisions – Reporting V.B.3 above must be submitted to the Central Valley Water Board and State Water Board prior to or together with any reports, information, or applications, to be signed by an authorized representative. (40 C.F.R. § 122.22(c).)
5. Any person signing a document under Standard Provisions – Reporting V.B.2 or V.B.3 above shall make the following certification:

“I certify under penalty of law that this document and all attachments were prepared under my direction or supervision in accordance with a system designed to assure that qualified personnel properly gather and evaluate the information submitted. Based on my inquiry of the person or persons who manage the system or those persons directly

responsible for gathering the information, the information submitted is, to the best of my knowledge and belief, true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment for knowing violations.” (40 C.F.R. § 122.22(d).)

**C. Monitoring Reports**

1. Monitoring results shall be reported at the intervals specified in the Monitoring and Reporting Program (Attachment E) in this Order. (40 C.F.R. § 122.41(l)(4).)
2. Monitoring results must be reported on a Discharge Monitoring Report (DMR) form or forms provided or specified by the Central Valley Water Board or State Water Board for reporting results of monitoring of sludge use or disposal practices. (40 C.F.R. § 122.41(l)(4)(i).)
3. If the Discharger monitors any pollutant more frequently than required by this Order using test procedures approved under 40 C.F.R. part 136, or another method required for an industry-specific waste stream under 40 C.F.R. subchapters N or O, the results of such monitoring shall be included in the calculation and reporting of the data submitted in the DMR or sludge reporting form specified by the Central Valley Water Board. (40 C.F.R. § 122.41(l)(4)(ii).)
4. Calculations for all limitations, which require averaging of measurements, shall utilize an arithmetic mean unless otherwise specified in this Order. (40 C.F.R. § 122.41(l)(4)(iii).)

**D. Compliance Schedules**

Reports of compliance or noncompliance with, or any progress reports on, interim and final requirements contained in any compliance schedule of this Order, shall be submitted no later than 14 days following each schedule date. (40 C.F.R. § 122.41(l)(5).)

**E. Twenty-Four Hour Reporting**

1. The Discharger shall report any noncompliance that may endanger health or the environment. Any information shall be provided orally within 24 hours from the time the Discharger becomes aware of the circumstances. A written submission shall also be provided within five (5) days of the time the Discharger becomes aware of the circumstances. The written submission shall contain a description of the noncompliance and its cause; the period of noncompliance, including exact dates and times, and if the noncompliance has not been corrected, the anticipated time it is expected to continue; and steps taken or planned to reduce, eliminate, and prevent reoccurrence of the noncompliance. (40 C.F.R. § 122.41(l)(6)(i).)
2. The following shall be included as information that must be reported within 24 hours under this paragraph (40 C.F.R. § 122.41(l)(6)(ii)):
  - a. Any unanticipated bypass that exceeds any effluent limitation in this Order. (40 C.F.R. § 122.41(l)(6)(ii)(A).)
  - b. Any upset that exceeds any effluent limitation in this Order. (40 C.F.R. § 122.41(l)(6)(ii)(B).)
3. The Central Valley Water Board may waive the above-required written report under this provision on a case-by-case basis if an oral report has been received within 24 hours. (40 C.F.R. § 122.41(l)(6)(iii).)

## **F. Planned Changes**

The Discharger shall give notice to the Central Valley Water Board as soon as possible of any planned physical alterations or additions to the permitted facility. Notice is required under this provision only when (40 C.F.R. § 122.41(l)(1)):

1. The alteration or addition to a permitted facility may meet one of the criteria for determining whether a facility is a new source in section 122.29(b) (40 C.F.R. § 122.41(l)(1)(i)); or
2. The alteration or addition could significantly change the nature or increase the quantity of pollutants discharged. This notification applies to pollutants that are not subject to effluent limitations in this Order. (40 C.F.R. § 122.41(l)(1)(ii).)

The alteration or addition results in a significant change in the Discharger's sludge use or disposal practices, and such alteration, addition, or change may justify the application of permit conditions that are different from or absent in the existing permit, including notification of additional use or disposal sites not reported during the permit application process or not reported pursuant to an approved land application plan. (40 C.F.R. § 122.41(l)(1)(iii).)

## **G. Anticipated Noncompliance**

The Discharger shall give advance notice to the Central Valley Water Board or State Water Board of any planned changes in the permitted facility or activity that may result in noncompliance with this Order's requirements. (40 C.F.R. § 122.41(l)(2).)

## **H. Other Noncompliance**

The Discharger shall report all instances of noncompliance not reported under Standard Provisions – Reporting V.C, V.D, and V.E above at the time monitoring reports are submitted. The reports shall contain the information listed in Standard Provision – Reporting V.E above. (40 C.F.R. § 122.41(l)(7).)

## **I. Other Information**

When the Discharger becomes aware that it failed to submit any relevant facts in a permit application, or submitted incorrect information in a permit application or in any report to the Central Valley Water Board, State Water Board, or U.S. EPA, the Discharger shall promptly submit such facts or information. (40 C.F.R. § 122.41(l)(8).)

# **VI. STANDARD PROVISIONS – ENFORCEMENT**

- A. The Central Valley Water Board is authorized to enforce the terms of this permit under several provisions of the Water Code, including, but not limited to, sections 13385, 13386, and 13387.

# **VII. ADDITIONAL PROVISIONS – NOTIFICATION LEVELS**

## **I. Publicly-Owned Treatment Works (POTW's)**

All POTW's shall provide adequate notice to the Central Valley Water Board of the following (40 C.F.R. § 122.42(b)):

1. Any new introduction of pollutants into the POTW from an indirect discharger that would be subject to sections 301 or 306 of the CWA if it were directly discharging those pollutants (40 C.F.R. § 122.42(b)(1)); and

2. Any substantial change in the volume or character of pollutants being introduced into that POTW by a source introducing pollutants into the POTW at the time of adoption of the Order. (40 C.F.R. § 122.42(b)(2).)
3. Adequate notice shall include information on the quality and quantity of effluent introduced into the POTW as well as any anticipated impact of the change on the quantity or quality of effluent to be discharged from the POTW. (40 C.F.R. § 122.42(b)(3).)



## ATTACHMENT E – MONITORING AND REPORTING PROGRAM

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### **ATTACHMENT E – MONITORING AND REPORTING PROGRAM (MRP)**

The Code of Federal Regulations (40 C.F.R. § 122.48) requires that all NPDES permits specify monitoring and reporting requirements. Water Code sections 13267 and 13383 also authorize the Central Valley Water Board to require technical and monitoring reports. This MRP establishes monitoring and reporting requirements that implement federal and California regulations.

#### **I. GENERAL MONITORING PROVISIONS**

- A.** Samples and measurements taken as required herein shall be representative of the volume and nature of the monitored discharge. All samples shall be taken at the monitoring locations specified below and, unless otherwise specified, before the monitored flow joins or is diluted by any other waste stream, body of water, or substance. Monitoring locations shall not be changed without notification to and the approval of the Central Valley Water Board.
- B.** Effluent samples shall be taken downstream of the last addition of wastes to the treatment or discharge works where a representative sample may be obtained prior to mixing with the receiving waters. Samples shall be collected at such a point and in such a manner to ensure a representative sample of the discharge.
- C.** Chemical, bacteriological, and bioassay analyses of any material required by this Order shall be conducted by a laboratory certified for such analyses by the State Water Resources Control Board (State Water Board), Division of Drinking Water (DDW; formerly the Department of Public Health). Laboratories that perform sample analyses must be identified in all monitoring reports submitted to the Central Valley Water Board. In the event a certified laboratory is not available to the Discharger for any onsite field measurements such as pH, dissolved oxygen (DO), turbidity, temperature, and residual chlorine, such analyses performed by a noncertified laboratory will be accepted provided a Quality Assurance-Quality Control Program is instituted by the laboratory. A manual containing the steps followed in this program for any onsite field measurements such as pH, DO, turbidity, temperature, and residual chlorine must be kept onsite in the treatment facility laboratory and shall be available for inspection by Central Valley Water Board staff. The Discharger must demonstrate sufficient capability (qualified and trained employees, properly calibrated and maintained field instruments, etc.) to adequately perform these field measurements. The Quality Assurance-Quality Control Program must conform to USEPA guidelines or to procedures approved by the Central Valley Water Board.
- D.** Appropriate flow measurement devices and methods consistent with accepted scientific practices shall be selected and used to ensure the accuracy and reliability of measurements of the volume of monitored discharges. All monitoring instruments and devices used by the Discharger to fulfill the prescribed monitoring program shall be properly maintained and calibrated as necessary, at least yearly, to ensure their continued accuracy. All flow measurement devices shall be calibrated at least once per year to ensure continued accuracy of the devices.
- E.** Monitoring results, including noncompliance, shall be reported at intervals and in a manner specified in this Monitoring and Reporting Program.
- F.** Laboratories analyzing monitoring samples shall be certified by DDW, in accordance with the provision of Water Code section 13176, and must include quality assurance/quality control data with their reports.
- G.** The Discharger shall ensure that the results of the Discharge Monitoring Report-Quality Assurance (DMR-QA) Study or the most recent Water Pollution Performance Evaluation Study are submitted annually to the State Water Resources Control Board at the following address:

State Water Resources Control Board Quality Assurance Program Officer  
Office of Information Management and Analysis  
State Water Resources Control Board  
1001 I Street, Sacramento, CA 95814

- H. The Discharger shall file with the Central Valley Water Board technical reports on self-monitoring performed according to the detailed specifications contained in this Monitoring and Reporting Program.
- I. The results of all monitoring required by this Order shall be reported to the Central Valley Water Board, and shall be submitted in such a format as to allow direct comparison with the limitations and requirements of this Order. Unless otherwise specified, discharge flows shall be reported in terms of the monthly average and the daily maximum discharge flows.

## II. MONITORING LOCATIONS

The Discharger shall establish the following monitoring locations to demonstrate compliance with the effluent limitations, discharge specifications, and other requirements in this Order:

**Table E-1. Monitoring Station Locations**

| Discharge Point Name | Monitoring Location Name | Monitoring Location Description                                                                                                                                                                                                                                     |
|----------------------|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --                   | <b>INF-001</b>           | A location where a representative sample of the influent into the Facility can be collected.                                                                                                                                                                        |
| <b>001</b>           | <b>EFF-001</b>           | A location where a representative sample of the effluent from the Facility can be collected after all treatment processes and prior to commingling with other waste streams or being discharged into Laguna Creek.<br>Latitude: 38°17'56 N, Longitude: 121°19'46" W |
| --                   | <b>RSW-001</b>           | A location on Laguna Creek approximately 4000 feet upstream from the confluence of the remanant channel of Skunk Creek and Laguna Creek.<br>Latitude: 38°18'37 N, Longitude: 121°19'41" W                                                                           |
| --                   | <b>RSW-002</b>           | A location on Laguna Creek approximately 600 feet downstream from the confluence of the remanant channel of Skunk Creek and Laguna creek.                                                                                                                           |
| --                   | <b>FIL-001</b>           | A location where a representative sample of wastewater can be collected down stream of the filtration system and upstream of the ultraviolet light (UV) disinfection system                                                                                         |
| --                   | <b>UVS-001</b>           | A location where a representative sample of wastewater can be collected immediately downstream of the (UV) disinfection system                                                                                                                                      |
| --                   | <b>SPL-001</b>           | Municipal Water Supply.                                                                                                                                                                                                                                             |

The North latitude and West longitude information in Table E-1 are approximate for administrative purposes.

### III. INFLUENT MONITORING REQUIREMENTS

#### A. Monitoring Location INF-001

1. The Discharger shall monitor influent to the Facility at INF-001 as follows:

**Table E-2. Influent Monitoring**

| Parameter                               | Units | Sample Type                  | Minimum Sampling Frequency | Required Analytical Test Method |
|-----------------------------------------|-------|------------------------------|----------------------------|---------------------------------|
| Flow                                    | MGD   | Meter                        | Continuous                 | --                              |
| Biochemical Oxygen Demand(5-day @ 20°C) | mg/L  | 24-hr Composite <sup>1</sup> | 1/Week                     | <sup>2</sup>                    |
| Total Suspended Solids                  | mg/L  | 24-hr Composite <sup>1</sup> | 1/Week                     | <sup>2</sup>                    |

<sup>1</sup> 24-hour flow proportional composite.

<sup>2</sup> Pollutants shall be analyzed using the analytical methods described in 40 CFR Part 136 or by methods approved by the Central Valley Water Board or the State Water Board.

### IV. EFFLUENT MONITORING REQUIREMENTS

#### A. Monitoring Location EFF-001

1. The Discharger shall monitor treatment plant effluent at EFF-001 as follows. Sampling is not required during periods when no effluent is discharged to surface water, however, the Discharger must clearly state in the monthly self-monitoring report to the Central Valley Water Board that there was no discharge to surface water during the specified period. If more than one analytical test method is listed for a given parameter, the Discharger must select from the listed methods and corresponding Minimum Level:

**Table E-3. Effluent Monitoring**

| Parameter                                             | Units            | Sample Type                    | Minimum Sampling Frequency | Required Analytical Test Method |
|-------------------------------------------------------|------------------|--------------------------------|----------------------------|---------------------------------|
| Flow                                                  | MGD              | Meter                          | Continuous                 | --                              |
| <b>Conventional Pollutants</b>                        |                  |                                |                            |                                 |
| Biochemical Oxygen Demand (5-day @ 20° C)             | mg/L             | 24-hr Composite <sup>2</sup>   | 1/Week                     | <sup>1</sup>                    |
|                                                       | lbs/day          | Calculate                      | 1/Week                     | --                              |
| Total Suspended Solids                                | mg/L             | 24-hr Composite <sup>2</sup>   | 1/Week                     | <sup>1</sup>                    |
|                                                       | lbs/day          | Calculate                      | 1/Week                     | --                              |
| pH                                                    | standard units   | Grab                           | 1/Week <sup>3, 4</sup>     | <sup>1</sup>                    |
| <b>Priority Pollutants</b>                            |                  |                                |                            |                                 |
| Arsenic, Total Recoverable                            | µg/L             | 24-hour Composite <sup>2</sup> | 1/Month                    | <sup>1</sup>                    |
| Mercury, Total Recoverable                            | µg/L             | Grab                           | 1/Month                    | <sup>8</sup>                    |
|                                                       | lbs/year         | Calculate                      | 1/Year                     | --                              |
| Zinc, Total Recoverable                               | µg/L             | 24-hour Composite <sup>2</sup> | 1/Month                    | <sup>1</sup>                    |
| Priority Pollutants and Other Constituents of Concern | See Section IX.D | See Section IX.D               | See Section IX.D           | <sup>1, 5</sup>                 |
| <b>Non-Conventional Pollutants</b>                    |                  |                                |                            |                                 |
| Ammonia Nitrogen, Total (as N)                        | mg/L             | Grab                           | 1/Week <sup>3, 6</sup>     | <sup>1</sup>                    |
|                                                       | lbs/day          | Calculate                      | 1/Week                     | --                              |
| Electrical Conductivity @ 25°C                        | µmhos/cm         | Grab                           | 1/Week                     | <sup>1</sup>                    |
| Hardness, Total (as CaCO <sub>3</sub> )               | mg/L             | 24-hour Composite <sup>2</sup> | 1/Month <sup>7</sup>       | <sup>1</sup>                    |

| Parameter                          | Units | Sample Type                    | Minimum Sampling Frequency | Required Analytical Test Method |
|------------------------------------|-------|--------------------------------|----------------------------|---------------------------------|
| Nitrate Nitrogen, Total (as N)     | mg/L  | Grab                           | 1/Month <sup>9</sup>       | 1                               |
| Nitrite Nitrogen, Total (as N)     | mg/L  | Grab                           | 1/Month <sup>9</sup>       | 1                               |
| Nitrate plus Nitrite, Total (as N) | mg/L  | Grab                           | 1/Month                    |                                 |
| Standard Minerals <sup>10</sup>    | mg/L  | 24-hour Composite <sup>2</sup> | 1/Year                     | 1                               |
| Temperature                        | °C    | Grab                           | 1/Week <sup>3,4</sup>      | 1                               |
| Total Dissolved Solids             | mg/L  | Grab                           | 1/Month                    | 1                               |

<sup>1</sup> Pollutants shall be analyzed using the analytical methods described in 40 CFR Part 136 or by methods approved by the Central Valley Water Board or the State Water Board.

<sup>2</sup> 24-hour flow proportional composite.

<sup>3</sup> pH and temperature shall be recorded at the time of ammonia sample collection.

<sup>4</sup> A hand-held field meter may be used, provided the meter utilizes a USEPA-approved algorithm/method and is calibrated and maintained in accordance with the manufacturer's instructions. A calibration and maintenance log for each meter used for monitoring required by this Monitoring and Reporting Program shall be maintained at the Facility.

<sup>5</sup> For priority pollutant constituents the reporting level shall be consistent with Sections 2.4.2 and 2.4.3 of the *Policy for Implementation of Toxics Standards for Inland Surface Waters, Enclosed Bays, and Estuaries of California* (See Attachment E, Table E-8).

<sup>6</sup> Concurrent with whole effluent toxicity monitoring.

<sup>7</sup> Hardness samples shall be collected concurrently with metals samples.

<sup>8</sup> Unfiltered total mercury samples shall be taken using clean hands/dirty hands procedures, as described in U.S. EPA method 1669: Sampling Ambient Water for Trace Metals at EPA Water Quality Criteria Levels, for collection of equipment blanks (section 9.4.4.2), and shall be analyzed by U.S. EPA method 1630/1631 (Revision E) with a reporting limit of 0.05 ng/L for total mercury.

<sup>9</sup> Monitoring for nitrite and nitrate shall be conducted concurrently.

<sup>10</sup> Standard minerals shall include the following: boron, calcium, iron, magnesium, potassium, sodium, chloride, manganese, phosphorus, and total alkalinity (including alkalinity series), and include verification that the analysis is complete (i.e., cation/anion balance).

## V. WHOLE EFFLUENT TOXICITY TESTING REQUIREMENTS

**A. Acute Toxicity Testing.** The Discharger shall conduct acute toxicity testing to determine whether the effluent is contributing acute toxicity to the receiving water. The Discharger shall meet the following acute toxicity testing requirements:

- Monitoring Frequency** – The Discharger shall perform annual acute toxicity testing, concurrent with effluent ammonia sampling.
- Sample Types** – The Discharger may use flow-through or static renewal testing. For static renewal testing, the samples shall be grab samples and shall be representative of the volume and quality of the discharge. The effluent samples shall be taken at Monitoring Location EFF-001.
- Test Species** – Test species shall be fathead minnows (*Pimephales promelas*).
- Methods** – The acute toxicity testing samples shall be analyzed using EPA-821-R-02-012, Fifth Edition. Temperature and pH shall be recorded at the time of sample collection. No pH adjustment may be made unless approved by the Executive Officer.
- Test Failure** – If an acute toxicity test does not meet all test acceptability criteria, as specified in the test method, the Discharger must re-sample and re-test as soon as possible, not to exceed 7 days following notification of test failure.

**B. Chronic Toxicity Testing.** The Discharger shall conduct three species chronic toxicity testing to determine whether the effluent is contributing chronic toxicity to the receiving water. The Discharger shall meet the following chronic toxicity testing requirements:

1. **Monitoring Frequency** – The Discharger shall perform quarterly three species chronic toxicity testing. Chronic toxicity testing is not required during quarters when no effluent is discharged to surface water, however, the Discharger must clearly state in the quarterly self-monitoring report to the Central Valley Water Board that there was no discharge to surface water during the specified quarter.
2. **Sample Types** – Effluent samples shall be flow proportional 24-hour composites and shall be representative of the volume and quality of the discharge. The effluent samples shall be taken at Monitoring Location EFF-001. The receiving water control shall be a grab sample obtained from Monitoring Location RSW-001 as identified in this Monitoring and Reporting Program.
3. **Sample Volumes** – Adequate sample volumes shall be collected to provide renewal water to complete the test in the event that the discharge is intermittent.
4. **Test Species** – Chronic toxicity testing measures sublethal (e.g., reduced growth, reproduction) and/or lethal effects to test organisms exposed to an effluent compared to that of the control organisms. The Discharger shall conduct chronic toxicity tests with:
  - a. The cladoceran, water flea, *Ceriodaphnia dubia* (survival and reproduction test);
  - b. The fathead minnow, *Pimephales promelas* (larval survival and growth test); and
  - c. The green alga, *Selenastrum capricornutum* (growth test).
5. **Methods** – The presence of chronic toxicity shall be estimated as specified in *Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to Freshwater Organisms, Fourth Edition*, EPA/821-R-02-013, October 2002.
6. **Reference Toxicant** – As required by the SIP, all chronic toxicity tests shall be conducted with concurrent testing with a reference toxicant and shall be reported with the chronic toxicity test results.
7. **Dilutions** – For routine and accelerated chronic toxicity monitoring, it is not necessary to perform the test using a dilution series. The test may be performed using 100% effluent and one control. For TRE monitoring, the chronic toxicity testing shall be performed using the dilution series identified in Table E-4, below, unless an alternative dilution series is detailed in the submitted TRE Action Plan. A receiving water control or laboratory water control may be used as the diluent.

**Table E-4. Chronic Toxicity Testing Dilution Series**

| Sample          | Dilutions <sup>a</sup> (%) |    |    |    |      | Control |
|-----------------|----------------------------|----|----|----|------|---------|
|                 | 100                        | 75 | 50 | 25 | 12.5 |         |
| % Effluent      | 100                        | 75 | 50 | 25 | 12.5 | 0       |
| % Control Water | 0                          | 25 | 50 | 75 | 87.5 | 100     |

<sup>a</sup> Receiving water control or laboratory water control may be used as the diluent.

8. **Test Failure** – The Discharger must re-sample and re-test as soon as possible, but no later than fourteen (14) days after receiving notification of a test failure. A test failure is defined as follows:

- a. The reference toxicant test or the effluent test does not meet all test acceptability criteria as specified in the *Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to Freshwater Organisms, Fourth Edition*, EPA/821-R-02-013, October 2002 (Method Manual), and its subsequent amendments or revisions; or
  - b. The percent minimum significant difference (PMSD) measured for the test exceeds the upper PMSD bound variability criterion in Table 6 on page 52 of the Method Manual. (A retest is only required in this case if the test results do not exceed the monitoring trigger specified in the Special Provision at section VI. 2.a.iii. of the Order.)
- C. WET Testing Notification Requirements.** The Discharger shall notify the Central Valley Water Board within 24-hours after the receipt of test results exceeding the monitoring trigger during regular or accelerated monitoring, or an exceedance of the acute toxicity effluent limitation.
- D. WET Testing Reporting Requirements.** All toxicity test reports shall include the contracting laboratory's complete report provided to the Discharger and shall be in accordance with the appropriate "Report Preparation and Test Review" sections of the method manuals. At a minimum, whole effluent toxicity monitoring shall be reported as follows:
  1. **Chronic WET Reporting.** Regular chronic toxicity monitoring results shall be reported to the Central Valley Water Board with the next monthly self monitoring report after the results are received from the laboratory, and shall contain, at minimum:
    - a. The results expressed in TUC, measured as 100/NOEC, and also measured as 100/LC50, 100/EC25, 100/IC25, and 100/IC50, as appropriate.
    - b. The statistical methods used to calculate endpoints;
    - c. The statistical output page, which includes the calculation of the percent minimum significant difference (PMSD);
    - d. The dates of sample collection and initiation of each toxicity test; and
    - e. The results compared to the numeric toxicity monitoring trigger.

Additionally, the monthly self-monitoring reports shall contain an updated chronology of chronic toxicity test results expressed in TUC, and organized by test species, type of test (survival, growth or reproduction), and monitoring frequency, i.e., either quarterly, monthly, accelerated, or Toxicity Reduction Evaluation (TRE).
  2. **Acute WET Reporting.** Acute toxicity test results shall be submitted with the monthly discharger self-monitoring reports and reported as percent survival.
  3. **TRE Reporting.** Reports for TREs shall be submitted in accordance with the schedule contained in the Discharger's approved TRE Workplan, or as amended by the Discharger's TRE Action Plan.
  4. **Quality Assurance (QA).** The Discharger must provide the following information for QA purposes:
    - a. Results of the applicable reference toxicant data with the statistical output page giving the species, NOEC, LOEC, type of toxicant, dilution water used, concentrations used, PMSD, and dates tested.
    - b. The reference toxicant control charts for each endpoint, which include summaries of reference toxicant tests performed by the contracting laboratory.

- c. Any information on deviations or problems encountered and how they were dealt with.

## **VI. LAND DISCHARGE MONITORING REQUIREMENTS – NOT APPLICABLE**

## **VII. RECYCLING MONITORING REQUIREMENTS – NOT APPLICABLE**

## **VIII. RECEIVING WATER MONITORING REQUIREMENTS**

The Discharger shall implement the Receiving Water Monitoring Requirements in Attachment E, Section VIII.A of this Order. However, the Central Valley Water Board hereby authorizes the Discharger to participate in the Delta Regional Monitoring Program or other Regional Monitoring Program in lieu of conducting the individual monitoring specified in Attachment E, Section VIII.A of this Order (including visual observations). The Discharger may choose to conduct all or part of the receiving water monitoring through the Regional Monitoring Program, as approved by the Executive Officer. If the Discharger elects to cease all or part of the individual receiving water monitoring and instead participates in the a Regional Monitoring Program, the Discharger shall submit a letter signed by an authorized representative informing the Board that the Discharger will participate in a Regional Monitoring Program, and the date on which individual receiving water monitoring required under Attachment E, Section VIII.A will cease, or be modified, and specific monitoring locations and constituent combinations that will no longer be conducted individually. To ensure consistency with this Order, discontinuing part or all of individual receiving water monitoring requires the Executive Officer's prior written approval of the Discharger's request. However, approval by the Executive Officer is not required prior to participating in a Regional Monitoring Program.

If the Discharger participates in a Regional Monitoring Program in lieu of conducting individual receiving water monitoring, the Discharger shall continue to participate in the Regional Monitoring Program until such time as the Discharger informs the Board that participation in the Regional Monitoring Program will cease and individual monitoring is reinstituted. After receiving written approval from the Executive Officer, receiving water monitoring under Attachment E, Section VIII.A, is not required under this Order so long as the Discharger adequately supports the Regional Monitoring Program. If the Discharger fails to adequately support the Regional Monitoring Program, as defined by the Regional Monitoring Program Steering Committee, the Discharger shall reinstitute individual receiving water monitoring under Attachment E, Section VIII.A, upon written notice from the Executive Officer. During participation in the Delta Regional Monitoring Program, the Discharger may conduct and submit any or part of the receiving water monitoring included in this Monitoring and Reporting Program that is deemed appropriate by the Discharger.

Regional Monitoring Program data is not intended to be used directly to represent either upstream or downstream water quality for purposes of determining compliance with this Permit. Regional Monitoring Program monitoring stations are established generally as "integrator sites" to evaluate the combined impacts on water quality of multiple discharges into the water body; Regional Monitoring Program monitoring stations would not normally be able to identify the source of any specific constituent, but would be used to identify water quality issues needing further evaluation. Regional Monitoring Program monitoring data, along with individual Discharger data, may be used to help establish background receiving water quality for reasonable potential analyses in an NPDES permit after evaluation of the applicability of the data for that purpose. Regional Monitoring Program data, as with all environmental monitoring data, can provide an assessment of water quality at a specific place and time that can be used in conjunction with other information, such as other receiving water monitoring data, spatial and temporal distribution and trends of receiving water data, effluent data from the Discharger's discharge and other point and non-point source



discharges, receiving water flow volume, speed and direction, and other information to determine the likely source or sources of a constituent that resulted in exceedance of a receiving water quality objective.

During the period of participation in the Regional Monitoring Program, the Discharger shall continue to report any individually conducted receiving water monitoring data in the Electronic Self-Monitoring Reports (eSMR) according to the Monitoring and Reporting Program. In addition, 1) with each submitted eSMR, the Discharger's eSMR cover letter shall state that the Discharger is participating in the Regional Monitoring Program in lieu of conducting the individual receiving water monitoring program required by the permit, and 2) with each annual report, the Discharger shall attach a copy of the letter originally submitted to the Central Valley Water Board describing the monitoring location(s) and constituent combinations that will no longer be conducted individually.

**A. Monitoring Location RSW-001 and RSW-002**

1. When discharging to surface water, the Discharger shall monitor Laguna Creek at RSW-001 and RSW-002 as required in Table E-5 below. Sampling is not required during periods when no effluent is discharged to Laguna Creek. Furthermore, when no existing flow is present in Laguna Creek, monitoring at RSW-001 is not required. However, the Discharger must clearly state in the monthly self-monitoring report to the Central Valley Water Board that there was no discharge to Laguna Creek or that there was no existing flow in Laguna Creek during the specified period.

**Table E-5. Receiving Water Monitoring Requirements**

| Parameter                        | Units          | Sample Type | Minimum Sampling Frequency | Required Analytical Test Method |
|----------------------------------|----------------|-------------|----------------------------|---------------------------------|
| Dissolved Oxygen                 | mg/L           | Grab        | 1/Week                     | 1, 2                            |
| pH <sup>3</sup>                  | standard units | Grab        | 1/Week                     | 1, 2                            |
| Turbidity                        | NTU            | Grab        | 1/Week                     | 1, 2                            |
| Temperature <sup>3</sup>         | °F             | Grab        | 1/Week                     | 1, 2                            |
| Electrical Conductivity @ 25°C   | µmhos/cm       | Grab        | 1/Week                     | 1, 2                            |
| Hardness (as CaCO <sub>3</sub> ) | mg/L           | Grab        | 1/Month                    | 2                               |

<sup>1</sup> A hand-held field meter may be used, provided the meter utilizes a USEPA-approved algorithm/method and is calibrated and maintained in accordance with the manufacturer's instructions. A calibration and maintenance log for each meter used for monitoring required by this Monitoring and Reporting Program shall be maintained at the Facility.

<sup>2</sup> Pollutants shall be analyzed using the analytical methods described in 40 CFR Part 136 or by methods approved by the Central Valley Water Board or the State Water Board.

<sup>3</sup> pH and temperature shall be determined on the same day as sample collection for effluent ammonia.

## IX. OTHER MONITORING REQUIREMENTS

### A. Municipal Water Supply

#### 1. Monitoring Location SPL-001

- a. The Discharger shall monitor the municipal water supply at SPL-001 as follows. A sampling station shall be established where a representative sample of the municipal water supply can be obtained. Municipal water supply samples shall be collected at approximately the same time as effluent samples.

**Table E-6. Municipal Water Supply Monitoring Requirements**

| Parameter                                   | Units    | Sample Type | Minimum Sampling Frequency | Required Analytical Test Method |
|---------------------------------------------|----------|-------------|----------------------------|---------------------------------|
| Total Dissolved Solids <sup>1</sup>         | mg/L     | Grab        | 1/Year                     | <sup>2</sup>                    |
| Electrical Conductivity @ 25°C <sup>1</sup> | µmhos/cm | Grab        | 1/Year                     | <sup>2</sup>                    |
| Standard Minerals <sup>3</sup>              | mg/L     | Grab        | 1/Year                     | <sup>2</sup>                    |

<sup>1</sup> If the water supply is from more than one source, the total dissolved solids and electrical conductivity shall be reported as a weighted average and include copies of supporting calculations.

<sup>2</sup> Pollutants shall be analyzed using the analytical methods described in 40 C.F.R. part 136 or by methods approved by the Central Valley Water Board or the State Water Board.

<sup>3</sup> Standard minerals shall include all major cations and anions and include verification that the analysis is complete (i.e., cation/anion balance).

### B. Filtration System and Ultraviolet Light (UV) Disinfection System

#### 1. Monitoring Locations UVS-001.

- a. When discharging to Laguna Creek, the Discharger shall monitor the filtration system and UV disinfection system at Monitoring Location FIL-001 and UVS-001 as follows:

**Table E-7. Filtration System and UV Disinfection System Monitoring Requirements**

| Parameter                       | Units              | Sample Type | Monitoring Location | Minimum Sampling Frequency |
|---------------------------------|--------------------|-------------|---------------------|----------------------------|
| Flow                            | MGD                | Meter       | FIL-001             | Continuous <sup>1</sup>    |
| Turbidity                       | NTU                | Meter       | FIL-001             | Continuous <sup>1,2</sup>  |
| Number of UV banks in operation | Number             | Observation | N/A                 | Continuous <sup>1,3</sup>  |
| UV Transmittance                | Percent (%)        | Meter       | FIL-001             | Continuous <sup>1,4</sup>  |
| UV Dose                         | mJ/cm <sup>2</sup> | Calculated  | N/A                 | Continuous <sup>1,5</sup>  |
| Total Coliform Organisms        | MPN/100mL          | Grab        | UVS-001             | 2/Week                     |

<sup>1</sup> For continuous analyzers, the Discharger shall report documented routine meter maintenance activities including date, time of day, and duration, in which the analyzer(s) is not in operation. If analyzer(s) fail to provide continuous monitoring for more than two hours and influent and/or effluent from the disinfection process is not diverted for retreatment or irrigation use, the Discharger shall obtain and report hourly manual and/or grab sample results. The Discharger shall not decrease power settings or reduce the number of UV lamp banks in operation while the continuous analyzers are out of service and water is being disinfected.

<sup>2</sup> Report daily average and maximum turbidity.

<sup>3</sup> Report daily minimum and daily maximum number of UV banks in operation.

<sup>4</sup> Report daily minimum hourly average (e.g., 1:00 PM to 1:59 PM) UV transmittance and daily average UV transmittance. The minimum hourly average transmittance shall consist of lowest average transmittance recorded over an hour of day when flow is being discharged to Laguna Creek. If the system does not operate

for an entire daily hour interval or if effluent flow is not discharged to Laguna Creek for the entire hour, the transmittance will be averaged based on the actual operation time when discharges to Laguna Creek occurred during that hour of the day.

- 5 Report daily minimum hourly average UV dose and daily average UV dose. The minimum hourly average dose shall consist of lowest hourly average dose provided in any channel that had at least one bank of lamps operating during the hour interval. For channels that did not operate for the entire hour interval or when effluent flow is not discharged to Laguna Creek for the entire hour, the dose will be averaged based on the actual operation time when discharges to Laguna Creek occurred. **Within in 6 months of the effective date of this Order**, the Discharger shall install flow monitoring devices in each channel to allow the calculation of UV dose for each channel. The reporting during this initial 6 month period shall be the UV dose for entire UV system. As an alternative to the installation of flow monitoring devices, the Discharger may conduct a flow study to estimate the flow split between channels. Upon Executive Officer approval, the Discharger may calculate the UV dose for each channel based on the estimated flow split to meet this reporting requirement.

### C. Effluent and Receiving Water Characterization

1. **Quarterly Monitoring.** Quarterly samples shall be collected from the effluent and upstream receiving water (Monitoring Locations EFF-001 and RSW-001) and analyzed for the constituents listed in Table E-8, below. Quarterly monitoring shall be conducted during 2018 and the results of such monitoring be submitted to the Central Valley Water Board with the quarterly self-monitoring reports. Each individual monitoring event shall provide representative sample results for the effluent and upstream receiving water.
2. **Concurrent Sampling.** Effluent and receiving water sampling shall be performed at approximately the same time, on the same date.
3. **Sample Type.** All receiving water samples shall be taken as grab samples. Effluent samples shall be taken as described in **Table E-8**, below.

**Table E-8. Effluent and Receiving Water Characterization Monitoring**

| Parameter                     | Units | Effluent Sample Type | Maximum Reporting Level <sup>1</sup> |
|-------------------------------|-------|----------------------|--------------------------------------|
| 2- Chloroethyl vinyl ether    | µg/L  | Grab                 | 1                                    |
| Acrolein                      | µg/L  | Grab                 | 2                                    |
| Acrylonitrile                 | µg/L  | Grab                 | 2                                    |
| Benzene                       | µg/L  | Grab                 | 0.5                                  |
| Bromoform                     | µg/L  | Grab                 | 0.5                                  |
| Carbon Tetrachloride          | µg/L  | Grab                 | 0.5                                  |
| Chlorobenzene                 | µg/L  | Grab                 | 0.5                                  |
| Chloroethane                  | µg/L  | Grab                 | 0.5                                  |
| Chloroform                    | µg/L  | Grab                 | 2                                    |
| Chloromethane                 | µg/L  | Grab                 | 2                                    |
| Dibromochloromethane          | µg/L  | Grab                 | 0.5                                  |
| Dichlorobromomethane          | µg/L  | Grab                 | 0.5                                  |
| Dichloromethane               | µg/L  | Grab                 | 2                                    |
| Ethylbenzene                  | µg/L  | Grab                 | 2                                    |
| Hexachlorobenzene             | µg/L  | Grab                 | 1                                    |
| Hexachlorobutadiene           | µg/L  | Grab                 | 1                                    |
| Hexachloroethane              | µg/L  | Grab                 | 1                                    |
| Methyl bromide (Bromomethane) | µg/L  | Grab                 | 1                                    |
| Naphthalene                   | µg/L  | Grab                 | 10                                   |
| Parachlorometa cresol         | µg/L  | Grab                 |                                      |
| Tetrachloroethene             | µg/L  | Grab                 | 0.5                                  |
| Toluene                       | µg/L  | Grab                 | 2                                    |

| Parameter                             | Units | Effluent Sample Type | Maximum Reporting Level <sup>1</sup> |
|---------------------------------------|-------|----------------------|--------------------------------------|
| trans-1,2-Dichloroethylene            | µg/L  | Grab                 | 1                                    |
| Trichloroethene                       | µg/L  | Grab                 | 2                                    |
| Vinyl chloride                        | µg/L  | Grab                 | 0.5                                  |
| Methyl-tert-butyl ether (MTBE)        | µg/L  | Grab                 |                                      |
| Trichlorofluoromethane                | µg/L  | Grab                 |                                      |
| 1,1,1-Trichloroethane                 | µg/L  | Grab                 | 0.5                                  |
| 1,1-dichloroethane                    | µg/L  | Grab                 | 0.5                                  |
| 1,1-dichloroethylene                  | µg/L  | Grab                 | 0.5                                  |
| 1,2-dichloropropane                   | µg/L  | Grab                 | 0.5                                  |
| 1,3-dichloropropylene                 | µg/L  | Grab                 | 0.5                                  |
| 1,1,2,2-tetrachloroethane             | µg/L  | Grab                 | 0.5                                  |
| 1,1,2-Trichloro-1,2,2-Trifluoroethane | µg/L  | Grab                 | 0.5                                  |
| 1,2,4-trichlorobenzene                | µg/L  | Grab                 | 1                                    |
| 1,2-dichloroethane                    | µg/L  | Grab                 | 0.5                                  |
| 1,2-dichlorobenzene                   | µg/L  | Grab                 | 0.5                                  |
| 1,3-dichlorobenzene                   | µg/L  | Grab                 | 0.5                                  |
| 1,4-dichlorobenzene                   | µg/L  | Grab                 | 0.5                                  |
| Styrene                               | µg/L  | Grab                 |                                      |
| Xylenes                               | µg/L  | Grab                 |                                      |
| 1,2-Benzanthracene                    | µg/L  | Grab                 | 5                                    |
| 1,2-Diphenylhydrazine                 | µg/L  | Grab                 | 1                                    |
| 2-Chlorophenol                        | µg/L  | Grab                 | 5                                    |
| 2,4-Dichlorophenol                    | µg/L  | Grab                 | 5                                    |
| 2,4-Dimethylphenol                    | µg/L  | Grab                 | 2                                    |
| 2,4-Dinitrophenol                     | µg/L  | Grab                 | 5                                    |
| 2,4-Dinitrotoluene                    | µg/L  | Grab                 | 5                                    |
| 2,4,6-Trichlorophenol                 | µg/L  | Grab                 | 10                                   |
| 2,6-Dinitrotoluene                    | µg/L  | Grab                 | 5                                    |
| 2-Nitrophenol                         | µg/L  | Grab                 | 10                                   |
| 2-Chloronaphthalene                   | µg/L  | Grab                 | 10                                   |
| 3,3'-Dichlorobenzidine                | µg/L  | Grab                 | 5                                    |
| 3,4-Benzofluoranthene                 | µg/L  | Grab                 | 10                                   |
| 4-Chloro-3-methylphenol               | µg/L  | Grab                 | 5                                    |
| 4,6-Dinitro-2-methylphenol            | µg/L  | Grab                 | 10                                   |
| 4-Nitrophenol                         | µg/L  | Grab                 | 10                                   |
| 4-Bromophenyl phenyl ether            | µg/L  | Grab                 | 10                                   |
| 4-Chlorophenyl phenyl ether           | µg/L  | Grab                 | 5                                    |
| Acenaphthene                          | µg/L  | Grab                 | 1                                    |
| Acenaphthylene                        | µg/L  | Grab                 | 10                                   |
| Anthracene                            | µg/L  | Grab                 | 10                                   |
| Benzidine                             | µg/L  | Grab                 | 5                                    |
| Benzo(a)pyrene (3,4-Benzopyrene)      | µg/L  | Grab                 | 2                                    |
| Benzo(g,h,i)perylene                  | µg/L  | Grab                 | 5                                    |
| Benzo(k)fluoranthene                  | µg/L  | Grab                 | 2                                    |
| Bis(2-chloroethoxy) methane           | µg/L  | Grab                 | 5                                    |
| Bis(2-chloroethyl) ether              | µg/L  | Grab                 | 1                                    |
| Bis(2-chloroisopropyl) ether          | µg/L  | Grab                 | 10                                   |
| Bis(2-ethylhexyl) phthalate           | µg/L  | Grab                 | 5                                    |
| Butyl benzyl phthalate                | µg/L  | Grab                 | 10                                   |
| Chrysene                              | µg/L  | Grab                 | 5                                    |

| Parameter                         | Units | Effluent Sample Type | Maximum Reporting Level <sup>1</sup> |
|-----------------------------------|-------|----------------------|--------------------------------------|
| Di-n-butylphthalate               | µg/L  | Grab                 | 10                                   |
| Di-n-octylphthalate               | µg/L  | Grab                 | 10                                   |
| Dibenzo(a,h)-anthracene           | µg/L  | Grab                 | 0.1                                  |
| Diethyl phthalate                 | µg/L  | Grab                 | 10                                   |
| Dimethyl phthalate                | µg/L  | Grab                 | 10                                   |
| Fluoranthene                      | µg/L  | Grab                 | 10                                   |
| Fluorene                          | µg/L  | Grab                 | 10                                   |
| Hexachlorocyclopentadiene         | µg/L  | Grab                 | 5                                    |
| Indeno(1,2,3-c,d)pyrene           | µg/L  | Grab                 | 0.05                                 |
| Isophorone                        | µg/L  | Grab                 | 1                                    |
| N-Nitrosodiphenylamine            | µg/L  | Grab                 | 1                                    |
| N-Nitrosodimethylamine            | µg/L  | Grab                 | 5                                    |
| N-Nitrosodi-n-propylamine         | µg/L  | Grab                 | 5                                    |
| Nitrobenzene                      | µg/L  | Grab                 | 10                                   |
| Pentachlorophenol                 | µg/L  | Grab                 | 1                                    |
| Phenanthrene                      | µg/L  | Grab                 | 5                                    |
| Phenol                            | µg/L  | Grab                 | 1                                    |
| Pyrene                            | µg/L  | Grab                 | 10                                   |
| Aluminum                          | µg/L  | 24-hr Composite      |                                      |
| Antimony                          | µg/L  | 24-hr Composite      | 5                                    |
| Arsenic                           | µg/L  | 24-hr Composite      | 10                                   |
| Asbestos                          | µg/L  | 24-hr Composite      |                                      |
| Barium                            | µg/L  | 24-hr Composite      |                                      |
| Beryllium                         | µg/L  | 24-hr Composite      | 2                                    |
| Cadmium                           | µg/L  | 24-hr Composite      | 0.5                                  |
| Chromium (III)                    | µg/L  | 24-hr Composite      | 50                                   |
| Chromium (VI)                     | µg/L  | 24-hr Composite      | 10                                   |
| Copper                            | µg/L  | 24-hr Composite      | 0.5                                  |
| Cyanide                           | µg/L  | 24-hr Composite      | 5                                    |
| Fluoride                          | µg/L  | 24-hr Composite      |                                      |
| Iron                              | µg/L  | 24-hr Composite      |                                      |
| Lead                              | µg/L  | 24-hr Composite      | 0.5                                  |
| Mercury                           | µg/L  | Grab                 | 0.5                                  |
| Manganese                         | µg/L  | 24-hr Composite      |                                      |
| Molybdenum                        | µg/L  | 24-hr Composite      |                                      |
| Nickel                            | µg/L  | 24-hr Composite      | 20                                   |
| Selenium                          | µg/L  | 24-hr Composite      | 5                                    |
| Silver                            | µg/L  | 24-hr Composite      | 0.25                                 |
| Thallium                          | µg/L  | 24-hr Composite      | 1                                    |
| Tributyltin                       | µg/L  | 24-hr Composite      |                                      |
| Zinc                              | µg/L  | 24-hr Composite      | 20                                   |
| 4,4'-DDD                          | µg/L  | 24-hr Composite      | 0.05                                 |
| 4,4'-DDE                          | µg/L  | 24-hr Composite      | 0.05                                 |
| 4,4'-DDT                          | µg/L  | 24-hr Composite      | 0.01                                 |
| alpha-Endosulfan                  | µg/L  | 24-hr Composite      | 0.02                                 |
| alpha-Hexachlorocyclohexane (BHC) | µg/L  | 24-hr Composite      | 0.01                                 |
| Alachlor                          | µg/L  | 24-hr Composite      |                                      |
| Aldrin                            | µg/L  | 24-hr Composite      | 0.005                                |
| beta-Endosulfan                   | µg/L  | 24-hr Composite      | 0.01                                 |
| beta-Hexachlorocyclohexane        | µg/L  | 24-hr Composite      | 0.005                                |

| Parameter                             | Units     | Effluent Sample Type | Maximum Reporting Level <sup>1</sup> |
|---------------------------------------|-----------|----------------------|--------------------------------------|
| Chlordane                             | µg/L      | 24-hr Composite      | 0.1                                  |
| delta-Hexachlorocyclohexane           | µg/L      | 24-hr Composite      | 0.005                                |
| Dieldrin                              | µg/L      | 24-hr Composite      | 0.01                                 |
| Endosulfan sulfate                    | µg/L      | 24-hr Composite      | 0.01                                 |
| Endrin                                | µg/L      | 24-hr Composite      | 0.01                                 |
| Endrin Aldehyde                       | µg/L      | 24-hr Composite      | 0.01                                 |
| Heptachlor                            | µg/L      | 24-hr Composite      | 0.01                                 |
| Heptachlor Epoxide                    | µg/L      | 24-hr Composite      | 0.02                                 |
| Lindane (gamma-Hexachlorocyclohexane) | µg/L      | 24-hr Composite      | 0.5                                  |
| PCB-1016                              | µg/L      | 24-hr Composite      | 0.5                                  |
| PCB-1221                              | µg/L      | 24-hr Composite      | 0.5                                  |
| PCB-1232                              | µg/L      | 24-hr Composite      | 0.5                                  |
| PCB-1242                              | µg/L      | 24-hr Composite      | 0.5                                  |
| PCB-1248                              | µg/L      | 24-hr Composite      | 0.5                                  |
| PCB-1254                              | µg/L      | 24-hr Composite      | 0.5                                  |
| PCB-1260                              | µg/L      | 24-hr Composite      | 0.5                                  |
| Toxaphene                             | µg/L      | 24-hr Composite      |                                      |
| Atrazine                              | µg/L      | 24-hr Composite      |                                      |
| Bentazon                              | µg/L      | 24-hr Composite      |                                      |
| Carbofuran                            | µg/L      | 24-hr Composite      |                                      |
| 2,4-D                                 | µg/L      | 24-hr Composite      |                                      |
| Dalapon                               | µg/L      | 24-hr Composite      |                                      |
| 1,2-Dibromo-3-chloropropane (DBCP)    | µg/L      | 24-hr Composite      |                                      |
| Di(2-ethylhexyl)adipate               | µg/L      | 24-hr Composite      |                                      |
| Dinoseb                               | µg/L      | 24-hr Composite      |                                      |
| Diquat                                | µg/L      | 24-hr Composite      |                                      |
| Endothal                              | µg/L      | 24-hr Composite      |                                      |
| Ethylene Dibromide                    | µg/L      | 24-hr Composite      |                                      |
| Methoxychlor                          | µg/L      | 24-hr Composite      |                                      |
| Molinate (Ordram)                     | µg/L      | 24-hr Composite      |                                      |
| Oxamyl                                | µg/L      | 24-hr Composite      |                                      |
| Picloram                              | µg/L      | 24-hr Composite      |                                      |
| Simazine (Princep)                    | µg/L      | 24-hr Composite      |                                      |
| Thiobencarb                           | µg/L      | 24-hr Composite      |                                      |
| 2,4,5-TP (Silvex)                     | µg/L      | 24-hr Composite      |                                      |
| Diazinon                              | µg/L      | 24-hr Composite      |                                      |
| Chlorpyrifos                          | µg/L      | 24-hr Composite      |                                      |
| Ammonia (as N)                        | mg/L      | 24-hr Composite      |                                      |
| Boron                                 | µg/L      | 24-hr Composite      |                                      |
| Chloride                              | mg/L      | 24-hr Composite      |                                      |
| Flow                                  | MGD       | Meter                |                                      |
| Hardness (as CaCO <sub>3</sub> )      | mg/L      | 24-hr Composite      |                                      |
| Foaming Agents (MBAS)                 | µg/L      | 24-hr Composite      |                                      |
| Mercury, Methyl                       | ng/L      | Grab                 |                                      |
| Nitrate (as N)                        | mg/L      | 24-hr Composite      |                                      |
| Nitrite (as N)                        | mg/L      | 24-hr Composite      |                                      |
| pH                                    | Std Units | Grab                 |                                      |
| Phosphorus, Total (as P)              | mg/L      | 24-hr Composite      |                                      |
| Specific conductance (EC)             | µmhos/cm  | 24-hr Composite      |                                      |

| Parameter                     | Units | Effluent Sample Type | Maximum Reporting Level <sup>1</sup> |
|-------------------------------|-------|----------------------|--------------------------------------|
| Sulfate                       | mg/L  | 24-hr Composite      |                                      |
| Sulfide (as S)                | mg/L  | 24-hr Composite      |                                      |
| Sulfite (as SO <sub>3</sub> ) | mg/L  | 24-hr Composite      |                                      |
| Temperature                   | °C    | Grab                 |                                      |
| Total Dissolved Solids (TDS)  | mg/L  | 24-hr Composite      |                                      |

The reporting levels required in this table for priority pollutant constituents are established based on Section 2.4.2 and Appendix 4 of the SIP.

- <sup>2</sup> In order to verify if bis (2-ethylhexyl) phthalate is truly present, the Discharger shall take steps to assure that sample containers, sampling apparatus, and analytical equipment are not sources of the detected contaminant.
- <sup>3</sup> The Discharger is not required to conduct effluent monitoring for constituents that have already been sampled in a given month, as required in Table E-3, except for hardness, pH, and temperature, which shall be conducted concurrently with the effluent sampling.
- <sup>4</sup> 24-hour flow proportional composite.

## X. REPORTING REQUIREMENTS

### A. General Monitoring and Reporting Requirements

1. The Discharger shall comply with all Standard Provisions (Attachment D) related to monitoring, reporting, and recordkeeping.
2. Upon written request of the Central Valley Water Board, the Discharger shall submit a summary monitoring report. The report shall contain both tabular and graphical summaries of the monitoring data obtained during the previous year(s).
3. **Compliance Time Schedules.** For compliance time schedules included in the Order, the Discharger shall submit to the Central Valley Water Board, on or before each compliance due date, the specified document or a written report detailing compliance or noncompliance with the specific date and task. If noncompliance is reported, the Discharger shall state the reasons for noncompliance and include an estimate of the date when the Discharger will be in compliance. The Discharger shall notify the Central Valley Water Board by letter when it returns to compliance with the compliance time schedule.
4. The Discharger shall report to the Central Valley Water Board any toxic chemical release data it reports to the State Emergency Response Commission within 15 days of reporting the data to the Commission pursuant to section 313 of the "*Emergency Planning and Community Right to Know Act*" of 1986.

### B. Self-Monitoring Reports (SMRs)

1. The Discharger shall electronically submit SMRs using the State Water Board's California Integrated Water Quality System (CIWQS) Program Web site (<http://www.waterboards.ca.gov/ciwqs/index.html>). The CIWQS Web site will provide additional information for SMR submittal in the event there will be a planned service interruption for electronic submittal.
2. The Discharger shall report in the SMR the results for all monitoring specified in this MRP under sections III through IX. The Discharger shall submit monthly, quarterly, and annual SMRs including the results of all required monitoring using U.S. EPA-approved test methods or other test methods specified in this Order. SMRs are to include all new monitoring results obtained since the last SMR was submitted. If the Discharger monitors any pollutant more frequently than required by this Order, the results of this monitoring shall be included in the calculations and reporting of the data submitted in the SMR.

3. Monitoring periods and reporting for all required monitoring shall be completed according to the following schedule:

**Table E-9. Monitoring Periods and Reporting Schedule**

| Sampling Frequency | Monitoring Period Begins On... | Monitoring Period                                                                                                     | SMR Due Date                                                    |
|--------------------|--------------------------------|-----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| Continuous         | Permit effective date          | All                                                                                                                   | Submit with monthly SMR                                         |
| 1/Day              | Permit effective date          | (Midnight through 11:59 PM) or any 24-hour period that reasonably represents a calendar day for purposes of sampling. | Submit with monthly SMR                                         |
| 1/Week             | Permit effective date          | Sunday through Saturday                                                                                               | Submit with monthly SMR                                         |
| 1/Month            | Permit effective date          | 1 <sup>st</sup> day of calendar month through last day of calendar month                                              | First day of second calendar month following month of sampling  |
| 1/Quarter          | Permit effective date          | 1 January through 31 March<br>1 April through 30 June<br>1 July through 30 September<br>1 October through 31 December | 1 May<br>1 August<br>1 November<br>1 February of following year |
| 1/Year             | Permit effective date          | 1 January through 31 December                                                                                         | 1 February of following year                                    |

4. **Reporting Protocols.** The Discharger shall report with each sample result the applicable Reporting Level (RL) and the current laboratory's Method Detection Limit (MDL), as determined by the procedure in 40 C.F.R. part 136.

The Discharger shall report the results of analytical determinations for the presence of chemical constituents in a sample using the following reporting protocols:

- Sample results greater than or equal to the RL shall be reported as measured by the laboratory (i.e., the measured chemical concentration in the sample).
- Sample results less than the RL, but greater than or equal to the laboratory's MDL, shall be reported as "Detected, but Not Quantified," or DNQ. The estimated chemical concentration of the sample shall also be reported.

For the purposes of data collection, the laboratory shall write the estimated chemical concentration next to DNQ. The laboratory may, if such information is available, include numerical estimates of the data quality for the reported result. Numerical estimates of data quality may be percent accuracy ( $\pm$  a percentage of the reported value), numerical ranges (low to high), or any other means considered appropriate by the laboratory.

- Sample results less than the laboratory's MDL shall be reported as "Not Detected," or ND.
- Dischargers are to instruct laboratories to establish calibration standards so that the Minimum Level (ML) value (or its equivalent if there is differential treatment of samples relative to calibration standards) is the lowest calibration standard. At no



time is the Discharger to use analytical data derived from extrapolation beyond the lowest point of the calibration curve.

5. **Multiple Sample Data.** When determining compliance with an AMEL, AWEL, or MDEL for priority pollutants and more than one sample result is available, the Discharger shall compute the arithmetic mean unless the data set contains one or more reported determinations of “Detected, but Not Quantified” (DNQ) or “Not Detected” (ND). In those cases, the Discharger shall compute the median in place of the arithmetic mean in accordance with the following procedure:
  - a. The data set shall be ranked from low to high, ranking the reported ND determinations lowest, DNQ determinations next, followed by quantified values (if any). The order of the individual ND or DNQ determinations is unimportant.
  - b. The median value of the data set shall be determined. If the data set has an odd number of data points, then the median is the middle value. If the data set has an even number of data points, then the median is the average of the two values around the middle unless one or both of the points are ND or DNQ, in which case the median value shall be the lower of the two data points where DNQ is lower than a value and ND is lower than DNQ.
6. The Discharger shall submit SMRs in accordance with the following requirements:
  - a. The Discharger shall arrange all reported data in a tabular format. The data shall be summarized to clearly illustrate whether the facility is operating in compliance with interim and/or final effluent limitations. The Discharger is not required to duplicate the submittal of data that is entered in a tabular format within CIWQS. When electronic submittal of data is required and CIWQS does not provide for entry into a tabular format within the system, the Discharger shall electronically submit the data in a tabular format as an attachment.
  - b. The Discharger shall attach a cover letter to the SMR. The information contained in the cover letter shall clearly identify violations of the WDR’s; discuss corrective actions taken or planned; and the proposed time schedule for corrective actions. Identified violations must include a description of the requirement that was violated and a description of the violation.
  - c. The Discharger shall attach all laboratory analysis sheets, including quality assurance/quality control information, with all its SMRs for which sample analyses were performed.
7. The Discharger shall submit in the SMRs calculations and reports in accordance with the following requirements:
  - a. **Mercury.** The Discharger shall calculate and report effluent total annual mass loading of total mercury in the December SMR. The total annual mass loading shall be calculated as specified in Section VII.B of the Limitations and Discharge Requirements.
  - b. **Average Dry Weather Flow.** The Discharger shall calculate and report the average dry weather flow for the effluent. The average dry weather flow shall be calculated as specified in Section VII.C and reported in the December SMR.
  - c. **Mass Loading Limitations.** For BOD<sub>5</sub>, TSS, and ammonia, the Discharger shall calculate and report the mass loading (lbs/day) in the SMRs. The mass loading shall be calculated as follows:

$$\text{Mass Loading (lbs/day)} = \text{Flow (MGD)} \times \text{Concentration (mg/L)} \times 8.34$$

When calculating daily mass loading, the daily average flow and constituent concentration shall be used. For weekly average mass loading, the weekly average flow and constituent concentration shall be used. For monthly average mass loading, the monthly average flow and constituent concentration shall be used.

- d. **Removal Efficiency (BOD<sub>5</sub> and TSS).** The Discharger shall calculate and report the percent removal of BOD<sub>5</sub> and TSS in the SMRs. The percent removal shall be calculated as specified in Section VII.A. of the Limitations and Discharge Requirements.
- e. **Total Coliform Organisms Effluent Limitations.** The Discharger shall calculate and report the 7-day median of total coliform organisms for the effluent. The 7 day median of total coliform organisms shall be calculated as specified in Section VII.D of the Limitations and Discharge Requirements.
- f. **Turbidity Receiving Water Limitations.** The Discharger shall calculate and report the turbidity change in the receiving water applicable to the natural turbidity condition specified in Section V.A.17.a-e. of the Limitations and Discharge Requirements.
- g. **Temperature Receiving Water Limitations.** The Discharger shall calculate and report the temperature change in the receiving water based on the difference in temperature at Monitoring Locations RSW-001 and RSW-002.

#### C. Discharge Monitoring Reports (DMR's)

- 1. The Discharger shall electronically submit DMRs together with SMRs using Electronic Self-Monitoring Reports module eSMR 2.5 or any upgraded version. Electronic submittal of DMRs will be in addition to electronic submittal of SMRs. Information about electronic submittal of DMRs is provided by the Discharge Monitoring Report website as follows: ([http://www.waterboards.ca.gov/water\\_issues/programs/discharge\\_monitoring/](http://www.waterboards.ca.gov/water_issues/programs/discharge_monitoring/)).

#### D. Other Reports

- 1. The Discharger shall report the results of any TRE required by Special Provisions – VI.C.2.a. The Discharger shall report the progress in satisfaction of compliance schedule dates specified in Special Provisions – VI.C.7. The Discharger shall submit reports with the first monthly SMR scheduled to be submitted on or immediately following the report due date.
- 2. Within **60 days** of permit adoption, the Discharger shall submit a report outlining reporting levels (RL's), method detection limits (MDL's), and analytical methods for the constituents listed in tables E-2, E-3, E-5, E-6, and E-7. In addition, no less than 6 months prior to conducting the effluent and receiving water characterization monitoring required in Section IX. D, the Discharger shall submit a report outlining RL's, MDL's, and analytical methods for the constituents listed in Table E-8. The Discharger shall comply with the monitoring and reporting requirements for CTR constituents as outlined in section 2.3 and 2.4 of the SIP. The maximum required reporting levels for priority pollutant constituents shall be based on the Minimum Levels (ML's) contained in Appendix 4 of the SIP, determined in accordance with Section 2.4.2 and Section 2.4.3 of the SIP. In accordance with Section 2.4.2 of the SIP, when there is more than one ML value for a given substance, the Central Valley Water Board shall include as RL's, in the permit, all ML values, and their associated analytical methods, listed in Appendix 4 that are below the calculated effluent limitation. The Discharger may select any one of those cited analytical methods for compliance determination. If no ML value is below the effluent limitation, then the Central Valley Water Board shall select as the RL, the lowest ML value, and its associated analytical method, listed in Appendix 4 for inclusion in the

permit. Table E-8 provides required maximum reporting levels in accordance with the SIP.

3. **Annual Operations Report.** By 30 January of each year, the Discharger shall submit a written report to the Executive Officer containing the following:
- a. The names, certificate grades, and general responsibilities of all persons employed at the Facility.
  - b. The names and telephone numbers of persons to contact regarding the plant for emergency and routine situations.
  - c. A statement certifying when the flow meter(s) and other monitoring instruments and devices were last calibrated, including identification of who performed the calibration.
  - d. A statement certifying whether the current operation and maintenance manual, and contingency plan, reflect the wastewater treatment plant as currently constructed and operated, and the dates when these documents were last revised and last reviewed for adequacy.

The Discharger may also be requested to submit an annual report to the Central Valley Water Board with both tabular and graphical summaries of the monitoring data obtained during the previous year. Any such request shall be made in writing. The report shall discuss the compliance record. If violations have occurred, the report shall also discuss the corrective actions taken and planned to bring the discharge into full compliance with the waste discharge requirements.

## ATTACHMENT F – FACT SHEET

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## ATTACHMENT F – FACT SHEET

As described in section II.B of this Order, the Central Valley Water Board incorporates this Fact Sheet as findings of the Central Valley Water Board supporting the issuance of this Order. This Fact Sheet includes the legal requirements and technical rationale that serve as the basis for the requirements of this Order.

This Order has been prepared under a standardized format to accommodate a broad range of discharge requirements for Dischargers in California. Only those sections or subsections of this Order that are specifically identified as “not applicable” have been determined not to apply to this Discharger. Sections or subsections of this Order not specifically identified as “not applicable” are fully applicable to this Discharger.

### I. PERMIT INFORMATION

The following table summarizes administrative information related to the Facility.

**Table F-1. Facility Information**

|                                                     |                                                                  |
|-----------------------------------------------------|------------------------------------------------------------------|
| <b>WDID</b>                                         | 5B340101001                                                      |
| <b>CIWQS Facility Place ID</b>                      | 226690                                                           |
| <b>Discharger</b>                                   | City of Galt                                                     |
| <b>Name of Facility</b>                             | City of Galt Wastewater Treatment Plant and Reclamation Facility |
| <b>Facility Address</b>                             | 10059 Twin Cities Road                                           |
|                                                     | Galt, CA 95632                                                   |
|                                                     | Sacramento County                                                |
| <b>Facility Contact, Title and Phone</b>            | Mark A. Clarkson, Utilities Manager, (209) 366-7260              |
| <b>Authorized Person to Sign and Submit Reports</b> | Mark A. Clarkson, Utilities Manager, (209) 366-7260              |
| <b>Mailing Address</b>                              | 495 Industrial Drive, Galt, CA 95632                             |
| <b>Billing Address</b>                              | 495 Industrial Drive, Galt, CA 95632                             |
| <b>Type of Facility</b>                             | Publicly Owned Treatment Works (POTW)                            |
| <b>Major or Minor Facility</b>                      | Major                                                            |
| <b>Threat to Water Quality</b>                      | 1                                                                |
| <b>Complexity</b>                                   | A                                                                |
| <b>Pretreatment Program</b>                         | N                                                                |
| <b>Recycling Requirements</b>                       | Producer                                                         |
| <b>Facility Permitted Flow</b>                      | 4.5 million gallons per day (MGD)                                |
| <b>Facility Design Flow</b>                         | Currently 3.0 MGD, planned: 4.5 MGD                              |
| <b>Watershed</b>                                    | Lower Cosumnes River – Lower Mokelumne                           |
| <b>Receiving Water</b>                              | Laguna Creek                                                     |
| <b>Receiving Water Type</b>                         | Inland Surface Water                                             |

- A. The City of Galt (hereinafter Discharger) is the owner and operator of The City of Galt Wastewater Treatment Plant and Reclamation Facility (hereinafter Facility), a Domestic wastewater Publicly Owned Treatment Works.

For the purposes of this Order, references to the “discharger” or “permittee” in applicable federal and state laws, regulations, plans, or policy are held to be equivalent to references to the Discharger herein.

- B.** The Facility discharges tertiary treated wastewater to Laguna Creek a water of the United States, which is tributary to the Cosumnes River within the Lower Cosumnes River – Lower Mokelumne River Watershed. The discharge to Laguna Creek is via a remanant channel of Skunk Creek. Skunk Creek was routed around the Facility when the Facility was originally constructed. The discharge flows in the remanant channel of Skunk Creek approximately 1000 meters before discharging to Laguna Creek. The Discharger was previously regulated by Order R5-2010-0099 and National Pollutant Discharge Elimination System (NPDES) Permit No. CA0081434 adopted on 23 September 2010 and expired on 1 September 2015. Attachment B provides a map of the area around the Facility. Attachment C provides a flow schematic of the Facility.

Prior to making any change in the point of discharge, place of use, or purpose of use of treated wastewater that results in a decrease of flow in any portion of a watercourse, the Discharger must file a petition with the State Water Board, Division of Water Rights, and receive approval for such a change. The State Water Board retains the jurisdictional authority to enforce such requirements under Water Code section 1211.

- C.** The Discharger filed a report of waste discharge (ROWD) and submitted an application for reissuance of its WDR's and NPDES permit on 3 March 2015. The application was deemed complete on 27 March 2015. In the March 2015 ROWD the Discharger requested separate permits to be issued by the Central Valley Water Board for the surface water and land discharges. Due to the complexities of the discharges to land for this Facility, separate permits have been issued for the surface water and land discharges. WDR Order R5-2015-0125 regulates the reuse of undisinfected secondary wastewater and the land application of biosolids on the Discharger's land application area. This Order regulates the wastewater treatment facility and discharges to Laguna Creek. A site visit was conducted on 4 June 2015 to observe operations and collect additional data to develop permit limitations and requirements for waste discharge.

## **II. FACILITY DESCRIPTION**

The Discharger owns and operates the City of Galt Wastewater Treatment Plant and Reclamation Facility (Facility), a Publicly-Owned Treatment Works servicing a population of approximately 24,000. The Facility is currently a 3.0 million gallon per day (mgd) average dry weather flow (ADWF) facility that provides a tertiary level of treatment for municipal wastewater from the City of Galt.

### **A. Description of Wastewater and Biosolids Treatment and Controls**

The major wastewater treatment facilities at the Facility have been in place since 1991. The facilities include coarse bar screening, grit removal, extended aeration in two oxidation ditches, solids settling in two secondary clarifiers, cloth media filtration, and ultraviolet (UV) disinfection. Recent major improvement projects have included the addition of filtration and UV light disinfection systems in 2011. The wastewater treatment process schematic and solids schematic can be found in Attachment C of this order.

The Discharger is in the process of constructing nitrification/denitrification improvements, which began in 2014 and will be completed in 2016. These improvements include construction of a third oxidation ditch and clarifier and improvements to all three oxidation ditches to allow for reliable simultaneous nitrification/denitrification. The immediate improvements also include minor upgrades to improve overall facility performance and reliability. One key improvement will be the installation of automated diversion facilities that will allow the Discharger to utilize the lined Auxiliary Basin for biosolids, dewatering filtrate, sludge lagoon decant, sludge dewatering bed decant, and influent flow equalization.

Solids are aerobically digested within the oxidation ditches, which are operated with a mean cell residence time (MCRT) of at least 25 days to stabilize the wastewater solids. A portion of the stabilized sludge is removed (“wasted”) from the process on a daily basis to maintain the desired mixed liquor suspended solids concentration in the oxidation ditches. This portion of the solids stream is called waste activated sludge (WAS) and is pumped into the two existing, high-density polyethylene (HDPE) membrane-lined storage lagoons.

Sludge is directed to one lagoon at a time. As the lagoons are filling, supernatant is decanted from the sludge and is returned to the oxidation ditches for further treatment. Once full, solids in the lagoons are blended using the mixers installed in each basin to create a uniformly mixed sludge that is sent to the Facility’s dewatering beds. Supplemental mechanical dewatering equipment is also being installed as part of the improvement project to increase the dewatering capacity of the Facility.

The combination of (1) stabilization of solids within the oxidation ditches and the storage lagoons, and (2) the Discharger’s disposal practices qualifies the biosolids as “Class B” biosolids in accordance with the USEPA’s regulations as established in Code of Federal Regulations, Title 40 Section 503.

A portion of the Discharger’s effluent is used to irrigate the land application area, which consists of approximately 164 acres of agricultural fields located adjacent to the main Facility site. Animal feed crops are grown in the land application area, and “Undisinfected Secondary Recycled Water” is used for irrigation. With the recent construction of the dewatering system, the Discharger has begun applying solids via surface spreading, followed by tilling, to incorporate the biosolids into the soil of the land application area fields. The application of undisinfected secondary recycled water and biosolids to the Discharge’s land application area is regulated by separate WDR Order R5-2015-0125.

Undisinfected secondary effluent reused for irrigation is typically directed from the filter influent pump station to the Effluent Storage Reservoir where it is pumped to the land application area. The Discharger can also direct disinfected tertiary effluent to the Effluent Storage Reservoir for irrigation. The Effluent Storage Reservoir is the primary storage facility, and currently has approximately 220 acre-feet of available storage capacity, with 2 feet of freeboard. Flows from the Effluent Storage Reservoir may also be directed to three onsite ponds for additional storage prior to being returned to the Effluent Storage Reservoir. (The Discharger historically operated four storage ponds; however, Pond 2 has recently been repurposed for a solar facility.) Because the Discharger has the ability to discharge any flows not needed for irrigation to surface water under this Order, the storage volume available in the Effluent Storage Reservoir is more than adequate to meet the needs of the land application area, so the storage ponds are rarely used. Furthermore, in the event of an emergency the Effluent Storage Reservoir and storage ponds may be used to store untreated or partially treated wastewater. Under these circumstances all wastewater within the ponds is returned to the Facility headworks for treatment.

## **B. Discharge Points and Receiving Waters**

1. The Facility is located in Section 9, T5N, R6E, MDB&M, as shown in Attachment B, a part of this Order.
2. Treated municipal wastewater is discharged at Discharge Point No. 001 to Laguna Creek, a water of the United States and tributary to the Cosumnes River at a point latitude 38° 18’ 14.88” N and longitude 121° 19’ 55.87” W.



### C. Summary of Existing Requirements and Self-Monitoring Report (SMR) Data

Effluent limitations contained in Order R5-2010-0099 for discharges from Discharge Point No. 001 (Monitoring Location EFF-001) and representative monitoring data from the term of Order R5-2010-0099 are as follows:

**Table F-2. Historic Effluent Limitations and Monitoring Data**

| Parameter                                | Units                | Effluent Limitation |                  |                                     | Monitoring Data<br>(April 2010 – April 2014) |                                  |                         |
|------------------------------------------|----------------------|---------------------|------------------|-------------------------------------|----------------------------------------------|----------------------------------|-------------------------|
|                                          |                      | Average Monthly     | Average Weekly   | Maximum Daily                       | Highest Average Monthly Discharge            | Highest Average Weekly Discharge | Highest Daily Discharge |
| Biochemical Oxygen Demand (5-day @ 25°C) | mg/L                 | 10                  | 15               | 20                                  | 5.6                                          | 5.6                              | 5.6                     |
|                                          | lbs/day <sup>1</sup> | 375                 | 560              | 750                                 | 65.0                                         | 92.30                            | 92.30                   |
|                                          | lbs/day <sup>2</sup> | 250 <sup>3</sup>    | 375 <sup>3</sup> | 500 <sup>3</sup>                    |                                              |                                  |                         |
|                                          | % Removal            | 85                  | --               | --                                  | 98.8 <sup>4</sup>                            | --                               | --                      |
| Total Suspended Solids                   | mg/L                 | 10                  | 15               | 20                                  | 2.0                                          | 5.0                              | 5.0                     |
|                                          | lbs/day <sup>1</sup> | 375                 | 560              | 750                                 | 35.0                                         | 54.2                             | 54.2                    |
|                                          | lbs/day <sup>2</sup> | 250 <sup>3</sup>    | 375 <sup>3</sup> | 500 <sup>3</sup>                    |                                              |                                  |                         |
|                                          | % Removal            | 85                  | --               | --                                  | 97.8 <sup>4</sup>                            | --                               | --                      |
| pH                                       | standard units       | --                  | --               | 6.5 – 8.2                           | --                                           | --                               | 7.1 – 8.9               |
| Ammonia Nitrogen, Total (as N)           | mg/L                 | 1.1                 | --               | 2.1                                 | 6.7                                          | --                               | 19.0                    |
|                                          | mg/L                 | --                  | --               | 14 <sup>3</sup>                     |                                              |                                  |                         |
|                                          | lbs/day <sup>1</sup> | 183                 | --               | 350                                 | 133                                          | --                               | 380                     |
| Aluminum, Total Recoverable              | µg/L                 | 200 <sup>5</sup>    | --               | --                                  | 54 <sup>6</sup>                              | --                               | --                      |
| Arsenic                                  | mg/L                 | 10                  | --               | --                                  | 20.8                                         | --                               | 27.0                    |
| Bis(2-ethylhexyl)phthalate               | µg/L                 | 1.8                 | --               | 3.6                                 | 1.0                                          | --                               | 1.4                     |
| Carbon Tetrachloride                     | µg/L                 | 0.25                | --               | 0.5                                 | <0.077                                       | --                               | <0.077                  |
| Chlorodibromomethane                     | µg/L                 | 0.41                | --               | 0.83                                | <0.049                                       | --                               | <0.049                  |
| Copper, Total Recoverable                | µg/L                 | 3.1                 | --               | 4.3                                 | 10.0                                         | --                               | 17.0                    |
| Cyanide, Total (as CN)                   | µg/L                 | 3.4                 | --               | 9.6                                 | <0.031                                       | --                               | <0.031                  |
| Dichlorobromomethane                     | µg/L                 | 0.56                | --               | 1.3                                 | 41.9                                         | --                               | 41.9                    |
| Iron, Total Recoverable                  | µg/L                 | 300 <sup>5</sup>    | --               | --                                  | 56 <sup>6</sup>                              | --                               | --                      |
| Lead, Total Recoverable                  | µg/L                 | 0.6                 | --               | 1.0                                 | 0.34                                         | --                               | 0.34                    |
| Manganese, Total Recoverable             | µg/L                 | 50 <sup>5</sup>     | --               | --                                  | 7.0 <sup>6</sup>                             | --                               | --                      |
| Mercury, Total Recoverable               | lbs/year             | 0.05 <sup>7</sup>   | --               | --                                  | 0.03 <sup>8</sup>                            | --                               | --                      |
| Nitrate plus Nitrite (as N)              | mg/L                 | 10                  | --               | --                                  | 28                                           | --                               | --                      |
| Total Coliform Organisms                 | MPN/100 mL           | --                  | 2.2 <sup>9</sup> | 23 <sup>10</sup> /240 <sup>11</sup> | --                                           | --                               | 2.0                     |

| Parameter                   | Units         | Effluent Limitation |                   |                                    | Monitoring Data<br>(April 2010 – April 2014) |                                           |                                     |
|-----------------------------|---------------|---------------------|-------------------|------------------------------------|----------------------------------------------|-------------------------------------------|-------------------------------------|
|                             |               | Average<br>Monthly  | Average<br>Weekly | Maximum<br>Daily                   | Highest<br>Average<br>Monthly<br>Discharge   | Highest<br>Average<br>Weekly<br>Discharge | Highest<br>Daily<br>Discharge       |
| Acute Toxicity              | %<br>Survival | --                  | --                | 70 <sup>12</sup> /90 <sup>13</sup> | --                                           | --                                        | 95 <sup>14</sup> /100 <sup>15</sup> |
| Total Residual Chlorine     | mg/L          | 0.011 <sup>16</sup> | --                | 0.019 <sup>17</sup>                | --                                           | --                                        | --                                  |
| Average Dry Weather<br>Flow | MGD           |                     |                   | 4.5 <sup>18</sup>                  | --                                           | --                                        | 3.5 <sup>20</sup>                   |
|                             |               | --                  | --                | 3.0 <sup>3, 19</sup>               |                                              |                                           |                                     |

1. Based on a design flow of 4.5 MGD.
2. Based on a design flow of 3.0 MGD.
3. Interim effluent limitation.
4. Represents the minimum reported percent removal based on data from December 2011 through November 2014.
5. Applied as an annual average effluent limitation.
6. Represents the maximum observed annual average concentration.
7. The total annual mass discharge of total mercury shall not exceed 0.05 pounds.
8. Represents the maximum observed total annual mercury discharged.
9. Applied as a 7-day median effluent limitation.
10. Not to be exceeded more than once in any 30-day period.
11. Applied as an instantaneous maximum effluent limitation.
12. Minimum reported for any one bioassay.
13. Median for any three consecutive bioassays.
14. Represents lowest reported minimum for one bioassay.
15. Represents lowest reported median for three consecutive bioassays.
16. Applied as a 4-day average effluent limitation.
17. Applied as a 1-hour average effluent limitation.
18. The average dry weather flow shall not exceed 4.5 MGD.
19. The average dry weather flow shall not exceed 3.0 MGD
20. Maximum average dry weather flow.

#### D. Compliance Summary

The Discharger was issued Administrative Civil Liability Complaint (ACLC) R5-2013-0564 while regulated by WDR Order No. R5-2010-0099 for violations occurring between 12 November 2010 and 30 November 2012. This ACLC addressed violations of effluent limitations for arsenic, copper, cyanide, and pH, and was settled through payment. The Discharger was also issued Time Schedule Order R5-2015-0900 that contains compliance schedules for arsenic, copper and nitrate plus nitrite.

#### E. Planned Changes

##### Arsenic Removal Improvements.

Arsenic is naturally present in the City of Galt's groundwater based drinking water supply. The Discharger has historically employed multiple wells to supply drinking water to the residents and businesses of Galt and surrounding communities. In 2009, the Discharger employed well head filtration treatment at three of the City's well sites to reduce arsenic concentrations in the drinking water supply. The wellhead filtration systems remove arsenic from the potable water via media filtration, which requires regular backwashing of the media to ensure optimal performance. Backwash from the well head filtration system is currently conveyed to the Facility through discharge to the sanitary sewer system. Upon arrival to the Facility, it was originally expected that the arsenic would be sufficiently removed with the installation of tertiary filtration. However,

anaerobic conditions in the collection system cause the arsenic to resolubilize during transport, and the tertiary filtration cannot remove dissolved arsenic effectively.

The Discharger is planning to install a new deep well that will eliminate the need for wellhead filtration systems, except under peak flow/emergency conditions. In addition, the Discharger intends to evaluate hauling of the arsenic-laden solids contained in the backwash directly to the Facility which would reduce discharge of the solids to the sanitary sewer system. The Discharger has a compliance schedule implemented in TSO R5-2015-0124 in order to evaluate and implement alternatives for reducing arsenic levels in the influent that results in long-term permit compliance.

The Discharger is exploring the following options for well backwash introduction into the Facility's treatment train:

- a. Discharge backwash directly to Facility headworks or the Auxiliary Basin (for gradual/continuous discharge to headworks).
- b. Mix backwash with the Facility's biosolids in a day mixing tank or the Sludge lagoon and dewater the combined solids. Combined solids would then be land applied in accordance with the Discharger's current biosolids land application practices as well as biosolids limitations and loading rates.
- c. Discharge backwash to the Facility's Effluent Storage Reservoir, which is used to hold Facility effluent for subsequent agricultural irrigation on the land application area.

This Order allows the discharge of well head filter backwash to the above listed portions of the Facility's treatment train through the expiration of the compliance schedule for arsenic implemented in TSO R5-2015-0124.

### **III. APPLICABLE PLANS, POLICIES, AND REGULATIONS**

The requirements contained in this Order are based on the requirements and authorities described in this section.

#### **A. Legal Authorities**

This Order serves as WDR's pursuant to article 4, chapter 4, division 7 of the California Water Code (commencing with section 13260). This Order is also issued pursuant to section 402 of the federal Clean Water Act (CWA) and implementing regulations adopted by the U.S. EPA and chapter 5.5, division 7 of the Water Code (commencing with section 13370). It shall serve as an NPDES permit for point source discharges from this Facility to surface waters.

#### **B. California Environmental Quality Act (CEQA)**

Under Water Code section 13389, this action to adopt an NPDES permit is exempt from the provisions of Chapter 3 of CEQA, (commencing with section 21100) of Division 13 of the Public Resources Code.

#### **C. State and Federal Laws, Regulations, Policies, and Plans**

1. **Water Quality Control Plans.** Requirements of this Order specifically implement the applicable Water Quality Control Plans.
  - a. **Basin Plan.** The Central Valley Water Board adopted a Water Quality Control Plan for the Water Quality Control Plan, Fourth Edition (Revised October 2011), for the Sacramento and San Joaquin River Basins (hereinafter Basin Plan) that designates beneficial uses, establishes water quality objectives, and contains implementation

programs and policies to achieve those objectives for all waters addressed through the plan. Requirements in this Order implement the Basin Plan.

The Basin Plan at II-2.00 states that the beneficial uses of any specifically identified water body generally apply to its tributary streams. The Basin Plan in Table II-1, Section II, does not specifically identify beneficial uses for Laguna Creek, but does identify present and potential uses for the Cosumnes River, to which Laguna Creek is tributary. In addition, the Basin Plan implements State Water Board Resolution 88-63, which established state policy that all waters, with certain exceptions, should be considered suitable or potentially suitable for municipal or domestic supply. Thus, beneficial uses applicable to Laguna Creek are as follows:

**Table F-3. Basin Plan Beneficial Uses**

| Discharge Point | Receiving Water Name | Beneficial Use(s)                                                                                                                                                                                                                                                   |
|-----------------|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 001             | Laguna Creek         | <u>Existing:</u><br>Municipal and domestic water supply (MUN), agricultural irrigation and stock watering (AGR), contact (REC-1) and non-contact (REC-2) water recreation, warm freshwater habitat (WARM); cold freshwater habitat (COLD); wildlife habitat (WILD). |

2. **National Toxics Rule (NTR) and California Toxics Rule (CTR).** U.S. EPA adopted the NTR on 22 December 1992, and later amended it on 4 May 1995 and 9 November 1999. About forty criteria in the NTR applied in California. On 18 May 2000, U.S. EPA adopted the CTR. The CTR promulgated new toxics criteria for California and, in addition, incorporated the previously adopted NTR criteria that were applicable in the state. The CTR was amended on 13 February 2001. These rules contain federal water quality criteria for priority pollutants.
3. **State Implementation Policy.** On 2 March 2000, the State Water Board adopted the Policy for Implementation of Toxics Standards for Inland Surface Waters, Enclosed Bays, and Estuaries of California (State Implementation Policy or SIP). The SIP became effective on 28 April 2000, with respect to the priority pollutant criteria promulgated for California by the U.S. EPA through the NTR and to the priority pollutant objectives established by the Central Valley Water Board in the Basin Plan. The SIP became effective on 18 May 2000, with respect to the priority pollutant criteria promulgated by the U.S. EPA through the CTR. The State Water Board adopted amendments to the SIP on 24 February 2005, that became effective on 13 July 2005. The SIP establishes implementation provisions for priority pollutant criteria and objectives and provisions for chronic toxicity control. Requirements of this Order implement the SIP.
4. **Antidegradation Policy.** Federal regulation 40 C.F.R. section 131.12 requires that the state water quality standards include an antidegradation policy consistent with the federal policy. The State Water Board established California's antidegradation policy in State Water Board Resolution 68-16 ("Statement of Policy with Respect to Maintaining High Quality of Waters in California"). Resolution 68-16 is deemed to incorporate the federal antidegradation policy where the federal policy applies under federal law. Resolution 68-16 requires that existing water quality be maintained unless degradation is justified based on specific findings. The Central Valley Water Board's Basin Plan implements, and incorporates by reference, both the State and federal antidegradation policies. The permitted discharge must be consistent with the antidegradation provision of 40 C.F.R. section 131.12 and State Water Board Resolution 68-16.

5. **Anti-Backsliding Requirements.** Sections 402(o) and 303(d)(4) of the CWA and federal regulations at 40 C.F.R. section 122.44(l) restrict backsliding in NPDES permits. These anti-backsliding provisions require that effluent limitations in a reissued permit must be as stringent as those in the previous permit, with some exceptions in which limitations may be relaxed.
6. **Domestic Water Quality.** In compliance with Water Code section 106.3, it is the policy of the State of California that every human being has the right to safe, clean, affordable, and accessible water adequate for human consumption, cooking, and sanitary purposes. This Order promotes that policy by requiring discharges to meet maximum contaminant levels designed to protect human health and ensure that water is safe for domestic use.
7. **Endangered Species Act Requirements.** This Order does not authorize any act that results in the taking of a threatened or endangered species or any act that is now prohibited, or becomes prohibited in the future, under either the California Endangered Species Act (Fish and Game Code, §§ 2050 to 2097) or the Federal Endangered Species Act (16 U.S.C.A. §§ 1531 to 1544). This Order requires compliance with effluent limits, receiving water limits, and other requirements to protect the beneficial uses of waters of the state. The Discharger is responsible for meeting all requirements of the applicable Endangered Species Act.
8. **Emergency Planning and Community Right to Know Act.** Section 13263.6(a) of the Water Code, requires that *"the Regional Water Board shall prescribe effluent limitations as part of the waste discharge requirements of a POTW for all substances that the most recent toxic chemical release data reported to the state emergency response commission pursuant to Section 313 of the Emergency Planning and Community Right to Know Act of 1986 (42 U.S.C. Sec. 11023) (EPCRA) indicate as discharged into the POTW, for which the State Water Board or the Regional Water Board has established numeric water quality objectives, and has determined that the discharge is or may be discharged at a level which will cause, have the reasonable potential to cause, or contribute to, an excursion above any numeric water quality objective"*.

The most recent toxic chemical data report does not indicate any reportable off-site releases or discharges to the collection system for this Facility. Therefore, a reasonable potential analysis based on information from EPCRA cannot be conducted. Based on information from EPCRA, there is no reasonable potential to cause or contribute to an excursion above any numeric water quality objectives included within the Basin Plan or in any State Water Board plan, so no effluent limitations are included in this permit pursuant to Water Code section 13263.6(a).

However, as detailed elsewhere in this Order, available effluent data indicate that there are constituents present in the effluent that have a reasonable potential to cause or contribute to exceedances of water quality standards and require inclusion of effluent limitations based on federal and state laws and regulations.

9. **Storm Water Requirements.** USEPA promulgated federal regulations for storm water on 16 November 1990 in 40 C.F.R. parts 122, 123, and 124. The NPDES Industrial Storm Water Program regulates storm water discharges from wastewater treatment facilities. Wastewater treatment plants are applicable industries under the storm water program and are obligated to comply with the federal regulations. The State Water Resources Control Board Water Quality Order 2014-0057-DWQ, General Permit for Storm Water Discharges Associated with Industrial Activities (NPDES General Permit No. CAS000001), does not require facilities to obtain coverage if discharges of storm water are regulated under another individual or general NPDES permit adopted by the State Water Board or Regional Water Board (Finding I.B.20). All storm water from the

Facility area is diverted to, and contained within, the Discharger's storage facilities. In July 2015 the Discharger submitted a No Discharge Technical Report to the Central Valley Water Board. Accordingly, the State Water Board issued a Notice of Non-Applicability (NONA ID 5S34NNA000057) to the Discharger, documenting the Discharger's exemption from Order 2014-0057-DWQ. Therefore, coverage under the General Storm Water Permit is not required.

10. **Impaired Water Bodies on CWA 303(d) List.** Under section 303(d) of the 1972 CWA, states, territories and authorized tribes are required to develop lists of water quality limited segments. The waters on these lists do not meet water quality standards, even after point sources of pollution have installed the minimum required levels of pollution control technology. On 11 October 2011 U.S. EPA gave final approval to California's 2008-2010 section 303(d) List of Water Quality Limited Segments. The Basin Plan references this list of Water Quality Limited Segments (WQLSs), which are defined as "...those sections of lakes, streams, rivers or other fresh water bodies where water quality does not meet (or is not expected to meet) water quality standards even after the application of appropriate limitations for point sources (40 C.F.R. Part 130, et seq.)." The Basin Plan also states, "Additional treatment beyond minimum federal standards will be imposed on dischargers to [WQLSs]. Dischargers will be assigned or allocated a maximum allowable load of critical pollutants so that water quality objectives can be met in the segment." Laguan Creek is not listed on the 303(d) list, however, Laguna Creek is tributary to the Lower Cosumnes River, which is listed on the 303(d) list. The listing for the Lower Cosumnes River (below Michigan Bar; partly in Delta Waterways, eastern portion) includes: *Escherichia coli* (*E. coli*), invasive species, and sediment toxicity.
11. **Total Maximum Daily Loads (TMDL's).** U.S. EPA requires the Central Valley Water Board to develop TMDL's for each 303(d) listed pollutant and water body combination. Table F-5, below, identifies the 303(d) listings and the status of each TMDL.

**Table F-4. 303 (d) List for the Lower Cosumnes (below Michigan Bar; partly in Delta Waterways, eastern portion))**

| Pollutant         | Potential Sources | TMDL Completion <sup>1</sup> |
|-------------------|-------------------|------------------------------|
| E. coli           | Source Unknown    | (2021)                       |
| Invasive Species  | Source Unknown    | (2019)                       |
| Sediment Toxicity | Agriculture       | (2021)                       |

<sup>1</sup> Dates in parenthesis are proposed TMDL completion dates.

12. The 303(d) listings and TMDL's have been considered in the development of the Order. A pollutant-by-pollutant evaluation of each pollutant of concern is described in section IV.C.3 of this Fact Sheet.

#### IV. RATIONALE FOR EFFLUENT LIMITATIONS AND DISCHARGE SPECIFICATIONS

Effluent limitations and toxic and pretreatment effluent standards established pursuant to sections 301 (Effluent Limitations), 302 (Water Quality Related Effluent Limitations), 304 (Information and Guidelines), and 307 (Toxic and Pretreatment Effluent Standards) of the CWA and amendments thereto are applicable to the discharge.

The CWA mandates the implementation of effluent limitations that are as stringent as necessary to meet water quality standards established pursuant to state or federal law [33 U.S.C., §1311(b)(1)(C); 40 C.F.R. § 122.44(d)(1)]. NPDES permits must incorporate discharge limits necessary to ensure that water quality standards are met. This requirement applies to narrative

criteria as well as to criteria specifying maximum amounts of particular pollutants. Pursuant to federal regulations, 40 C.F.R. section 122.44(d)(1)(i), NPDES permits must contain limits that control all pollutants that “are or may be discharged at a level which will cause, have the reasonable potential to cause, or contribute to an excursion above any state water quality standard, including state narrative criteria for water quality.” Federal regulations, 40 C.F.R. section 122.44(d)(1)(vi), further provide that “[w]here a state has not established a water quality criterion for a specific chemical pollutant that is present in an effluent at a concentration that causes, has the reasonable potential to cause, or contributes to an excursion above a narrative criterion within an applicable State water quality standard, the permitting authority must establish effluent limits.”

The CWA requires point source dischargers to control the amount of conventional, non-conventional, and toxic pollutants that are discharged into the waters of the United States. The control of pollutants discharged is established through effluent limitations and other requirements in NPDES permits. There are two principal bases for effluent limitations in the Code of Federal Regulations: 40 C.F.R. section 122.44(a) requires that permits include applicable technology-based limitations and standards; and 40 C.F.R. section 122.44(d) requires that permits include WQBEL's to attain and maintain applicable numeric and narrative water quality criteria to protect the beneficial uses of the receiving water where numeric water quality objectives have not been established. The Basin Plan at page IV-17.00 contains an implementation policy, “Policy for Application of Water Quality Objectives” that specifies that the Central Valley Water Board “will, on a case-by-case basis, adopt numerical limitations in orders which will implement the narrative objectives.” This Policy complies with 40 C.F.R. section 122.44(d)(1). With respect to narrative objectives, the Central Valley Water Board must establish effluent limitations using one or more of three specified sources, including: (1) USEPA's published water quality criteria, (2) a proposed state criterion (i.e., water quality objective) or an explicit state policy interpreting its narrative water quality criteria (i.e., the Central Valley Water Board's “Policy for Application of Water Quality Objectives”)(40 C.F.R. § 122.44(d)(1)(vi)(A), (B) or (C)), or (3) an indicator parameter.

The Basin Plan includes numeric site-specific water quality objectives and narrative objectives for toxicity, chemical constituents, discoloration, radionuclides, and tastes and odors. The narrative toxicity objective states: “*All waters shall be maintained free of toxic substances in concentrations that produce detrimental physiological responses in human, plant, animal, or aquatic life.*” (Basin Plan at III-8.00) The Basin Plan states that material and relevant information, including numeric criteria, and recommendations from other agencies and scientific literature will be utilized in evaluating compliance with the narrative toxicity objective. The narrative chemical constituents objective states that waters shall not contain chemical constituents in concentrations that adversely affect beneficial uses. At minimum, “...*water designated for use as domestic or municipal supply (MUN) shall not contain concentrations of chemical constituents in excess of the maximum contaminant levels (MCLs)*” in Title 22 of CCR. The Basin Plan further states that, to protect all beneficial uses, the Central Valley Water Board may apply limits more stringent than MCLs. The narrative tastes and odors objective states: “*Water shall not contain taste- or odor-producing substances in concentrations that impart undesirable tastes or odors to domestic or municipal water supplies or to fish flesh or other edible products of aquatic origin, or that cause nuisance, or otherwise adversely affect beneficial uses.*”

#### **A. Discharge Prohibitions**

- 1. Prohibition III.A (No discharge or application of waste other than that described in this Order).** This prohibition is based on Water Code section 13260 that requires filing of a ROWD before discharges can occur. The Discharger submitted a ROWD for the discharges described in this Order; therefore, discharges not described in this Order are prohibited.

2. **Prohibition III.B (No bypasses or overflow of untreated wastewater, except under the conditions at CFR section 122.41(m)(4)).** As stated in section I.G of Attachment D, Standard Provisions, this Order prohibits bypass from any portion of the treatment facility. Federal regulations, 40 C.F.R. section 122.41(m), define “bypass” as the intentional diversion of waste streams from any portion of a treatment facility. This section of the federal regulations, 40 C.F.R. section 122.41(m)(4), prohibits bypass unless it is unavoidable to prevent loss of life, personal injury, or severe property damage. In considering the Regional Water Board’s prohibition of bypasses, the State Water Board adopted a precedential decision, Order No. WQO 2002-0015, which cites the federal regulations, 40 C.F.R. section 122.41(m), as allowing bypass only for essential maintenance to assure efficient operation.
3. **Prohibition III.C (No controllable condition shall create a nuisance).** This prohibition is based on Water Code section 13050 that requires water quality objectives established for the prevention of nuisance within a specific area. The Basin Plan prohibits conditions that create a nuisance
4. **Prohibition III.D (No inclusion of pollutant free wastewater shall cause improper operation of the Facility’s systems).** This prohibition is based on 40 C.F.R. section 122.41 et seq. that requires the proper design and operation of treatment facilities

## **B. Technology-Based Effluent Limitations**

### **1. Scope and Authority**

Section 301(b) of the CWA and implementing U.S. EPA permit regulations at 40 C.F.R. section 122.44 require that permits include conditions meeting applicable technology-based requirements at a minimum, and any more stringent effluent limitations necessary to meet applicable water quality standards. The discharge authorized by this Order must meet minimum federal technology-based requirements based on Secondary Treatment Standards at 40 C.F.R. part 133.

Regulations promulgated in 40 C.F.R. section 125.3(a)(1) require technology-based effluent limitations for municipal Dischargers to be placed in NPDES permits based on Secondary Treatment Standards or Equivalent to Secondary Treatment Standards.

The Federal Water Pollution Control Act Amendments of 1972 (PL 92-500) established the minimum performance requirements for POTW’s [defined in section 304(d)(1)]. Section 301(b)(1)(B) of that Act requires that such treatment works must, as a minimum, meet effluent limitations based on secondary treatment as defined by the U.S. EPA Administrator.

Based on this statutory requirement, U.S. EPA developed secondary treatment regulations, which are specified in 40 C.F.R. part 133. These technology-based regulations apply to all municipal wastewater treatment plants and identify the minimum level of effluent quality attainable by secondary treatment in terms of biochemical oxygen demand (BOD<sub>5</sub>), total suspended solids (TSS), and pH.

### **2. Applicable Technology-Based Effluent Limitations**

- a. **BOD<sub>5</sub> and TSS.** Federal regulations at 40 C.F.R. part 133, establish the minimum weekly and monthly average level of effluent quality attainable by secondary treatment for BOD<sub>5</sub> and TSS. A daily maximum effluent limitation for BOD<sub>5</sub> and TSS is also included in the Order to ensure that the treatment works are not organically overloaded and operate in accordance with design capabilities. In addition, 40 C.F.R. section 133.102, in describing the minimum level of effluent



quality attainable by secondary treatment, states that the 30-day average percent removal shall not be less than 85 percent. This Order contains a limitation requiring an average of 85 percent removal of BOD<sub>5</sub> and TSS over each calendar month. This Order requires Water Quality Based Effluent Limitations (WQBEL's) that are equal to or more stringent than the secondary technology-based treatment described in 40 CFR Part 133 (See section IV.C.3.d of the Fact Sheet for a discussion on Pathogens which includes WQBEL's for BOD<sub>5</sub> and TSS).

- b. **Flow.** The Facility was designed to provide a tertiary level of treatment for up to a design flow of 4.5 mgd. Therefore, this Order contains an average dry weather discharge flow effluent limit of 4.5 mgd.
- c. **pH.** The secondary treatment regulations at 40 C.F.R. part 133 also require that pH be maintained between 6.0 and 9.0 standard units. This Order, however, requires more stringent WQBEL's for pH to comply with the Basin Plan's water quality objectives for pH.

**Summary of Technology-based Effluent Limitations  
Discharge Point EFF-001**

**Table F-5. Summary of Technology-based Effluent Limitations**

| Parameter                                             | Units          | Effluent Limitations |                |               |                       |                       |
|-------------------------------------------------------|----------------|----------------------|----------------|---------------|-----------------------|-----------------------|
|                                                       |                | Average Monthly      | Average Weekly | Maximum Daily | Instantaneous Minimum | Instantaneous Maximum |
| Average Dry Weather Flow                              | MGD            | 3.0 <sup>1</sup>     | --             | --            | --                    | --                    |
|                                                       |                | 4.5 <sup>2</sup>     |                |               |                       |                       |
| Biochemical Oxygen Demand (5-Day @ 20°C) <sup>3</sup> | mg/L           | 30                   | 45             | --            | --                    | --                    |
| pH <sup>3</sup>                                       | standard units | --                   | --             | --            | 6.0                   | 9.0                   |
| Total Suspended Solids <sup>3</sup>                   | mg/L           | 30                   | 45             | --            | --                    | --                    |

<sup>1</sup> Effective until the Discharger demonstrates compliance with Special Provision VI.C.6.b of this Order, the average dry weather flow shall not exceed 3.0 MGD.

<sup>2</sup> Effective upon compliance with Special Provision VI.C.6.b of this Order, the average dry weather flow shall not exceed 4.5 MGD.

<sup>3</sup> More stringent WQBEL's are applicable to the discharge and are included in this Order, as described further in section IV.C.3 of this Fact Sheet.

**C. Water Quality-Based Effluent Limitations (WQBEL's)**

**1. Scope and Authority**

CWA Section 301(b) and 40 C.F.R. section 122.44(d) require that permits include limitations more stringent than applicable federal technology-based requirements where necessary to achieve applicable water quality standards. This Order contains requirements, expressed as a technology equivalence requirement, more stringent than secondary treatment requirements that are necessary to meet applicable water quality standards. The rationale for these requirements, which consist of tertiary treatment, is discussed in section IV.C.3. of this Fact Sheet.

Section 122.44(d)(1)(i) of 40 C.F.R. requires that permits include effluent limitations for all pollutants that are or may be discharged at levels that have the reasonable potential

to cause or contribute to an exceedance of a water quality standard, including numeric and narrative objectives within a standard. Where reasonable potential has been established for a pollutant, but there is no numeric criterion or objective for the pollutant, WQBEL's must be established using: (1) U.S. EPA criteria guidance under CWA section 304(a), supplemented where necessary by other relevant information; (2) an indicator parameter for the pollutant of concern; or (3) a calculated numeric water quality criterion, such as a proposed state criterion or policy interpreting the state's narrative criterion, supplemented with other relevant information, as provided in section 122.44(d)(1)(vi).

The process for determining reasonable potential and calculating WQBEL's when necessary is intended to protect the designated uses of the receiving water as specified in the Basin Plan, and achieve applicable water quality objectives and criteria that are contained in other state plans and policies, or any applicable water quality criteria contained in the CTR and NTR.

## **2. Applicable Beneficial Uses and Water Quality Criteria and Objectives**

The Basin Plan designates beneficial uses, establishes water quality objectives, and contains implementation programs and policies to achieve those objectives for all waters addressed through the plan. In addition, the Basin Plan implements State Water Board Resolution No. 88-63, which established state policy that all waters, with certain exceptions, should be considered suitable or potentially suitable for municipal or domestic supply.

The Basin Plan on page II-1.00 states: "*Protection and enhancement of existing and potential beneficial uses are primary goals of water quality planning...*" and with respect to disposal of wastewaters states that "*...disposal of wastewaters is [not] a prohibited use of waters of the State; it is merely a use which cannot be satisfied to the detriment of beneficial uses.*"

The federal CWA section 101(a)(2), states: "*it is the national goal that wherever attainable, an interim goal of water quality which provides for the protection and propagation of fish, shellfish, and wildlife, and for recreation in and on the water be achieved by July 1, 1983.*" Federal Regulations, developed to implement the requirements of the CWA, create a rebuttable presumption that all waters be designated as fishable and swimmable. Federal Regulations, 40 CFR sections 131.2 and 131.10, require that all waters of the State regulated to protect the beneficial uses of public water supply, protection and propagation of fish, shell fish and wildlife, recreation in and on the water, agricultural, industrial and other purposes including navigation. 40 C.F.R. section 131.3(e) defines existing beneficial uses as those uses actually attained after 28 November 1975, whether or not they are included in the water quality standards. Federal Regulation, 40 C.F.R. section 131.10 requires that uses be obtained by implementing effluent limitations, requires that all downstream uses be protected and states that in no case shall a state adopt waste transport or waste assimilation as a beneficial use for any waters of the United States.

- a. **Receiving Water and Beneficial Uses.** Refer to III.C.1. above for a complete description of the receiving water and beneficial uses.
- b. **Effluent and Ambient Background Data.** The reasonable potential analysis (RPA), as described in section IV.C.3 of this Fact Sheet, was based on data from December 2011 through November 2014, which includes effluent and ambient background data submitted in SMRs and the Report of Waste Discharge (ROWD).
- c. **Assimilative Capacity/Mixing Zone.** Laguna Creek is an ephemeral stream with little or no natural flow at times, therefore, no credit for receiving water dilution is

available. Dilution credits have not been allowed in this Order. The Central Valley Board finds that based on the available information and on the Discharger's application, that Laguna Creek, absent this and other NPDES discharges, is an ephemeral stream. The ephemeral nature of Laguna Creek means that the designated beneficial uses must be protected, but that no credit for receiving water dilution is available. Although the discharge, at times, maintains the aquatic habitat, constituents may not be discharged that may cause harm to aquatic life. At other times, natural flows within Laguna Creek help support the aquatic life. Both conditions may exist within a short time span, where Laguna Creek would be dry without the discharge or other NPDES discharges, and periods when sufficient background flows provide hydraulic continuity with the Cosumnes River. Dry conditions occur primarily in the summer months, but dry conditions may also occur throughout the year, particularly in low rainfall years. The lack of dilution results in more stringent effluent limitations to protect contact recreational uses, drinking water standards, agricultural water quality goals, and aquatic life. Significant dilution may occur during and immediately following high rainfall events.

- d. **Water Effects Ratio Study For Copper.** This Order allows for a site-specific water effects ratio (WER) of 15 to calculate the aquatic life criteria for copper based on the Discharger's *City of Galt Wastewater Treatment Plant Copper Water Effects Ratio Study Final Report* (February 2015). The Discharger's study followed U.S. EPA's 2001 *Streamlined Water-Effect Ratio Procedure for Discharges of Copper* (EPA 822-R-01-005). This Order calculates the aquatic life criteria for copper using a total recoverable WER of 15. A discussion on the calculation of the criteria for hardness dependent metals, such as copper, can be found in Section IV.C.2.f. below.
- e. **Conversion Factors.** The CTR contains aquatic life criteria for arsenic, cadmium, chromium III, chromium VI, copper, lead, nickel, silver, and zinc which are presented in dissolved concentrations. USEPA recommends conversion factors to translate dissolved concentrations to total concentrations. The default USEPA conversion factors contained in Appendix 3 of the SIP were used to convert the applicable dissolved criteria to total recoverable criteria.
- f. **Hardness-Dependent CTR Metals Criteria.** The CTR and the NTR contain water quality criteria for seven metals that vary as a function of hardness. The lower the hardness the lower the water quality criteria. The metals with hardness-dependent criteria include cadmium, copper, chromium III, lead, nickel, silver, and zinc.

This Order has established the criteria for hardness-dependent metals based on the hardness of the receiving water (actual ambient hardness) as required by the SIP<sup>1</sup> and the CTR<sup>2</sup>. The SIP and the CTR require the use of "receiving water" or "actual ambient" hardness, respectively, to determine effluent limitations for these metals. The CTR requires that the hardness values used shall be consistent with the design discharge conditions for design flows and mixing zones<sup>3</sup>. Where design flows for aquatic life criteria include the lowest one-day flow with an average reoccurrence frequency of once in ten years (1Q10) and the lowest average seven consecutive

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<sup>1</sup> The SIP does not address how to determine the hardness for application to the equations for the protection of aquatic life when using hardness-dependent metals criteria. It simply states, in Section 1.2, that the criteria shall be properly adjusted for hardness using the hardness of the receiving water.

<sup>2</sup> The CTR requires that, for waters with a hardness of 400 mg/L (as CaCO<sub>3</sub>), or less, the actual ambient hardness of the surface water must be used (40 C.F.R. § 131.38(c)(4)).

<sup>3</sup> 40 C.F.R. §131.38(c)(4)(ii)

day flow with an average reoccurrence frequency of once in ten years (7Q10).<sup>1</sup> This section of the CTR also indicates that the design conditions should be established such that the appropriate criteria are not exceeded more than once in a three year period on average.<sup>2</sup> The CTR requires that when mixing zones are allowed the CTR criteria apply at the edge of the mixing zone, otherwise the criteria apply throughout the water body including at the point of discharge.<sup>3</sup> The CTR does not define the term “ambient,” as applied in the regulations. Therefore, the Central Valley Water Board has considerable discretion to consider upstream and downstream ambient conditions when establishing the appropriate water quality criteria that fully complies with the CTR and SIP.

### ***Summary findings***

At design discharge conditions Laguna Creek is effluent dominated. Under these regularly occurring critical conditions the effluent is the receiving water that is used to define the ambient receiving water conditions to define the appropriate water quality criteria in accordance with the CTR and SIP. The ambient hardness for Laguna Creek is represented by the data in Figure F-1, below, which shows ambient hardness ranging from 28 mg/L to 140 mg/L based on all collected ambient data from October 2011 through February 2015. Given the high variability in ambient hardness values, there is no single hardness value that describes the ambient receiving water for all possible scenarios (e.g., minimum, maximum). Because of this variability, staff has determined that based on the ambient hardness concentrations measured in the receiving water, the Central Valley Water Board has discretion to select ambient hardness values within the range of 28 mg/L (minimum) up to 140 mg/L (maximum). Staff recommends that the Board use the ambient hardness values shown in Table F-6 for the following reasons.

- i. The ambient receiving water hardness values shown in Table F-6 are consistent with design discharge conditions and will result in criteria and effluent limitations that ensure protection of beneficial uses under all ambient receiving water conditions.
- ii. The Water Code mandates that the Central Valley Water Board establish permit terms that will ensure the reasonable protection of beneficial uses. In this case, using the lowest measured ambient hardness to calculate effluent limitations is not required to protect beneficial uses. Calculating effluent limitations based on the lowest measured ambient hardness is not required by the CTR or SIP, and is not reasonable as it would result in overly conservative limits that will impart substantial costs to the Discharger and ratepayers without providing any additional protection of beneficial uses. In compliance with applicable state and federal regulatory requirements, after considering the entire range of ambient hardness values, Board staff has used the ambient hardness values shown in Table F-6 to calculate the proposed effluent limitations for hardness-dependent metals. The proposed effluent limitations are protective of beneficial uses under all flow conditions.
- iii. Using an ambient hardness that is higher than the minimum observed ambient hardness will result in limits that may allow increased metals to be discharged to Laguna Creek, but such discharge is allowed under the antidegradation

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<sup>1</sup> 40 C.F.R. §131.38(c)(2)(iii) Table 4

<sup>2</sup> 40 C.F.R. §131.38(c)(2)(iii) Table 4, notes 1 and 2

<sup>3</sup> 40 C.F.R. §131.38(c)(2)(i)

policy (State Water Board Resolution 68-16). The Central Valley Water Board finds that this degradation is consistent with the antidegradation policy (see antidegradation findings in Section IV.D.4 of the Fact Sheet). The Antidegradation policy requires the Discharger to meet waste discharge requirements which will result in the best practicable treatment or control of the discharge necessary to assure that: a) a pollution or nuisance will not occur, and b) the highest water quality consistent with maximum benefit to the people of the State will be maintained.

- iv. Using the ambient hardness values shown in Table F-6 is consistent with the CTR and SIP's requirements for developing metals criteria.

**Table F-6. Summary of CTR Criteria for Hardness-dependent Metals**

| CTR Metals          | Ambient Hardness (mg/L) <sup>2</sup> | CTR Criteria (µg/L, total recoverable) <sup>1</sup> |         |
|---------------------|--------------------------------------|-----------------------------------------------------|---------|
|                     |                                      | acute                                               | chronic |
| Copper <sup>3</sup> | 37                                   | 82                                                  | 60      |
| Chromium III        | 37                                   | 770                                                 | 92      |
| Cadmium             | 37                                   | 1.5                                                 | 1.1     |
| Lead                | 37                                   | 23                                                  | 0.90    |
| Nickel              | 37                                   | 200                                                 | 22      |
| Silver              | 36                                   | 0.70                                                | --      |
| Zinc                | 37                                   | 52                                                  | 52      |

<sup>1</sup> Metal criteria rounded to two significant figures in accordance with the CTR (40 C.F.R. §131.38(b)(2)).

<sup>2</sup> The ambient hardness values in this table represent actual observed receiving water hardness measurements from the dataset shown in Figure F-1.

<sup>3</sup> Site-specific Water Effects Ratio of 15 used to calculate CTR criteria for copper.

### **Background**

The State Water Board provided direction regarding the selection of hardness in two precedential water quality orders; WQO 2008-0008 for the City of Davis Wastewater Treatment Plant (Davis Order) and WQO 2004-0013 for the Yuba City Wastewater Treatment Plant (Yuba City Order). The State Water Board recognized that the SIP and the CTR do not discuss the manner in which hardness is to be ascertained, thus regional water boards have considerable discretion in determining ambient hardness so long as the selected value is protective of water quality criteria under the given flow conditions. (Davis Order, p.10). The State Water Board explained that it is necessary that, "The [hardness] value selected should provide protection for all times of discharge under varying hardness conditions." (Yuba City Order, p. 8). The Davis Order also provides that, "Regardless of the hardness used, the resulting limits must always be protective of water quality criteria under all flow conditions." (Davis Order, p. 11)

The equation describing the total recoverable regulatory criterion, as established in the CTR, is as follows:

$$\text{CTR Criterion} = \text{WER} \times (e^{m[\ln(H)]+b}) \text{ (Equation 1)}$$

Where:

H = ambient hardness (as  $\text{CaCO}_3$ )<sup>1</sup>

WER = water-effect ratio

m, b = metal- and criterion-specific constants

The direction in the CTR regarding hardness selection is that it must be based on ambient hardness and consistent with design discharge conditions for design flows and mixing zones. Consistent with design discharge conditions and design flows means that the selected “design” hardness must result in effluent limitations under design discharge conditions that do not result in more than one exceedance of the applicable criteria in a three year period.<sup>2</sup> Where design flows for aquatic life criteria include the lowest one-day flow with an average reoccurrence frequency of once in ten years (1Q10) and the lowest average seven consecutive day flow with an average reoccurrence frequency of once in ten years (7Q10). Since Laguna Creek regularly contains no upstream flow, the critical design flow is zero.

***Ambient conditions***

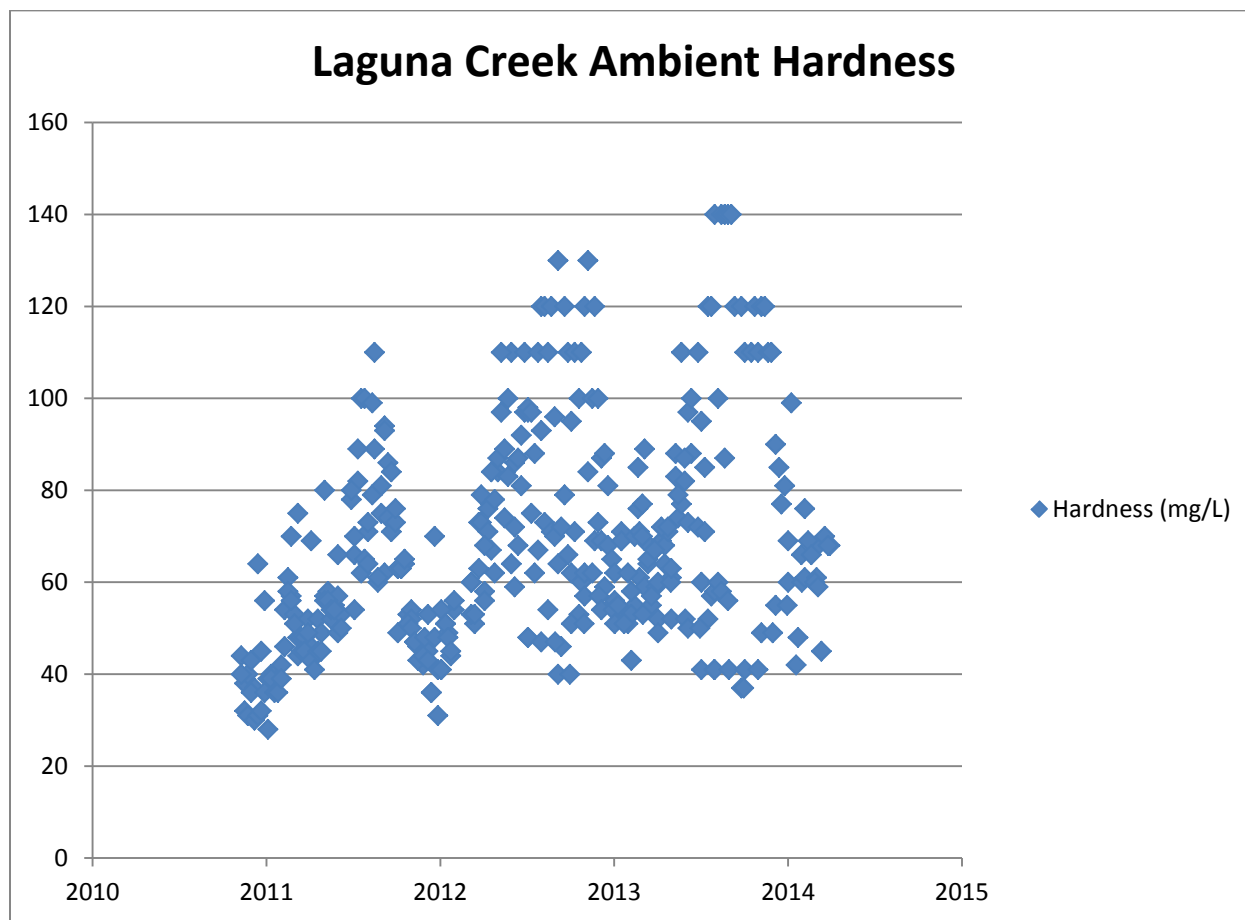
The ambient receiving water hardness varied from 28 mg/L to 140 mg/L, based on 367 samples from October 2011 through February 2015 (see Figure F-1).

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<sup>1</sup> For this discussion, all hardness values are expressed in mg/L as  $\text{CaCO}_3$ .

<sup>2</sup> 40 C.F.R. §131.38(c)(2)(iii) Table 4, notes 1 and 2

**Figure F-1. Ambient Hardness (October 2011 to February 2015)**



In this analysis, the entire range of ambient hardness concentrations shown in Figure F-1 were considered to determine the appropriate ambient hardness to calculate the CTR criteria and effluent limitations that are protective under all discharge conditions.

***Approach to derivation of criteria***

As shown above, ambient hardness is variable. Because of the variation, there is no single hardness value that describes the ambient receiving water for all possible scenarios (e.g., minimum, maximum, mid-point). While the hardness selected must be hardness of the ambient receiving water, selection of an ambient receiving water hardness that is too high would result in effluent limitations that do not protect beneficial uses. Also, the use of minimum ambient hardness would result in criteria that may not be representative considering the wide range of ambient conditions.

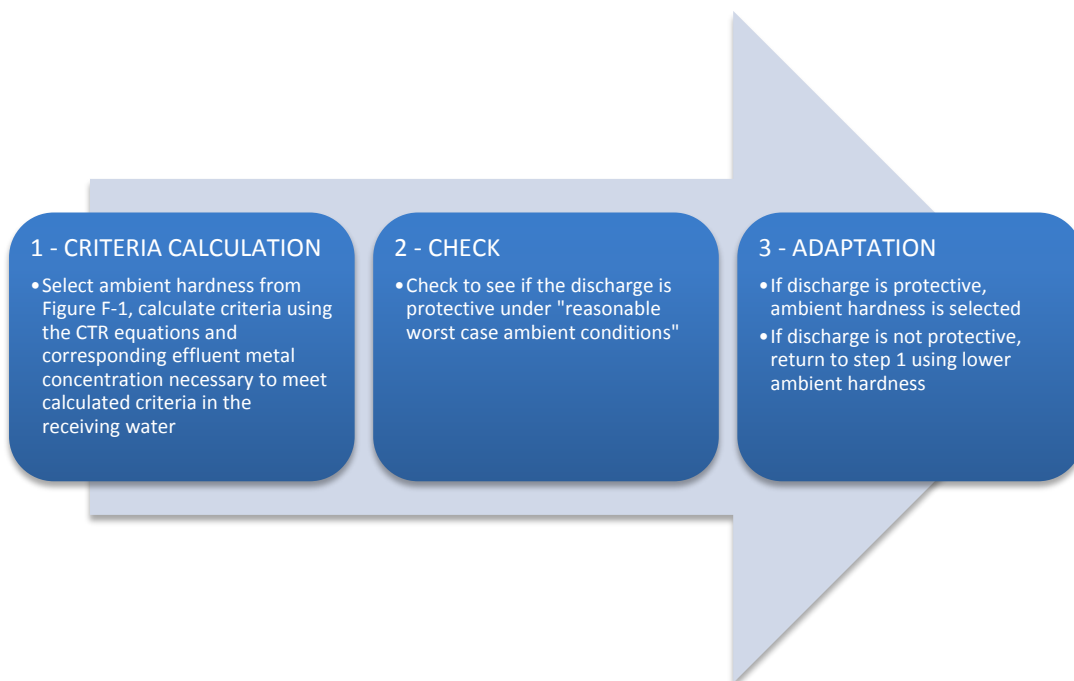
***Reasonable worst-case ambient conditions.*** To determine whether a selected ambient hardness value results in effluent limitations that are fully protective while complying with federal regulations and state policy, staff have conducted an analysis considering varying ambient hardness and flow conditions. To do this, the Central Valley Water Board has ensured that the receiving water hardness and criteria selected for effluent limitations are protective under "reasonable-worst case ambient conditions." These conditions represent the receiving water conditions under which derived effluent limitations would ensure protection of beneficial uses under all ambient flow and hardness conditions.

Reasonable worst-case ambient conditions:

- “Low receiving water flow.” CTR design discharge conditions (1Q10 and 7Q10) have been selected to represent reasonable worst case receiving water flow conditions.
- “High receiving water flow (maximum receiving water flow).” This additional flow condition has been selected consistent with the Davis Order, which required that the hardness selected be protective of water quality criteria under all flow conditions.
- “Low receiving water hardness.” The minimum ambient receiving water hardness condition of 28 mg/L was selected to represent the reasonable worst case receiving water hardness.
- “Background ambient metal concentration at criteria.” This condition assumes that the metal concentration in the background receiving water is equal to CTR criteria (upstream of the facility’s discharge). Based on the data in the record, this is a design condition that has not occurred in the receiving water and is used in this analysis to ensure that limits are protective of beneficial uses even in the situation where there is no assimilative capacity.

*Iterative approach.* An iterative analysis has been used to select the ambient hardness to calculate the criteria that will result in effluent limitations that protect beneficial uses under all flow conditions.

The iterative approach is summarized in the following algorithm and described below in more detail.





1. **CRITERIA CALCULATION.** CTR criteria are calculated using the CTR equations based on actual measured ambient hardness sample results, starting with the maximum observed ambient hardness of 140 mg/L. Effluent metal concentrations necessary to meet the above calculated CTR criteria in the receiving water are calculated in accordance with the SIP.<sup>1</sup> This should not be confused with an effluent limit. Rather, it is the Effluent Concentration Allowance (ECA), which is synonymous with the wasteload allocation defined by USEPA as “a definition of effluent water quality that is necessary to meet the water quality standards in the receiving water.”<sup>2</sup> If effluent limits are found to be needed, the limits are calculated to enforce the ECA considering effluent variability and the probability basis of the limit.
2. **CHECK.** USEPA’s simple mass balance equation<sup>3</sup> is used to evaluate if discharge at the computed ECA is protective. Resultant downstream metal concentrations are compared with downstream calculated CTR criteria under reasonable worst-case ambient conditions.
3. **ADAPT.** If step 2 results in:
  - (A) receiving water metal concentration that complies with CTR criteria under reasonable worst-case ambient conditions, then the hardness value is selected.
  - (B) receiving water metal concentration greater than CTR criteria, then return to bullet 1, selecting a lower ambient hardness value.

The CTR’s hardness dependent metals criteria equations contain metal-specific constants, so the criteria vary depending on the metal. Therefore, steps 1 through 3 must be repeated separately for each metal until ambient hardness values are determined that will result in criteria and effluent limitations that comply with the CTR and protect beneficial uses for all metals.

### ***Results of iterative analysis***

The above iterative analysis for each CTR hardness-dependent metal results in the selected ambient hardness values shown in Table F-6, above. Using these hardness values to calculate criteria, which are actual ambient sample results, will result in effluent limitations that are protective under all ambient flow conditions. Zinc and silver are used as examples below to illustrate the results of the analysis. Tables F-7 and F-8 below summarize the numeric results of the three step iterative approach for zinc and silver. As shown in the example tables, ambient hardness values of 37 mg/L (zinc) and 36 mg/L (silver) are used in the CTR equations to derive criteria and effluent limitations. Then under the “check” step, worst-case ambient receiving water conditions are used to test whether the discharge results in compliance with CTR criteria and protection of beneficial uses.

The results of the above analysis, summarized in the tables below, show that the ambient hardness values selected using the three-step iterative process results in protective effluent limitations that achieve CTR criteria under all flow conditions. Tables

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<sup>1</sup> SIP Section 1.4.B, Step 2, provides direction for calculating the Effluent Concentration Allowance.

<sup>2</sup> U.S. EPA Technical Support Document for Water Quality-based Toxics Control (TSD), pg. 96.

<sup>3</sup> U.S. EPA NPDES Permit Writers’ Handbook (EPA 833-K-10-001 September 2010, pg. 6-24)

F-7 and F-8 summarize the critical flow conditions. However, the analysis evaluated all flow conditions to ensure compliance with the CTR criteria at all times.

Table F-7. Verification of CTR Compliance for Zinc

| Receiving water hardness used to compute effluent limitations |                                                                                       |                     |                                                | 37 mg/L                     |
|---------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------|------------------------------------------------|-----------------------------|
| Effluent Concentration Allowance (ECA) for Zinc <sup>2</sup>  |                                                                                       |                     |                                                | 52 µg/L                     |
|                                                               | Downstream Ambient Concentrations Under Worst-Case Ambient Receiving Water Conditions |                     |                                                | Complies with CTR Criteria? |
|                                                               | Hardness                                                                              | CTR Criteria (µg/L) | Ambient Zinc Concentration <sup>1</sup> (µg/L) |                             |
| 1Q10                                                          | 37                                                                                    | 52                  | 52                                             | Yes                         |
| 7Q10                                                          | 37                                                                                    | 52                  | 52                                             | Yes                         |
| Max receiving water flow                                      | 28.5                                                                                  | 41                  | 41                                             | Yes                         |

<sup>1</sup> This concentration is derived using worst-case ambient conditions. These conservative assumptions will ensure that the receiving water always complies with CTR criteria.

<sup>2</sup> The ECA defines effluent quality necessary to meet the CTR criteria in the receiving water. There is no effluent limitation for zinc as it demonstrates no reasonable potential.

Table F-8. Verification of CTR Compliance for Silver

| Receiving water hardness used to compute effluent limitations  |                                                                                       |                     |                                                  | 36 mg/L                     |
|----------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------|--------------------------------------------------|-----------------------------|
| Effluent Concentration Allowance (ECA) for Silver <sup>2</sup> |                                                                                       |                     |                                                  | 0.70 µg/L                   |
|                                                                | Downstream Ambient Concentrations Under Worst-Case Ambient Receiving Water Conditions |                     |                                                  | Complies with CTR Criteria? |
|                                                                | Hardness                                                                              | CTR Criteria (µg/L) | Ambient Silver Concentration <sup>1</sup> (µg/L) |                             |
| 1Q10                                                           | 37                                                                                    | 0.73                | 0.73                                             | Yes                         |
| 7Q10                                                           | 37                                                                                    | 0.73                | 0.73                                             | Yes                         |
| Max receiving water flow                                       | 28.5                                                                                  | 0.47                | 0.47                                             | Yes                         |

<sup>1</sup> This concentration is derived using worst-case ambient conditions. These conservative assumptions will ensure that the receiving water always complies with CTR criteria.

<sup>2</sup> The ECA defines effluent quality necessary to meet the CTR criteria in the receiving water. There is no effluent limitation for silver as it demonstrates no reasonable potential.

### 3. Determining the Need for WQBEL's

- a. **Constituents with No Reasonable Potential.** WQBEL's are not included in this Order for constituents that do not demonstrate reasonable potential (i.e. constituents were not detected in the effluent or receiving water); however, monitoring for those pollutants is established in this Order as required by the SIP. If the results of effluent monitoring demonstrate reasonable potential, this Order may be reopened and modified by adding an appropriate effluent limitation.

Most constituents with no reasonable potential are not discussed in this Order. However, the following constituents were found to have no reasonable potential after assessment of the data:

i. **Aluminum**

Aluminum is the third most abundant element in the earth's crust and is ubiquitous in both soils and aquatic sediments. When mobilized in surface waters, aluminum has been shown to be toxic to various fish species. However, the potential for aluminum toxicity in surface waters is directly related to the chemical form of aluminum present, and the chemical form is highly dependent on water quality characteristics that ultimately determine the mechanism of aluminum toxicity. Surface water characteristics, including pH, temperature, colloidal material, fluoride and sulfate concentrations, and total organic carbon, all influence aluminum speciation and its subsequent bioavailability to aquatic life. Calcium [hardness] concentrations in surface water may also reduce aluminum toxicity by competing with monomeric aluminum ( $\text{Al}^{3+}$ ) binding to negatively charged fish gills.

- (a) **WQO.** The State Water Board Division of Drinking Water (DDW; formerly the Department of Public Health) has established Secondary MCL's to assist public drinking water systems in managing their drinking water for aesthetic conditions such as taste, color, and odor. The Secondary MCL for aluminum is 200 µg/L for protection of the MUN beneficial use. Title 22 requires compliance with Secondary MCL's on an annual average basis.

Title 40 of the Code of Federal Regulations promulgated criteria for priority toxic pollutants for California's surface waters as part of section 131.38. Establishment of Numeric Criteria for Priority Toxic Pollutants for the State of California (California Toxics Rule or CTR), including metals criteria. However, aluminum criteria were not promulgated as part of the CTR. Absent numeric aquatic life criteria for aluminum, WQBEL's in the Central Valley Region's NPDES permits are based on the Basin Plans' narrative toxicity objective. The Basin Plans' *Policy for Application of Water Quality Objectives* requires the Central Valley Water Board to consider, "on a case-by-case basis, direct evidence of beneficial use impacts, all material and relevant information submitted by the discharger and other interested parties, and relevant numerical criteria and guidelines developed and/or published by other agencies and organizations. In considering such criteria, the Board evaluates whether the specific numerical criteria which are available through these sources and through other information supplied to the Board, are relevant and appropriate to the situation at hand and, therefore, should be used in determining compliance with the narrative objective." Relevant information includes, but is not limited to (1) U.S. EPA Ambient Water Quality Criteria (NAWQC) and subsequent Correction, (2) site-specific conditions of Laguna Creek, the receiving water, and (3) site-specific aluminum studies conducted by dischargers within the Central Valley Region. (Basin Plan, p. IV.-17.00; see also, 40 C.F.R. §122.44(d)(vi).)

**U.S. EPA NAWQC.** U.S. EPA recommended the NAWQC aluminum acute criterion at 750 µg/L based on test waters with a pH of 6.5 to 9.0. U.S. EPA also recommended the NAWQC aluminum chronic criterion at

87 µg/L based upon the following two toxicity tests. All test waters contained hardness at 12 mg/L as CaCO<sub>3</sub>.

- (1) Acute toxicity tests at various aluminum doses were conducted in various acidic waters (pH 6.0 – 6.5) on 159- and 160-day old striped bass. The 159-day old striped bass showed no mortality in waters with pH at 6.5 and aluminum doses at 390 µg/L, and the 160-day old striped bass showed 58% mortality at a dose of 174.4 µg/L in same pH waters. However, the 160-day old striped bass showed 98% mortality at an aluminum dose of 87.2 µg/L in waters with pH at 6.0, which is U.S. EPA's basis for the 87 µg/L chronic criterion. The varied results draw into question this study and the applicability of the NAWQC chronic criterion of 87 µg/L.
- (2) Chronic toxicity effects on 60-day old brook trout were evaluated in circumneutral pH waters (6.5-6.9 pH) in five cells at various aluminum doses (4, 57, 88, 169, and 350 µg/L). Chronic evaluation started upon hatching of eyed eggs of brook trout, and their weight and length were measured after 45 days and 60 days. The 60-day old brook trout showed 24% weight loss at 169 µg/L of aluminum and 4% weight loss at 88 µg/L of aluminum, which is the basis for U.S. EPA's chronic criteria. Though this test study shows chronic toxic effects of 4% reduction in weight after exposure for 60-days, the chronic criterion is based on 4-day exposure; so again, the applicability of the NAWQC chronic criterion of 87 µg/L is questionable.

**Site-specific Conditions.** U.S. EPA advises that a water effects ratio (WER) may be more appropriate to better reflect the actual toxicity of aluminum to aquatic organisms when the pH and hardness conditions of the receiving water are not similar to that of the test conditions<sup>1</sup>. Effluent and Laguna Creek monitoring data indicate that the pH and hardness values are not similar to the low pH and hardness conditions under which the chronic criterion for aluminum was developed, as shown in the table below, and therefore, the Central Valley Water Board does not expect aluminum to be as toxic in Laguna Creek as in the previously described toxicity tests. The pH of Laguna Creek, the receiving water, ranged from 6.6 to 9.7 with a median of 7.7 based on 76 monitoring results obtained between May 2011 and April 2014. These water conditions typically are circumneutral pH where aluminum is predominately in the form of Al(OH)<sub>3</sub> and non-toxic to aquatic life. The hardness of Laguna Creek ranged from 28 mg/L to 700 mg/L, based on 148 samples from December 2011 and November 2014, which is above the conditions, and thus less toxic, than the tests used to develop the chronic criterion.

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<sup>1</sup> "The value of 87 micro-g/L is based on a toxicity test with striped bass in water with pH = 6.5-6.6 and hardness < 10 mg/L. Data in [a 1994 Study] indicate that aluminum is substantially less toxic at higher pH and hardness, but the effects of pH and hardness are not well quantified at this time." U.S. EPA 1999 NAWQC Correction, Footnote L.

| Parameter                               | Units          | Test Conditions for Applicability of Chronic Criterion | Effluent  | Receiving Water |
|-----------------------------------------|----------------|--------------------------------------------------------|-----------|-----------------|
| pH                                      | standard units | 6.0 – 6.5                                              | 7.1 – 8.9 | 6.4 – 9.0       |
| Hardness, Total (as CaCO <sub>3</sub> ) | mg/L           | 12                                                     | 37 - 470  | 28 - 700        |
| Aluminum, Total Recoverable             | µg/L           | 87.2 - 390                                             | <14 – 240 | 66 – 1100       |

**Local Environmental Conditions and Studies.** Twenty-one site-specific aluminum toxicity tests have been conducted within the Central Valley Region. The pH and hardness of Laguna Creek are similar, as shown in the table below, and thus the results of these site-specific aluminum toxicity tests are relevant and appropriate for Laguna Creek. As shown in the following table, all EC<sub>50</sub><sup>1</sup> toxicity study result values are at concentrations of aluminum above 5,000 µg/L. Thus, the toxic effects of aluminum in these surface waters and in Laguna creek, is less toxic (or less reactive) to aquatic species than demonstrated in the toxicity tests that U.S. EPA used for the basis of establishing the chronic criterion of 87 µg/L. This new information, and review of the toxicity tests U.S. EPA used to establish the chronic criterion, indicates that 87 µg/L is overly stringent and not applicable to the Laguna Creek.

**Central Valley Region Site-Specific Aluminum Toxicity Data**

| Discharger                                        | Test Waters            | Hardness Value       | Total Aluminum EC <sub>50</sub> Value | pH        | WER   |
|---------------------------------------------------|------------------------|----------------------|---------------------------------------|-----------|-------|
| <b><i>Oncorhynchus mykiss</i> (rainbow trout)</b> |                        |                      |                                       |           |       |
| Manteca                                           | Surface Water/Effluent | 124                  | >8600                                 | 9.14      | N/C   |
| Auburn                                            | Surface Water          | 16                   | >16500                                | 7.44      | N/C   |
| Modesto                                           | Surface Water/Effluent | 120/156              | >34250                                | 8.96      | >229  |
| Yuba City                                         | Surface Water/Effluent | 114/164 <sup>1</sup> | >8000                                 | 7.60/7.46 | >53.5 |
| <b><i>Ceriodaphnia dubia</i> (water flea)</b>     |                        |                      |                                       |           |       |
| Auburn                                            | Effluent               | 99                   | >5270                                 | 7.44      | >19.3 |
|                                                   | Surface Water          | 16                   | >5160                                 | 7.44      | >12.4 |
| Manteca                                           | Surface Water/Effluent | 124                  | >8800                                 | 9.14      | N/C   |
|                                                   | Effluent               | 117                  | >8700                                 | 7.21      | >27.8 |
|                                                   | Surface Water          | 57                   | 7823                                  | 7.58      | 25.0  |
|                                                   | Effluent               | 139                  | >9500                                 | 7.97      | >21.2 |
|                                                   | Surface Water          | 104                  | >11000                                | 8.28      | >24.5 |
|                                                   | Effluent               | 128                  | >9700                                 | 7.78      | >25.0 |
|                                                   | Surface Water          | 85                   | >9450                                 | 7.85      | >25.7 |
|                                                   | Effluent               | 106                  | >11900                                | 7.66      | >15.3 |
|                                                   | Surface Water          | 146                  | >10650                                | 7.81      | >13.7 |
| Modesto                                           | Surface Water/Effluent | 120/156              | 31604                                 | 8.96      | 211   |
| Yuba City                                         | Surface Water/Effluent | 114/164 <sup>1</sup> | >8000                                 | 7.60/7.46 | >53.5 |
| Placer County (SMD 1)                             | Effluent               | 150                  | >5000                                 | 7.4 – 8.7 | >13.7 |

<sup>1</sup>The effect concentration is a point estimate of the toxicant concentration that would cause an observable adverse effect (e.g. death, immobilization, or serious incapacitation) in a given percent of the test organisms, calculated from a continuous model (e.g., Probit Model). EC<sub>50</sub> is a point estimate of the toxicant concentration that would cause an observable adverse effect in 50 percent of the test organisms. The EC<sub>50</sub> is used in toxicity testing to determine the appropriate chronic criterion.

| Discharger                               | Test Waters            | Hardness Value       | Total Aluminum EC <sub>50</sub> Value | pH        | WER   |
|------------------------------------------|------------------------|----------------------|---------------------------------------|-----------|-------|
| <b><i>Daphnia magna</i> (water flea)</b> |                        |                      |                                       |           |       |
| Manteca                                  | Surface Water/Effluent | 124                  | >8350                                 | 9.14      | N/C   |
| Modesto                                  | Surface Water/Effluent | 120/156              | >11900                                | 8.96      | >79.6 |
| Yuba City                                | Surface Water/Effluent | 114/164 <sup>1</sup> | >8000                                 | 7.60/7.46 | >53.5 |

**Applicable WQOs.** This Order implements the Secondary MCL of 200 µg/L as an annual average for the protection of MUN and implements the Basin Plan's narrative toxicity objective for the protection of aquatic life using an acute (1-hour) criterion and chronic (4-day) criterion of 750 µg/L based on U.S. EPA's NAWQC and the discussion above. Order R5-2010-0099 included effluent limitations for aluminum based on the Secondary MCL and the NAWQC acute criterion.

- (b) **RPA Results.** For priority pollutants, the SIP dictates the procedures for conducting the RPA. Aluminum is not a priority pollutant. Therefore, the Central Valley Water Board is not restricted to one particular RPA method. Due to the site-specific conditions of the discharge, the Central Valley Water Board has used its judgment in determining the appropriate method for conducting the RPA for this non-priority pollutant constituent. The most stringent objective is the Secondary MCL, which is derived from human welfare considerations (e.g., taste, odor, laundry staining), not for toxicity. Secondary MCL's are drinking water standards contained in Title 22 of the California Code of Regulations. Title 22 requires compliance with these standards on an annual average basis, when sampling at least quarterly. To be consistent with how compliance with the standards is determined, the RPA was conducted based on the calendar year annual average effluent aluminum concentrations.

The maximum observed effluent annual average aluminum concentration was 54 µg/L based on 38 samples collected between December 2011 and November 2014. Therefore, the Central Valley Water Board finds the discharge does not have reasonable potential to cause or contribute to an exceedance in the receiving water and the Facility is adequately controlling the discharge of aluminum. Since the discharge does not demonstrate reasonable potential, the effluent limitations for aluminum have not been retained in this Order. Removal of these effluent limitations is in accordance with federal anti-backsliding regulations (see section IV.D.3 of the Fact Sheet).

ii. **Bis(2-ethylhexyl)phthalate.**

- (a) **WQO.** The CTR includes a criterion of 1.8 µg/L for bis(2-ethylhexyl) phthalate (bis-2) for the protection of human health.
- (b) **RPA Results.** The maximum effluent concentration (MEC) for bis-2 was 1.4 µg/L based on 53 samples collected between December 2011 and November 2011. The Discharger sampled for bis-2 in the receiving water 10 times during their priority pollutant scan occurring between August 2013 and September 2014 with a maximum background concentration of 2.0 µg/L. The minimum level required in Appendix 4 of the SIP is 5 µg/L for bis-2. Therefore, the detected receiving water sample is not reliable data.

In addition, the concentration of 2.0 µg/L of bis-2 in the receiving water is not characteristic of Laguna Creek as there are no potential sources of bis-2 upstream of the Facility's discharge point. Therefore, the detections of bis-2 in the receiving water are suspected to be the result of sample contamination. Therefore, bis-2 in the discharge does not demonstrate reasonable potential to cause or contribute to an in-stream excursion above the CTR Criterion of 1.8 µg/L, and the effluent limitations for bis-2 have not been retained in this Order. Removal of these effluent limitations is in accordance with federal anti-backsliding regulations (see section IV.D.3 of the Fact Sheet).

iii. **Carbon Tetrachloride.**

- (a) **WQO.** The CTR includes a criterion of 0.25 µg/L for carbon tetrachloride for the protection of human health protection for waters from which both water and organisms are consumed.
- (b) **RPA Results.** Carbon tetrachloride is a commonly know byproduct of chlorine disinfection practices in wastewater treatment. The Discharger converted from chlorine disinfection to UV disinfection upon commencement of facility upgrades, which went online 2011. Since chlorine is no longer used at the Facility, the presence of carbon tetrachloride as a disinfection byproduct is not expected in the Facility's effluent. Furthermore, carbon tetrachloride was not detected in the effluent at a (MDL of 0.077 µg/L) in 47 samples collected between December 2011 and October 2014, and was also not detected in the upstream receiving water based on 10 samples collected between August 2011 and July 2014 (MDL of 0.0077 µg/L). Therefore, the Central Valley Water Board finds the discharge does not have reasonable potential to cause or contribute to an exceedance in the receiving water. Since the discharge does not demonstrate reasonable potential, the effluent limitations for carbon tetrachloride have not been retained in this Order. Removal of these effluent limitations is in accordance with federal anti-backsliding regulations (see section IV.D.3 of the Fact Sheet).

iv. **Chlorine Residual**

- (a) **WQO.** USEPA developed NAWQC for protection of freshwater aquatic life for chlorine residual. The recommended 4-day average (chronic) and 1-hour average (acute) criteria for chlorine residual are 0.011 µg/L and 0.019 µg/L, respectively. Order R5-2007-0039 included effluent limitations for chlorine residual based on the NAWQC criteria.
- (b) **RPA Results.** The Discharger converted from chlorine disinfection to UV disinfection upon commencement of facility upgrades , which went online 2011. Order R5-2010-0099 required the Discharger to monitor for chlorine only when chlorine was being used at the Facility, thus no chlorine monitoring data is available. Since chlorine is no longer used at the Facility, the discharge does not demonstrate reasonable potential to cause or contribute to an in-stream excursion above the NAWQC for chlorine and the effluent limitations for chlorine have not been retained in this Order. Removal of these effluent limitations is in accordance with federal anti-backsliding regulations (see section IV.D.3 of the Fact Sheet)

v. **Chlorodibromomethane.**

- (a) **WQO.** The CTR includes a criterion of 0.41 µg/L for chlorodibromomethane(CDBM) for the protection of human health protection for waters from which both water and organisms are consumed.
- (b) **RPA Results.** CDBM was not detected in the Facility's effluent in 34 samples conducted at a MDL of 0.049 µg/L and collected between December 2011 and November 2014. Therefore, CDBM in the discharge does not exhibit reasonable potential to cause or contribute to an in-stream excursion above the CTR criterion and the effluent limitation for CDBM has not been retained in this Order. Removal of the effluent limitations is in accordance with federal anti-backsliding regulations (see section IV.D.3 of the Fact Sheet).

vi. **Copper.**

- (a) **WQO.** The CTR includes hardness-dependent criteria for the protection of freshwater aquatic life for copper. The criteria for copper are presented in dissolved concentrations. USEPA recommends conversion factors to translate dissolved concentration to total concentrations. The USEPA default conversion factors for copper in freshwater of 0.96 for both the acute and the chronic criteria were used for the discharge.
- (b) **RPA Results.** As discussed above in Section IV.C.2.d of this Fact Sheet, the Discharger submitted a Copper Water Effects Ratio (WER) study along with their Report of Waste Discharge (ROWD) in March 2015 that calculated a site specific WER of 15 that is applicable to the effluent. Therefore, the applicable copper chronic criterion (maximum 4-day average concentration) is 60 µg/L and the applicable acute criterion (maximum (1-hour concentration) is 80 µg/L, as total recoverable, (see Table F-6, above). The MEC for total copper was 17 µg/L, based on data collected between December 2011 and November 2014. Upstream total copper concentration varied from 2.9 µg/L to 8.5 µg/L. Using paired hardness and copper data, the maximum ambient receiving water concentration did not exceed the applicable CTR criteria for copper. Based on this information, copper in the discharge does not exhibit reasonable potential to cause or contribute to an in-stream excursion above the CTR criterion for the protection of freshwater aquatic life. Therefore, effluent limitations for copper have not been retained in this Order. Removal of the effluent limitations is in accordance with federal anti-backsliding regulations (see section IV.D.3 of the Fact Sheet).

vii. **Cyanide.**

- (a) **WQO.** The CTR includes maximum 1-hour average and 4-day average criteria of 5.2 µg/L and 22 µg/L, respectively, for cyanide, for the protection of freshwater aquatic life. Order R5-2010-0099 included effluent limitations for cyanide based on the CTR freshwater aquatic life criteria.
- (b) **RPA Results.** The MEC for cyanide was 5.0 µg/L based on 65 samples collected between December 2011 and November 2014, while cyanide was not detected in the receiving water in 10 samples with a MDL of 4.0 µg/L. Therefore, cyanide in the discharge does not have a reasonable potential to cause or contribute to an in-stream excursion above the CTR



criterion for the protection of freshwater aquatic life, and the effluent limitation for cyanide has not been retained in this Order. Removal of the effluent limitations is in accordance with federal anti-backsliding regulations (see section IV.D.3 of the Fact Sheet).

viii. **Dichlorobromomethane.**

- (a) **WQO.** The CTR includes a criterion of 0.56 µg/L for dichlorobromomethane (DCBM) for the protection of human health protection for waters from which both water and organisms are consumed.
- (b) **RPA Results.** DCBM was not detected in the Facility's effluent in 38 samples conducted at a MDL of 0.031 µg/L and collected between December 2011 and November 2014. Therefore, DCBM in the discharge does not exhibit reasonable potential to cause or contribute to an in-stream excursion above the CTR criterion and the effluent limitation for DCBM has not been retained in this Order. Removal of the effluent limitations is in accordance with federal anti-backsliding regulations (see section IV.D.3 of the Fact Sheet).

ix. **Iron**

- (a) **WQO.** The Secondary MCL – Consumer Acceptance Limit for iron is 300 µg/L, which is used to implement the Basin Plan's chemical constituent objective for the protection of municipal and domestic supply. Order R5-2010-0099 included an annual average effluent limitation of 300 µg/L based on the Secondary MCL.
- (b) **RPA Results.** For priority pollutants, the SIP dictates the procedures for conducting the RPA. Iron is not a priority pollutant. Therefore, the Central Valley Water Board is not restricted to one particular RPA method. Due to the site-specific conditions of the discharge, the Central Valley Water Board has used its judgment in determining the appropriate method for conducting the RPA for this non-priority pollutant constituent. The most stringent objective is the Secondary MCL, which is derived from human welfare considerations (e.g., taste, odor, laundry staining), not for toxicity. Secondary MCL's are drinking water standards contained in Title 22 of the California Code of Regulations. Title 22 requires compliance with these standards on an annual average basis, when sampling at least quarterly. To be consistent with how compliance with the standards is determined, the RPA was conducted based on the calendar year annual average effluent iron concentrations.

The maximum annual average effluent concentration for iron was 56 µg/L based on 19 samples collected between October 2011 and September 2014. Therefore, the Central Valley Water Board finds the discharge does not have reasonable potential to cause or contribute to an exceedance in the receiving water and the Facility is adequately controlling the discharge of iron. Since the discharge does not demonstrate reasonable potential, the effluent limitation for iron has not been retained in this Order. Removal of this effluent limitation is in accordance with federal anti-backsliding regulations (see section IV.D.3 of the Fact Sheet).

x. **Lead.**

- (a) **WQO.** The CTR includes hardness-dependent criteria for the protection of freshwater aquatic life for lead. The criteria for lead are presented in dissolved concentrations. USEPA recommends conversion factors to translate dissolved concentration to total concentrations. The USEPA default conversion factors for lead were used for the discharge.
- (b) **RPA Results.** The applicable lead chronic criterion (maximum 4-day average concentration) is 0.87 µg/L and the applicable acute criterion (maximum (1-hour concentration) is 22 µg/L, as total recoverable, (see Table F- 6, above). The MEC for lead was 0.34 µg/L based on 33 samples collected between December 2011 and November 2014. . Upstream total lead concentration varied from 0.23 µg/L to 0.78 µg/L. Using paired hardness and lead data, the maximum ambient receiving water concentration did not exceed the applicable CTR criteria for lead. Based on this information, lead in the discharge does not exhibit reasonable potential to cause or contribute to an in-stream excursion above the CTR criterion for the protection of freshwater aquatic life. Therefore, effluent limitations for lead have not been retained in this Order. Removal of the effluent limitations is in accordance with federal anti-backsliding regulations (see section IV.D.3 of the Fact Sheet).

xi. **Manganese**

- (a) **WQO.** The Secondary MCL – Consumer Acceptance Limit for manganese is 50 µg/L, which is used to implement the Basin Plan’s chemical constituent objective for the protection of municipal and domestic supply. Order R5-2010-0099 included an annual average effluent limitation of 50 µg/L based on the Secondary MCL.
- (b) **RPA Results.** For priority pollutants, the SIP dictates the procedures for conducting the RPA. Manganese is not a priority pollutant. Therefore, the Central Valley Water Board is not restricted to one particular RPA method. Due to the site-specific conditions of the discharge, the Central Valley Water Board has used its judgment in determining the appropriate method for conducting the RPA for this non-priority pollutant constituent. The most stringent objective is the Secondary MCL, which is derived from human welfare considerations (e.g., taste, odor, laundry staining), not for toxicity. Secondary MCL’s are drinking water standards contained in Title 22 of the California Code of Regulations. Title 22 requires compliance with these standards on an annual average basis, when sampling at least quarterly. To be consistent with how compliance with the standards is determined, the RPA was conducted based on the calendar year annual average effluent manganese concentrations.

The maximum annual average effluent concentration for manganese was 7.0 µg/L based on 20 samples collected between October 2011 and September 2014. Therefore, the Central Valley Water Board finds the discharge does not have reasonable potential to cause or contribute to an exceedance in the receiving water and the Facility is adequately controlling the discharge of manganese. Since the discharge does not demonstrate reasonable potential, the effluent limitation for manganese has not been retained in this Order. Removal of this effluent limitation is in

accordance with federal anti-backsliding regulations (see section IV.D.3 of the Fact Sheet).

xii. **Salinity**

- (a) **WQO.** The Basin Plan contains a chemical constituent objective that incorporates state MCLs, contains a narrative objective, and contains numeric water quality objectives for certain specified water bodies for electrical conductivity, total dissolved solids, sulfate, and chloride. The USEPA Ambient Water Quality Criteria for Chloride recommends acute and chronic criteria for the protection of aquatic life. There are no USEPA water quality criteria for the protection of aquatic life for electrical conductivity, total dissolved solids, and sulfate. Additionally, there are no USEPA numeric water quality criteria for the protection of agricultural, live stock, and industrial uses. Numeric values for the protection of these uses are typically based on site specific conditions and evaluations to determine the appropriate constituent threshold necessary to interpret the narrative chemical constituent Basin Plan objective. The Central Valley Water Board must determine the applicable numeric limit to implement the narrative objective for the protection of agricultural supply. The Central Valley Water Board is currently implementing the CV-SALTS initiative to develop a Basin Plan Amendment that will establish a salt and nitrate Management Plan for the Central Valley. Through this effort the Basin Plan will be amended to define how the narrative water quality objective is to be interpreted for the protection of agricultural use. All studies conducted through this Order to establish an agricultural limit to implement the narrative objective will be reviewed by and consistent with the efforts currently underway by CV-SALTS.

**Table F-9. Salinity Water Quality Criteria/Objectives**

| Parameter       | Agricultural WQ Objective <sup>1</sup> | Secondary MCL <sup>2</sup> | USEPA NAWQC           | Effluent             |            |
|-----------------|----------------------------------------|----------------------------|-----------------------|----------------------|------------|
|                 |                                        |                            |                       | Average <sup>3</sup> | Maximum    |
| EC (µmhos/cm)   | Varies <sup>2</sup>                    | 900, 1600, 2200            | N/A                   | <b>508</b>           | <b>590</b> |
| TDS (mg/L)      | Varies                                 | 500, 1000, 1500            | N/A                   | <b>361</b>           | <b>510</b> |
| Sulfate (mg/L)  | Varies                                 | 250, 500, 600              | N/A                   | <b>37</b>            | <b>221</b> |
| Chloride (mg/L) | Varies                                 | 250, 500, 600              | 860 1-hr<br>230 4-day | <b>44</b>            | <b>50</b>  |

<sup>1</sup> Narrative chemical constituent objective of the Basin Plan. Procedures for establishing the applicable numeric limitation to implement the narrative objective can be found in the Policy for Application of Water Quality, Chapter IV, Section 8 of the Basin Plan., However, the Basin Plan does not require improvement over naturally occurring background concentrations. In cases where the natural background concentration of a particular constituent exceeds an applicable water quality objective, the natural background concentration will be considered to comply with the objective.

<sup>2</sup> The secondary MCLs are stated as a recommended level, upper level, and a short-term maximum level.

<sup>3</sup> Maximum calendar annual average.

- (1) **Chloride.** The Secondary MCL for chloride is 250 mg/L, as a recommended level, 500 mg/L as an upper level, and 600 mg/L as a short-term maximum.
- (2) **Electrical Conductivity.** The Secondary MCL for EC is 900  $\mu$ mhos/cm as a recommended level, 1600  $\mu$ mhos/cm as an upper level, and 2200  $\mu$ mhos/cm as a short-term maximum
- (3) **Sulfate.** The Secondary MCL for sulfate is 250 mg/L as a recommended level, 500 mg/L as an upper level, and 600 mg/L as a short-term maximum.
- (4) **Total Dissolved Solids.** The Secondary MCL for TDS is 500 mg/L as a recommended level, 1000 mg/L as an upper level, and 1500 mg/L as a short-term maximum.

(b) **RPA Results.**

- (1) **Chloride.** Chloride concentrations in the effluent ranged from 39 mg/L to 50 mg/L, with an average of 44 mg/L. Background concentrations in Laguna Creek ranged from 7.1 mg/L to 36 mg/L, with an average of 19 mg/L, for 12 samples collected by the Discharger from August 2013 through September 2014. These levels do not exceed the Secondary MCL.
- (2) **Electrical Conductivity.** A review of the Discharger's monitoring reports shows an average effluent EC of 508  $\mu$ mhos/cm, with a range from 140  $\mu$ mhos/cm to 590  $\mu$ mhos/cm. The background receiving water EC averaged 253  $\mu$ mhos/cm. These levels do not exceed the Secondary MCL.
- (3) **Sulfate.** Sulfate concentrations in the effluent ranged from 15 mg/L to 221 mg/L, with an average of 37 mg/L. Background concentrations in Laguna Creek ranged from 7.0 mg/L to 25 mg/L, with an average of 16 mg/L. These levels do not exceed the Secondary MCL.
- (4) **Total Dissolved Solids.** The average TDS effluent concentration was 361 mg/L with concentrations ranging from 290 mg/L to 510 mg/L. The background receiving water TDS ranged from 76 mg/L to 350 mg/L, with an average of 183 mg/L. These levels do not exceed the Secondary MCL.

- b. **Constituents with Reasonable Potential.** The Central Valley Water Board finds that the discharge has a reasonable potential to cause or contribute to an in-stream excursion above a water quality standard for ammonia, arsenic, nitrate plus nitrite, pathogens, pH, and zinc. WQBEL's for these constituents are included in this Order. A summary of the RPA is provided in Attachment G, and a detailed discussion of the RPA for each constituent is provided below.

i. **Ammonia**

- (a) **WQO.** The 1999 USEPA National Ambient Water Quality Criteria (NAWQC) for the protection of freshwater aquatic life for total ammonia (the "1999 Criteria"), recommends acute (1-hour average; criteria maximum concentration or CMC) standards based on pH and chronic (30-

day average; criteria continuous concentration or CCC) standards based on pH and temperature. USEPA also recommends that no 4-day average concentration should exceed 2.5 times the 30-day CCC. USEPA found that as pH increased, both the acute and chronic toxicity of ammonia increased. Salmonids were more sensitive to acute toxicity effects than other species. However, while the acute toxicity of ammonia was not influenced by temperature, it was found that invertebrates and young fish experienced increasing chronic toxicity effects with increasing temperature.

The USEPA recently published national recommended water quality criteria for the protection of aquatic life from the toxic effects of ammonia in freshwater (the "2013 Criteria")<sup>1</sup>. The 2013 Criteria is an update to USEPA's 1999 Criteria, and varies based on pH and temperature. Although the 2013 Criteria reflects the latest scientific knowledge on the toxicity of ammonia to certain freshwater aquatic life, including new toxicity data on sensitive freshwater mussels in the Family Unionidae, the species tested for development of the 2013 Criteria may not be present in some Central Valley waterways. The 2013 Criteria document therefore states that, "*unionid mussel species are not prevalent in some waters, such as the arid west ...*" and provides that, "*In the case of ammonia, where a state demonstrates that mussels are not present on a site-specific basis, the recalculation procedure may be used to remove the mussel species from the national criteria dataset to better represent the species present at the site.*"

The Central Valley Water Board issued a 3 April 2014 *California Water Code Section 13267 Order for Information: 2013 Final Ammonia Criteria for Protection of Freshwater Aquatic Life* (13267 Order) requiring the Discharger to either participate in an individual or group study to determine the presence of mussels or submit a method of compliance for complying with effluent limitations calculated assuming mussels present using the 2013 Criteria. The Discharger submitted a letter to the Central Valley Water Board indicating their participation in the Central Valley Clean Water Association Freshwater Collaborative Mussel Study. Studies are currently underway to determine how the latest scientific knowledge on the toxicity of ammonia reflected in the 2013 Criteria can be implemented in the Central Valley Region as part of a Basin Planning effort to adopt nutrient and ammonia objectives. Until the Basin Planning process is completed, the Central Valley Water Board will continue to implement the 1999 Criteria to interpret the Basin Plan's narrative toxicity objective. The 1999 NAWQC for the protection of freshwater aquatic life for total ammonia, recommends acute (1-hour average; criteria maximum concentration or CMC) standards based on pH and chronic (30-day average; criteria continuous concentration or CCC) standards based on pH and temperature. USEPA also recommends that no 4-day average concentration should exceed 2.5 times the 30-day CCC. USEPA found that as pH increased, both the acute and chronic toxicity of ammonia increased. Salmonids were more sensitive to acute toxicity effects than other species. However, while the acute toxicity of ammonia was not

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<sup>1</sup> Aquatic Life Ambient Water Quality Criteria for Ammonia – Freshwater, published August 2013 [EPA 822-R-13-001]

influenced by temperature, it was found that invertebrates and young fish experienced increasing chronic toxicity effects with increasing temperature. Because Laguna Creek has a beneficial use of cold freshwater habitat and the presence of salmonids and early fish life stages in Laguna Creek is well-documented, the recommended criteria for waters where salmonids and early life stages are present were used.

Based on 261 effluent samples from October 2011 through November 2014 the effluent pH ranged from 7.1 – 8.9. In order to protect against the reasonable worst-case short-term exposure of an organism, a pH value of 8.2 (the upper bound of the effluent limitation range for pH) was used to derive the acute criterion. The resulting acute criterion is 3.83 mg/L. A chronic criterion was calculated for each day when paired temperature data and pH were measured using downstream receiving water data for temperature and pH. Rolling 30-day average criteria were calculated from downstream receiving water data using the criteria calculated for each day and the minimum observed 30-day average criterion was established as the applicable 30-day average chronic criterion. The most stringent 30-day CCC was 2.43 mg/L (as N) and the 4-day average concentration was 6.08 mg/L (as N).

- (b) **RPA Results.** The Facility is a POTW that treats domestic wastewater. Untreated domestic wastewater contains ammonia in concentrations that, without treatment, would be harmful to fish and would violate the Basin Plan narrative toxicity objective if discharged to the receiving water. Reasonable potential therefore exists and effluent limitations are required.

Federal regulations at 40 C.F.R. §122.44(d)(1)(i) requires that, “*Limitations must control all pollutants or pollutant parameters (either conventional, nonconventional, or toxic pollutants) which the Director determines are or may be discharged at a level which will cause, have the reasonable potential to cause, or contribute to an excursion above any State water quality standard, including State narrative criteria for water quality.*” For priority pollutants, the SIP dictates the procedures for conducting the RPA. Ammonia is not a priority pollutant. Therefore, the Central Valley Water Board is not restricted to one particular RPA method. Due to the site-specific conditions of the discharge, the Central Valley Water Board has used professional judgment in determining the appropriate method for conducting the RPA for this non-priority pollutant constituent.

U.S. EPA’s September 2010 NPDES Permit Writer’s Manual, page 6-30, states, “*State implementation procedures might allow, or even require, a permit writer to determine reasonable potential through a qualitative assessment process without using available facility-specific effluent monitoring data or when such data are not available...A permitting authority might also determine that WQBEL’s are required for specific pollutants for all facilities that exhibit certain operational or discharge characteristics (e.g., WQBEL’s for pathogens in all permits for POTW’s discharging to contact recreational waters).*” U.S. EPA’s TSD also recommends that factors other than effluent data should be considered in the RPA, “*When determining whether or not a discharge causes, has the reasonable potential to cause, or contributes to an excursion of a numeric or narrative water quality criterion for individual toxicants or for toxicity, the*

*regulatory authority can use a variety of factors and information where facility-specific effluent monitoring data are unavailable. These factors also should be considered with available effluent monitoring data.” With regard to POTW’s, U.S. EPA recommends that, “POTW’s should also be characterized for the possibility of chlorine and ammonia problems.” (TSD, p. 50).*

Nitrification is a biological process that converts ammonia to nitrite and nitrite to nitrate. Denitrification is a process that converts nitrate to nitrite or nitric oxide and then to nitrous oxide or nitrogen gas, which is then released to the atmosphere. The Discharger currently uses nitrification to remove ammonia from the waste stream. Inadequate or incomplete nitrification may result in the discharge of ammonia to the receiving stream. Ammonia is known to cause toxicity to aquatic organisms in surface waters. Discharges of ammonia in concentrations that produce detrimental physiological responses to human, plant, animal, or aquatic life would violate the Basin Plan narrative toxicity objective. Although the Discharger nitrifies the discharge, inadequate or incomplete nitrification creates the potential for ammonia to be discharged and provides the basis for the discharge to have a reasonable potential to cause or contribute to an in-stream excursion above the NAWQC. Therefore, the Central Valley Water Board finds the discharge has reasonable potential for ammonia and WQBEL’s are required.

- (c) **WQBEL’s.** The Central Valley Water Board calculates WQBEL’s in accordance with SIP procedures for non-CTR constituents, and ammonia is a non-CTR constituent. The SIP procedure assumes a 4-day averaging period for calculating the long-term average discharge condition (LTA). However, U.S. EPA recommends modifying the procedure for calculating permit limits for ammonia using a 30-day averaging period for the calculation of the LTA corresponding to the 30-day CCC. Therefore, while the LTAs corresponding to the acute and 4-day chronic criteria were calculated according to SIP procedures, the LTA corresponding to the 30-day CCC was calculated assuming a 30-day averaging period. The lowest LTA representing the acute, 4-day CCC, and 30-day CCC is then selected for deriving the average monthly effluent limitation (AMEL) and the average weekly effluent limitation (AWEL). The remainder of the WQBEL calculation for ammonia was performed according to the SIP procedures. This Order contains final AMEL and AWEL for ammonia of 1.3 mg/L and 2.9 mg/L ammonia (as N), respectively.
- (d) **Plant Performance and Attainability.** Based on 156 sample results for the effluent collected between October 2011 and November 2014, the maximum weekly effluent ammonia concentration was 19 mg/L. The effluent exceeded the applicable AWEL once, but did not exceed the applicable AMEL.. Typically, the Facility removes ammonia to concentrations that are below the applicable AWEL. Thus the Central Valley Water Board concludes that immediate compliance with these effluent limitations is feasible.

ii. **Arsenic.**

- (a) **WQO.** DPH has adopted a Primary MCL for arsenic of 10 µg/L, which is protective of the Basin Plan’s chemical constituent objective.

- (b) **RPA Results.** The MEC for arsenic was 91 µg/L while the maximum observed upstream receiving water concentration was 18 µg/L. Therefore, arsenic in the discharge has a reasonable potential to cause or contribute to an in-stream excursion above the Primary MCL.
- (c) **WQBEL's.** Due to no assimilative capacity, dilution credits are not allowed for development of the WQBEL's for arsenic. This Order contains an AMEL and MDEL for arsenic of 10 µg/L and 19.1 µg/L, respectively, based on the Basin Plan's narrative chemical constituents objective for protection of the MUN beneficial use.
- (d) **Plant Performance and Attainability.** As previously stated, the monitoring data shows that the MEC is greater than the applicable WQBEL, and thus, appears to put the Discharger in immediate non-compliance with the final arsenic effluent limitation. Therefore, TSO R5-2015-0124 provides a compliance schedule to achieve compliance with the final effluent limitations for nitrate plus nitrite by 1 September 2018.

iii. **Mercury**

- (a) **WQO.** The current NAWQC for protection of freshwater aquatic life, continuous concentration, for mercury is 0.77 µg/L (30-day average, chronic criteria). The CTR contains a human health criterion (based on a threshold dose level causing neurological effects in infants) of 0.050 µg/L for waters from which both water and aquatic organisms are consumed. Both values are controversial and subject to change. In 40 C.F.R. part 131, U.S. EPA acknowledges that the human health criteria may not be protective of some aquatic or endangered species and that "*...more stringent mercury limits may be determined and implemented through use of the State's narrative criterion.*" In the CTR, U.S. EPA reserved the mercury criteria for freshwater and aquatic life and may adopt new criteria at a later date.
- (b) **RPA Results.** The MEC for mercury was 0.0035 µg/L based on 35 samples collected between December 2011 and November 2014. The maximum observed upstream receiving water concentration for mercury was 0.010 µg/L based on 10 samples collected between August 2013 and September 2014. Therefore, the effluent does not have reasonable potential to cause or contribute to an exceedance of the CTR criteria for mercury. However, mercury bioaccumulates in fish tissue and, therefore, the discharge of mercury to the receiving water may contribute to exceedances of the narrative toxicity objective and impact beneficial uses. The discharge of mercury to surface waters below major dams in the Central Valley draining to the Sacramento – San Joaquin Delta are being limited in order to protect the beneficial uses of the Delta.
- (c) **WQBEL's.** Order R5-2010-0099 contained a mercury mass-based effluent limitation of 0.05 pounds per calendar year. This Order retains the mercury mass-based effluent limitation of 0.05 pounds per calendar year. This limitation ensures the mercury loading is continued to be maintained at the current level until a TMDL can be established and U.S. EPA develops mercury standards that are protective of human health and is not less stringent than the previous limit. If U.S. EPA develops new water



quality standards for mercury, this permit may be reopened and the effluent limitations adjusted.

- (d) **Plant Performance and Attainability.** The maximum annual mercury mass loading was 0.03 lbs/year. The Central Valley Water Board concludes, therefore, that immediate compliance with these effluent limitations is feasible.

iv. **Nitrate and Nitrite**

- (a) **WQO.** DPH has adopted Primary MCLs for the protection of human health for nitrite and nitrate that are equal to 1 mg/L and 10 mg/L (measured as nitrogen), respectively. DPH has also adopted a Primary MCL of 10 mg/L for the sum of nitrate and nitrite, measured as nitrogen.

USEPA has developed a primary MCL and an MCL goal of 1 mg/L for nitrite (as nitrogen). For nitrate, USEPA has developed Drinking Water Standards (10 mg/L as Primary MCL) and NAWQC for protection of human health (10 mg/L for non-cancer health effects).

- (b) **RPA Results.** The Facility is a POTW that treats domestic wastewater. Untreated domestic wastewater contains ammonia in concentrations that, if untreated, will be harmful to fish and will violate the Basin Plan's narrative toxicity objective. This Order, therefore, requires removal of ammonia (i.e., nitrification). Nitrification is a biological process that converts ammonia to nitrate and nitrite, and will result in effluent nitrate concentrations above the Primary MCL for nitrate plus nitrite. Nitrate concentrations in a drinking water supply above the Primary MCL threatens the health of human fetuses and newborn babies by reducing the oxygen-carrying capacity of the blood (methemoglobinemia). Reasonable potential for nitrate and nitrite therefore exists and WQBEL's are required.

Federal regulations at 40 C.F.R. section 122.44(d)(1)(i) requires that, *"Limitations must control all pollutants or pollutant parameters (either conventional, nonconventional, or toxic pollutants) which the Director determines are or may be discharged at a level which will cause, have the reasonable potential to cause, or contribute to an excursion above any State water quality standard, including State narrative criteria for water quality."* For priority pollutants, the SIP dictates the procedures for conducting the RPA. Nitrate and nitrite are not priority pollutants. Therefore, the Central Valley Water Board is not restricted to one particular RPA method. Due to the site-specific conditions of the discharge, the Central Valley Water Board has used professional judgment in determining the appropriate method for conducting the RPA for this non-priority pollutant constituent.

USEPA's September 2010 NPDES Permit Writer's Manual, page 6-30, states, *"State implementation procedures might allow, or even require, a permit writer to determine reasonable potential through a qualitative assessment process without using available facility-specific effluent monitoring data or when such data are not available...A permitting authority might also determine that WQBEL's are required for specific pollutants for all facilities that exhibit certain operational or discharge characteristics (e.g., WQBEL's for pathogens in all permits for POTW's*

*discharging to contact recreational waters).*” USEPA’s TSD also recommends that factors other than effluent data should be considered in the RPA, “When determining whether or not a discharge causes, has the reasonable potential to cause, or contributes to an excursion of a numeric or narrative water quality criterion for individual toxicants or for toxicity, the regulatory authority can use a variety of factors and information where facility-specific effluent monitoring data are unavailable. These factors also should be considered with available effluent monitoring data.” With regard to POTW’S, USEPA recommends that, “POTW’s should also be characterized for the possibility of chlorine and ammonia problems.” (TSD, p. 50)

The concentration of nitrogen in raw domestic wastewater is sufficiently high that the resultant treated wastewater has a reasonable potential to exceed or threaten to exceed the Primary MCL for nitrate plus nitrite unless the wastewater is treated for nitrogen removal, and therefore an effluent limit for nitrate plus nitrite is required. Denitrification is a process that converts nitrate to nitrite or nitric oxide and then to nitrous oxide or nitrogen gas, which is then released to the atmosphere. The Discharger currently uses nitrification/denitrification to remove ammonia, nitrite, and nitrate from the waste stream. Inadequate or incomplete denitrification may result in the discharge of nitrate and/or nitrite to the receiving stream. Discharges of nitrate plus nitrite in concentrations that exceed the Primary MCL would violate the Basin Plan narrative chemical constituents objective. Although the Discharger denitrifies the discharge, inadequate or incomplete denitrification creates the potential for nitrate and nitrite to be discharged and provides the basis for the discharge to have a reasonable potential to cause or contribute to an in-stream excursion above the Primary MCL. Therefore, the Central Valley Water Board finds the discharge has reasonable potential for nitrate plus nitrite and WQBEL’s are required.

- (c) **WQBEL’s.** This Order contains a final AMEL and AWEL for nitrate plus nitrite of 10 mg/L (total as N) and 19.4 mg/L (total as N), respectively, based on the Primary MCL. This effluent limitation is included in this Order to assure the treatment process adequately nitrifies and denitrifies the waste stream to protect the beneficial use of municipal and domestic supply.
- (d) **Plant Performance and Attainability.** Analysis of the effluent data shows that the MEC for nitrate plus nitrite was 28 µg/L based on 132 samples collected between December 2011 and November 2014, which is greater than applicable WQBEL’s. Based on the sample results for the effluent, the limitations appear to put the Discharger in immediate non-compliance. TSO R5-2015-0124 provides a compliance schedule to achieve compliance with the final effluent limitations for nitrate plus nitrite by 1 September 2016.

v. **Pathogens**

- (a) **WQO.** DDW has developed reclamation criteria, CCR, Division 4, Chapter 3 (Title 22), for the reuse of wastewater. Title 22 requires that for spray irrigation of food crops, parks, playgrounds, schoolyards, and other areas

of similar public access, wastewater be adequately disinfected, oxidized, coagulated, clarified, and filtered, and that the effluent total coliform levels not exceed 2.2 MPN/100 mL as a 7-day median; 23 MPN/100 mL, not to be exceeded more than once in a 30-day period; and 240 MPN/100 mL, at any time.

Title 22 also requires that recycled water used as a source of water supply for non-restricted recreational impoundments be disinfected tertiary recycled water that has been subjected to conventional treatment. A non-restricted recreational impoundment is defined as “...an impoundment of recycled water, in which no limitations are imposed on body-contact water recreational activities.” Title 22 is not directly applicable to surface waters; however, the Central Valley Water Board finds that it is appropriate to apply an equivalent level of treatment to that required by the DDW'sreclamation criteria because the receiving water is used for irrigation of agricultural land and for contact recreation purposes. The stringent disinfection criteria of Title 22 are appropriate since the undiluted effluent may be used for the irrigation of food crops and/or for body-contact water recreation. Coliform organisms are intended as an indicator of the effectiveness of the entire treatment train and the effectiveness of removing other pathogens.

- (b) **RPA Results.** Raw domestic wastewater inherently contains human pathogens that threaten human health and life, and constitute a threatened pollution and nuisance under CWC Section 13050 if discharged untreated to the receiving water. Reasonable potential for pathogens therefore exists and WQBEL's are required.

Federal regulations at 40 C.F.R. §122.44(d)(1)(i) requires that, “*Limitations must control all pollutants or pollutant parameters (either conventional, nonconventional, or toxic pollutants) which the Director determines are or may be discharged at a level which will cause, have the reasonable potential to cause, or contribute to an excursion above any State water quality standard, including State narrative criteria for water quality.*” For priority pollutants, the SIP dictates the procedures for conducting the RPA. Pathogens are not priority pollutants. Therefore, the Central Valley Water Board is not restricted to one particular RPA method. Due to the site-specific conditions of the discharge, the Central Valley Water Board has used professional judgment in determining the appropriate method for conducting the RPA for this non-priority pollutant constituent.

USEPA's September 2010 NPDES Permit Writer's Manual, page 6-30, states, “*State implementation procedures might allow, or even require, a permit writer to determine reasonable potential through a qualitative assessment process without using available facility-specific effluent monitoring data or when such data are not available...A permitting authority might also determine that WQBEL's are required for specific pollutants for all facilities that exhibit certain operational or discharge characteristics (e.g., WQBEL's for pathogens in all permits for POTW's discharging to contact recreational waters).*” USEPA's TSD also recommends that factors other than effluent data should be considered in the RPA, “*When determining whether or not a discharge causes, has the reasonable potential to cause, or contributes to an excursion of a numeric*

*or narrative water quality criterion for individual toxicants or for toxicity, the regulatory authority can use a variety of factors and information where facility-specific effluent monitoring data are unavailable. These factors also should be considered with available effluent monitoring data.” (TSD, p. 50)*

The beneficial uses of Laguna Creek include municipal and domestic supply, water contact recreation, and agricultural irrigation supply, and there is, at times, less than 20:1 dilution. To protect these beneficial uses, the Central Valley Water Board finds that the wastewater must be disinfected and adequately treated to prevent disease. Although the Discharger provides disinfection, inadequate or incomplete disinfection creates the potential for pathogens to be discharged. Therefore, the Central Valley Water Board finds the discharge has reasonable potential for pathogens and WQBEL's are required.

- (c) **WQBEL's.** In accordance with the requirements of Title 22, this Order includes effluent limitations for total coliform organisms of 2.2 MPN/100 mL as a 7-day median; 23 MPN/100 mL, not to be exceeded more than once in a 30-day period; and 240 MPN/100 mL as an instantaneous maximum.

The tertiary treatment process, or equivalent, is capable of reliably treating wastewater to a turbidity level of 2 nephelometric turbidity units (NTU) as a daily average. Failure of the filtration system such that virus removal is impaired would normally result in increased particles in the effluent, which result in higher effluent turbidity. Turbidity has a major advantage for monitoring filter performance. Coliform testing, by comparison, is not conducted continuously and requires several hours, to days, to identify high coliform concentrations. Therefore, to ensure compliance with the DDW recommended Title 22 disinfection criteria, weekly average specifications are impracticable for turbidity. This Order includes operational specifications for turbidity of 2 NTU as a daily average; 5 NTU, not to be exceeded more than 5 percent of the time within a 24-hour period; and 10 NTU as an instantaneous maximum.

This Order contains effluent limitations for BOD<sub>5</sub>, total coliform organisms, and TSS and requires a tertiary level of treatment, or equivalent, necessary to protect the beneficial uses of the receiving water. The Central Valley Water Board has previously considered the factors in Water Code section 13241 in establishing these requirements.

Final WQBEL's for BOD<sub>5</sub> and TSS are based on the technical capability of the tertiary process, which is necessary to protect the beneficial uses of the receiving water. BOD<sub>5</sub> is a measure of the amount of oxygen used in the biochemical oxidation of organic matter. The tertiary treatment standards for BOD<sub>5</sub> and TSS are indicators of the effectiveness of the tertiary treatment process. The principal design parameter for wastewater treatment plants is the daily BOD<sub>5</sub> and TSS loading rates and the corresponding removal rate of the system. The application of tertiary treatment processes results in the ability to achieve lower levels for BOD<sub>5</sub> and TSS than the secondary standards currently prescribed. Therefore, this Order requires AMEL's for BOD<sub>5</sub> and TSS of 10 mg/L, which is technically based on the capability of a tertiary system. In addition to the

average weekly and average monthly effluent limitations, a daily maximum effluent limitation for BOD<sub>5</sub> and TSS is included in the Order to ensure that the treatment works are not organically overloaded and operate in accordance with design capabilities.

vi. **pH**

- (a) **WQO.** The Basin Plan includes a water quality objective for surface waters (except for Goose Lake) that the “...pH shall not be depressed below 6.5 nor raised above 8.5.”
- (b) **RPA Results.** Raw domestic wastewater inherently has variable pH. Additionally, some wastewater treatment processes can increase or decrease wastewater pH which if not properly controlled, would violate the Basin Plan’s numeric objective for pH in the receiving water. Therefore, reasonable potential exists for pH and WQBEL’s are required.

Federal regulations at 40 C.F.R. §122.44(d)(1)(i) requires that, “*Limitations must control all pollutants or pollutant parameters (either conventional, nonconventional, or toxic pollutants) which the Director determines are or may be discharged at a level which will cause, have the reasonable potential to cause, or contribute to an excursion above any State water quality standard, including State narrative criteria for water quality.*” For priority pollutants, the SIP dictates the procedures for conducting the RPA. pH is not a priority pollutant. Therefore, the Central Valley Water Board is not restricted to one particular RPA method. Due to the site-specific conditions of the discharge, the Central Valley Water Board has used professional judgment in determining the appropriate method for conducting the RPA for this non-priority pollutant constituent.

USEPA’s September 2010 NPDES Permit Writer’s Manual, page 6-30, states, “*State implementation procedures might allow, or even require, a permit writer to determine reasonable potential through a qualitative assessment process without using available facility-specific effluent monitoring data or when such data are not available...A permitting authority might also determine that WQBEL’s are required for specific pollutants for all facilities that exhibit certain operational or discharge characteristics (e.g., WQBEL’s for pathogens in all permits for POTW’s discharging to contact recreational waters).*” USEPA’s TSD also recommends that factors other than effluent data should be considered in the RPA, “*When determining whether or not a discharge causes, has the reasonable potential to cause, or contributes to an excursion of a numeric or narrative water quality criterion for individual toxicants or for toxicity, the regulatory authority can use a variety of factors and information where facility-specific effluent monitoring data are unavailable. These factors also should be considered with available effluent monitoring data.*” (TSD, p. 50)

The Facility is a POTW that treats domestic wastewater. Based on 253 samples taken from December 2011 to November 2014 the maximum pH reported was 8.9 and the minimum was 7.1. The Facility exceeded the instantaneous maximum effluent limitation one time on 31 July 2012 . Although the Discharger has proper pH controls in place, the pH for the Facility’s influent varies due to the nature of municipal sewage, which provides the basis for the discharge to have a reasonable potential to

cause or contribute to an in-stream excursion above the Basin Plan's numeric objective for pH in the receiving water. Therefore, WQBEL's for pH are required in this Order.

- (c) **WQBEL's.** Effluent limitations for pH of 6.5 as an instantaneous minimum and 8.2 as an instantaneous maximum are included in this Order based on protection of the Basin Plan objectives for pH.
- (d) **Plant Performance and Attainability.** Based on 253 samples obtained between December 2011 and November 2014, the maximum effluent pH was 8.9 and the minimum effluent pH was 7.1. Only on one occasion was the effluent pH above the instantaneous maximum or below the instantaneous minimum. Thus, the Central Valley Water Board concludes that immediate compliance with these effluent limitations is feasible

vii. **Zinc.**

- (a) **WQO.** The CTR includes hardness-dependent criteria for the protection of freshwater aquatic life for zinc. The criteria for zinc are presented in dissolved concentrations. USEPA recommends conversion factors to translate dissolved concentration to total concentrations. The USEPA default conversion factors for zinc were used for the discharge.
- (a) **RPA Results.** The applicable zinc chronic criterion (maximum 4-day average concentration) is 52 µg/L and the applicable acute criterion (maximum (1-hour concentration) is 52 µg/L, as total recoverable, (see Table F- 6, above). The MEC for total zinc was 53 µg/L, based on data collected between December 2011 and November 2014. Upstream total zinc concentrations varied from 2.6 µg/L to 11 µg/L. Using paired hardness and zinc data, the maximum ambient receiving water concentration did not exceed the applicable CTR criteria for zinc. Based on this information, zinc in the discharge does exhibit reasonable potential to cause or contribute to an in-stream excursion above the CTR criterion for the protection of freshwater aquatic life.
- (a) **WQBELs.** This Order contains a final AMEL and MDEL for zinc of 35 µg/L and 51 µg/L, respectively, based on the CTR criterion for the protection of freshwater aquatic life.
- (b) **Plant Performance and Attainability.** Based on the sample results for the effluent, the limitations appear to put the Discharger in immediate non-compliance. The Discharger submitted an Infeasibility to Comply Report and Request for a Time Schedule Order for zinc along with the ROWD. TSO R5-2015- 0124 contains a compliance schedule and interim effluent limitations for zinc.

**4. WQBEL Calculations**

- a. This Order includes WQBEL's for ammonia, arsenic, nitrate plus nitrite, pathogens, pH, and zinc. The general methodology for calculating WQBEL's based on the different criteria/objectives is described in subsections IV.C.5.b through e, below. See Attachment H for the WQBEL calculations.

- b. **Effluent Concentration Allowance.** For each water quality criterion/objective, the ECA is calculated using the following steady-state mass balance equation from Section 1.4 of the SIP:

$$\begin{aligned} ECA &= C + D(C - B) && \text{where } C > B, \text{ and} \\ ECA &= C && \text{where } C \leq B \end{aligned}$$

where:

ECA = effluent concentration allowance  
D = dilution credit  
C = the priority pollutant criterion/objective  
B = the ambient background concentration.

According to the SIP, the ambient background concentration (B) in the equation above shall be the observed maximum with the exception that an ECA calculated from a priority pollutant criterion/objective that is intended to protect human health from carcinogenic effects shall use the arithmetic mean concentration of the ambient background samples. For ECAs based on MCLs, which implement the Basin Plan's chemical constituents objective and are applied as annual averages, an arithmetic mean is also used for B due to the long-term basis of the criteria.

- c. **Basin Plan Objectives and MCLs.** For WQBEL's based on site-specific numeric Basin Plan objectives or MCLs, the effluent limitations are applied directly as the ECA as either an MDEL, AMEL, or average annual effluent limitations, depending on the averaging period of the objective.
- d. **Aquatic Toxicity Criteria.** WQBEL's based on acute and chronic aquatic toxicity criteria are calculated in accordance with Section 1.4 of the SIP. The ECAs are converted to equivalent long-term averages (i.e. LTA<sub>acute</sub> and LTA<sub>chronic</sub>) using statistical multipliers and the lowest LTA is used to calculate the AMEL and MDEL using additional statistical multipliers.
- e. **Human Health Criteria.** WQBEL's based on human health criteria, are also calculated in accordance with Section 1.4 of the SIP. The AMEL is set equal to ECA and a statistical multiplier was used to calculate the MDEL.

$$\begin{aligned} AMEL &= mult_{AMEL} \left[ \min \left( \overbrace{M_A ECA_{acute}, M_C ECA_{chronic}}^{LTA_{acute}} \right) \right] \\ MDEL &= mult_{MDEL} \left[ \min \left( \overbrace{M_A ECA_{acute}, M_C ECA_{chronic}}^{LTA_{chronic}} \right) \right] \\ MDEL_{HH} &= \left( \frac{mult_{MDEL}}{mult_{AMEL}} \right) AMEL_{HH} \end{aligned}$$

where:

$mult_{AMEL}$  = statistical multiplier converting minimum LTA to AMEL

$mult_{MDEL}$  = statistical multiplier converting minimum LTA to MDEL

$M_A$  = statistical multiplier converting acute ECA to  $LTA_{acute}$

$M_C$  = statistical multiplier converting chronic ECA to  $LTA_{chronic}$

### Summary of Water Quality-Based Effluent Limitations Discharge Point No. Eff-001

Table F-10. Summary of Water Quality-Based Effluent Limitations

| Parameter                      | Units                | Effluent Limitations |                  |                 |                       |                       |
|--------------------------------|----------------------|----------------------|------------------|-----------------|-----------------------|-----------------------|
|                                |                      | Average Monthly      | Average Weekly   | Maximum Daily   | Instantaneous Minimum | Instantaneous Maximum |
| Ammonia Nitrogen, Total (as N) | mg/L                 | 1.3                  | 2.9              | --              | --                    | --                    |
|                                | lbs/day <sup>1</sup> | 32                   | 75               | --              | --                    | --                    |
|                                | lbs/day <sup>2</sup> | 49                   | 109              | --              | --                    | --                    |
| Arsenic                        | µg/L                 | 10                   | --               | 19.1            | --                    | --                    |
| Mercury, Total Recoverable     | lbs/year             | 0.05 <sup>3</sup>    | --               | --              | --                    | --                    |
| Nitrate + Nitrite (as N)       | mg/L                 | 10                   | 19.4             | --              | --                    | --                    |
| pH                             | standard units       | --                   | --               | --              | 6.5                   | 8.2                   |
| Total Coliform Organisms       | MPN/100 mL           | --                   | 2.2 <sup>4</sup> | 23 <sup>5</sup> | --                    | 240                   |
| Zinc, Total Recoverable        | µg/L                 | 35                   | --               | 51              | --                    | --                    |

1. Based on an average dry weather flow of 3.0 million gallons per day (MGD). Effective immediately and until Executive Officer's written approval of flow increase (Special Provisions VI.C.6.b).
2. Based on an average dry weather flow of 4.5 MGD. Effective upon Executive Officer's written approval of flow increase (Special Provisions VI.C.6.b).
3. The effluent calendar year annual average total mercury load shall not exceed 0.05 pounds/year.
4. Applied as a 7-day median effluent limitation.
5. Not to be exceeded more than once in any 30-day period.

#### 5. Whole Effluent Toxicity (WET)

For compliance with the Basin Plan's narrative toxicity objective, this Order requires the Discharger to conduct whole effluent toxicity testing for acute and chronic toxicity, as specified in the Monitoring and Reporting Program (Attachment E section V.). This Order also contains effluent limitations for acute toxicity and requires the Discharger to implement best management practices to investigate the causes of, and identify corrective actions to reduce or eliminate effluent toxicity.

- a. **Acute Aquatic Toxicity.** The Basin Plan contains a narrative toxicity objective that states, "*All waters shall be maintained free of toxic substances in concentrations that produce detrimental physiological responses in human, plant, animal, or aquatic life.*" (Basin Plan at page III-8.00) The Basin Plan also states that, "...effluent limits based upon acute biotoxicity tests of effluents will be prescribed where appropriate...".



For priority pollutants, the SIP dictates the procedures for conducting the RPA. Acute toxicity is not a priority pollutant. Therefore, the Central Valley Water Board is not restricted to one particular RPA method. Acute whole effluent toxicity is not a priority pollutant. Therefore, due to the site-specific conditions of the discharge, the Central Valley Water Board has used professional judgment in determining the appropriate method for conducting the RPA. USEPA's September 2010 NPDES Permit Writer's Manual, page 6-30, states, "*State implementation procedures might allow, or even require, a permit writer to determine reasonable potential through a qualitative assessment process without using available facility-specific effluent monitoring data or when such data are not available...A permitting authority might also determine that WQBEL's are required for specific pollutants for all facilities that exhibit certain operational or discharge characteristics (e.g., WQBEL's for pathogens in all permits for POTW's discharging to contact recreational waters).*" Although the discharge has been consistently in compliance with the acute effluent limitations, the Facility is a POTW that treats domestic wastewater containing ammonia and other acutely toxic pollutants. Acute toxicity effluent limits are required to ensure compliance with the Basin Plan's narrative toxicity objective.

USEPA Region 9 provided guidance for the development of acute toxicity effluent limitations in the absence of numeric water quality objectives for toxicity in its document titled "Guidance for NPDES Permit Issuance", dated February 1994. In section B.2. "Toxicity Requirements" (pgs. 14-15) it states that, "*In the absence of specific numeric water quality objectives for acute and chronic toxicity, the narrative criterion 'no toxics in toxic amounts' applies. Achievement of the narrative criterion, as applied herein, means that ambient waters shall not demonstrate for acute toxicity: 1) less than 90% survival, 50% of the time, based on the monthly median, or 2) less than 70% survival, 10% of the time, based on any monthly median. For chronic toxicity, ambient waters shall not demonstrate a test result of greater than 1 TUc.*" Accordingly, effluent limitations for acute toxicity have been included in this Order as follows:

**Acute Toxicity.** Survival of aquatic organisms in 96-hour bioassays of undiluted waste shall be no less than:

|                                                 |     |
|-------------------------------------------------|-----|
| Minimum for any one bioassay -----              | 70% |
| Median for any three consecutive bioassays----- | 90% |

- b. **Chronic Aquatic Toxicity.** The Basin Plan contains a narrative toxicity objective that states, "*All waters shall be maintained free of toxic substances in concentrations that produce detrimental physiological responses in human, plant, animal, or aquatic life.*" (Basin Plan at page III-8.00.) As shown in the table below, based on chronic WET testing performed by the Discharger from December 2010 through October 2014, the discharge does not have reasonable potential to cause or contribute to an in-stream excursion above the Basin Plan's narrative toxicity objective.

**Table F-11. Whole Effluent Chronic Toxicity Testing Results**

| Date       | Fathead Minnow<br><i>Pimephales promelas</i> |                 | Water Flea<br><i>Ceriodaphnia dubia</i> |                       | Green Algae<br><i>Selenastrum capricornutum</i> |
|------------|----------------------------------------------|-----------------|-----------------------------------------|-----------------------|-------------------------------------------------|
|            | Survival<br>(TUC)                            | Growth<br>(TUC) | Survival<br>(TUC)                       | Reproduction<br>(TUC) | Growth<br>(TUC)                                 |
| 12/6/2010  | 1                                            | 1               | 1                                       | 1                     | 1                                               |
| 1/10/2011  | 1                                            | 1               | 1                                       | 1                     | 1                                               |
| 4/11/2011  | 1                                            | 1               | 1                                       | 1                     | 1                                               |
| 10/24/2011 | 1                                            | 1               | 1                                       | 1                     | 1                                               |
| 1/24/2012  | 1                                            | 1               | 1                                       | 1                     | 1                                               |
| 4/9/2012   | 1                                            | 1               | 1                                       | 1                     | 1                                               |
| 7/10/2012  | 11                                           | 1               | 1                                       | 1                     | 1                                               |
| 10/8/2012  | 1                                            | 1               | 1                                       | 1                     | 1                                               |
| 1/14/2013  | 1                                            | 1               | 1                                       | 1                     | 1                                               |
| 4/8/2013   | 1                                            | 1               | 1                                       | 1                     | 1                                               |
| 8/5/2013   | 1                                            | 1               | 1                                       | 1                     | 1                                               |
| 10/8/2013  | (a)                                          | (a)             | 1                                       | 1                     | 1                                               |
| 11/19/2013 | 1                                            | 1               | --                                      | --                    | --                                              |
| 1/6/2014   | 1                                            | 1               | 1                                       | 1                     | 1                                               |
| 5/6/2014   | 1                                            | 1               | 1                                       | 1                     | 1                                               |
| 7/8/2014   | 1                                            | 1               | 1                                       | 1                     | 1                                               |
| 10/7/2014  | 1                                            | 1               | 1                                       | 1                     | 1                                               |

(a) The 10/8/13 test was halted after day four because of excessive organism mortality in controls and all treatments. *C. dubia* survival and reproduction test was repeated on 11/19/2013.

The Monitoring and Reporting Program of this Order requires quarterly chronic WET monitoring for demonstration of compliance with the narrative toxicity objective. In addition to WET monitoring, the Special Provision in section VI.C.2.a of the Order requires the Discharger to submit to the Central Valley Water Board an Initial Investigative TRE Workplan for approval by the Executive Officer, to ensure the Discharger has a plan to immediately move forward with the initial tiers of a TRE, in the event effluent toxicity is encountered in the future. The Discharger currently has an approved TRE Work Plan in place. The provision also includes a numeric toxicity monitoring trigger, requirements for accelerated monitoring, and requirements for TRE initiation if toxicity is demonstrated.

Numeric chronic WET effluent limitations have not been included in this Order. The SIP contains implementation gaps regarding the appropriate form and implementation of chronic toxicity limits. This has resulted in the petitioning of a NPDES permit in the Los Angeles Region<sup>1</sup> that contained numeric chronic toxicity effluent limitations. To address the petition, the State Water Board adopted WQO 2003-012 directing its staff to revise the toxicity control provisions in the SIP. The State Water Board states the following in WQO 2003-012, "In reviewing this petition and receiving comments from numerous interested persons on the propriety of including numeric effluent limitations for chronic toxicity in NPDES permits for publicly-owned treatment works that

<sup>1</sup> In the Matter of the Review of Own Motion of Waste Discharge Requirements Order Nos. R4-2002-0121 [NPDES No. CA0054011] and R4-2002-0123 [NPDES NO. CA0055119] and Time Schedule Order Nos. R4-2002-0122 and R4-2002-0124 for Los Coyotes and Long Beach Wastewater Reclamation Plants Issued by the California Regional Water Quality Control Board, Los Angeles Region SWRCB/OCC FILES A-1496 AND 1496(a)

*discharge to inland waters, we have determined that this issue should be considered in a regulatory setting, in order to allow for full public discussion and deliberation. We intend to modify the SIP to specifically address the issue. We anticipate that review will occur within the next year. We therefore decline to make a determination here regarding the propriety of the final numeric effluent limitations for chronic toxicity contained in these permits.”* The process to revise the SIP is currently underway. Proposed changes include clarifying the appropriate form of effluent toxicity limits in NPDES permits and general expansion and standardization of toxicity control implementation related to the NPDES permitting process. Since the toxicity control provisions in the SIP are under revision it is infeasible to develop numeric effluent limitations for chronic toxicity. Therefore, this Order requires that the Discharger meet best management practices for compliance with the Basin Plan’s narrative toxicity objective, as allowed under 40 C.F.R. section 122.44(k).

To ensure compliance with the Basin Plan’s narrative toxicity objective, the Discharger is required to conduct chronic WET testing, as specified in the Monitoring and Reporting Program (Attachment E section V.). Furthermore, the Special Provision contained at VI.C.2.a. of this Order requires the Discharger to investigate the causes of, and identify and implement corrective actions to reduce or eliminate effluent toxicity. If the discharge demonstrates toxicity exceeding the numeric toxicity monitoring trigger, the Discharger is required to initiate a Toxicity Reduction Evaluation (TRE) in accordance with an approved TRE workplan. The numeric toxicity monitoring trigger is not an effluent limitation; it is the toxicity threshold at which the Discharger is required to perform accelerated chronic toxicity monitoring, as well as, the threshold to initiate a TRE if effluent toxicity has been demonstrated.

## **D. Final Effluent Limitation Considerations**

### **1. Mass-based Effluent Limitations**

40 C.F.R section 122.45(f)(1) requires effluent limitations be expressed in terms of mass, with some exceptions, and 40 C.F.R. section 122.45(f)(2) allows pollutants that are limited in terms of mass to additionally be limited in terms of other units of measurement. This Order includes effluent limitations expressed in terms of mass and concentration. In addition, pursuant to the exceptions to mass limitations provided in 40 C.F.R. section 122.45(f)(1), some effluent limitations are not expressed in terms of mass, such as pH and temperature, and when the applicable standards are expressed in terms of concentration (e.g., CTR criteria and MCL’s) and mass limitations are not necessary to protect the beneficial uses of the receiving water.

Mass-based effluent limitations were calculated based upon the design flow (Average Dry Weather Flow) permitted in section IV.A.1.e of this Order.

### **2. Averaging Periods for Effluent Limitations**

40 C.F.R. section 122.45 (d) requires average weekly and average monthly discharge limitations for POTW’s unless impracticable. However, for toxic pollutants and pollutant parameters in water quality permitting, USEPA recommends the use of a maximum daily effluent limitation in lieu of average weekly effluent limitations for two reasons. *“First, the basis for the 7-day average for POTW’s derives from the secondary treatment requirements. This basis is not related to the need for assuring achievement of water quality standards. Second, a 7-day average, which could comprise up to seven or more daily samples, could average out peak toxic concentrations and therefore the discharge’s potential for causing acute toxic effects*

*would be missed.*" (TSD, pg. 96) This Order uses maximum daily effluent limitations in lieu of average weekly effluent limitations for zinc as required by the SIP.

### 3. Satisfaction of Anti-Backsliding Requirements

The CWA specifies that a revised permit may not include effluent limitations that are less stringent than the previous permit unless a less stringent limitation is justified based on exceptions to the anti-backsliding provisions contained in CWA sections 402(o) or 303(d)(4), or, where applicable, 40 C.F.R. section 122.44(l).

The effluent limitations in this Order are at least as stringent as the effluent limitations in the previous Order, with the exception of effluent limitations for **aluminum, bis(2-ethylhexyl)phthalate, carbon tetrachloride, chlorodibromomethane, copper, cyanide, dichlorobromomethane, iron, lead, and manganese**. The effluent limitations for these pollutants are less stringent than those in Order R5-2010-0099. This relaxation of effluent limitations is consistent with the anti-backsliding requirements of the CWA and federal regulations.

- a. **CWA section 402(o)(1) and 303(d)(4).** CWA section 402(o)(1) prohibits the establishment of less stringent water quality-based effluent limits "*except in compliance with Section 303(d)(4).*" CWA section 303(d)(4) has two parts: paragraph (A) which applies to nonattainment waters and paragraph (B) which applies to attainment waters.
  - i. For waters where standards are not attained, CWA section 304(d)(4)(A) specifies that any effluent limit based on a TMDL or other WLA may be revised only if the cumulative effect of all such revised effluent limits based on such TMDL's or WLAs will assure the attainment of such water quality standards.
  - ii. For attainment waters, CWA section 303(d)(4)(B) specifies that a limitation based on a water quality standard may be relaxed where the action is consistent with the antidegradation policy.

Laguna Creek is considered an attainment water for aluminum, bis(2-ethylhexyl)phthalate, carbon tetrachloride, chlorodibromomethane, copper, cyanide, dichlorobromomethane, iron, lead, manganese, and residual chlorine because the receiving water is not listed as impaired on the 303(d) list for this constituent.<sup>1</sup> As discussed in section IV.D.4, below, removal of the effluent limits complies with federal and state antidegradation requirements. Thus, removal of the effluent limitations for aluminum, bis(2-ethylhexyl)phthalate, carbon tetrachloride, chlorodibromomethane, copper, cyanide, dichlorobromomethane, iron, lead, manganese, and residual chlorine from Order R5-2010-0099 meets the exception in CWA section 303(d)(4)(B).

- b. **CWA section 402(o)(2).** CWA section 402(o)(2) provides several exceptions to the anti-backsliding regulations. CWA 402(o)(2)(B)(i) allows a renewed, reissued, or modified permit to contain a less stringent effluent limitation for a pollutant if information is available which was not available at the time of permit issuance (other than revised regulations, guidance, or test methods) and which would have justified the application of a less stringent effluent limitation at the time of permit issuance.

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<sup>1</sup> "The exceptions in Section 303(d)(4) address both waters in attainment with water quality standards and those not in attainment, i.e. waters on the section 303(d) impaired waters list." State Water Board Order WQ 2008-0006, Berry Petroleum Company, Poso Creek/McVan Facility.

As described further in section IV.C.3.b of this Fact Sheet, updated information that was not available at the time Order R5-2010-0099 was issued indicates that aluminum, bis(2-ethylhexyl)phthalate, carbon tetrachloride, chlorodibromomethane, copper, cyanide, dichlorobromomethane, iron, lead, manganese, and residual chlorine do not exhibit reasonable potential to cause or contribute to an exceedance of water quality objectives in the receiving water. The updated information that supports the relaxation of effluent limitations for these constituents includes the following:

- i. **Aluminum.** Effluent monitoring data collected between December 2011 and November 2014 indicates that aluminum in the discharge does not exhibit reasonable potential to cause or contribute to an exceedance of the Secondary MCL. This is consistent with the federal antibacksliding regulations, because the new data represent new information that was not available at the time the previous Order was adopted.
- ii. **Bis (2-ethylhexyl) Phthalate.** Effluent and receiving water monitoring data collected between December 2011 and November 2014 for bis (2-ethylhexyl) phthalate indicates that the discharge does not exhibit reasonable potential to cause or contribute to an exceedance of the CTR human health criteria. This is consistent with the federal antibacksliding regulations, because the new data represent new information that was not available at the time the previous Order was adopted.
- iii. **Carbon Tetrachloride.** Effluent and receiving water monitoring data collected between December 2011 and November 2014 for carbon tetrachloride indicates that the discharge does not exhibit reasonable potential to cause or contribute to an exceedance of the CTR human health criteria. This is consistent with the federal antibacksliding regulations, because the new data represent new information that was not available at the time the previous Order was adopted.
- iv. **Chlorine Residual.** The Discharger converted from chlorine disinfection to UV disinfection upon completion of tertiary upgrades in 2011. Therefore, there is no reasonable potential to cause or contribute to an exceedance of the NAWQC criterion for chlorine. This is consistent with the federal antibacksliding regulations, because the new data represent new information that was not available at the time the previous Order was adopted.
- v. **Chlorodibromomethane.** The Discharger converted from chlorine disinfection to UV disinfection upon completion of tertiary upgrades in 2011. Therefore, there is no reasonable potential to cause or contribute to an exceedance of the NAWQC criterion for chlorodibromomethane. This is consistent with the federal antibacksliding regulations, because the new data represent new information that was not available at the time the previous Order was adopted.
- vi. **Copper.** Previous Order R5-2010-0099 included WQBEL's for copper. The Discharger submitted an approved WER study for copper in March 2015 that calculated a site specific WER for copper of 15. The application of the WER for copper to the Reasonable Potential Analysis and effluent and receiving water monitoring data collected between December 2011 and November 2014 for copper indicates that the discharge does not exhibit reasonable potential to cause or contribute to an exceedance of the CTR freshwater aquatic life criteria. Therefore, the effluent limits for copper have been removed from this Order. This is consistent with the federal antibacksliding regulations, because

the new data and WER study represent new information that was not available at the time the previous Order was adopted. See Section IV.D.4.a of this Fact Sheet for Antidegradation discussion.

- vii. **Cyanide.** Effluent and receiving water monitoring data collected between December 2011 and November 2014 for cyanide indicates that the discharge does not exhibit reasonable potential to cause or contribute to an exceedance of the CTR criteria for the protection of freshwater aquatic life. This is consistent with the federal antibacksliding regulations, because the new data represent new information that was not available at the time the previous Order was adopted.
- viii. **Dichlorobromomethane.** The Discharger converted from chlorine disinfection to UV disinfection upon completion of tertiary upgrades in 2011. Therefore, there is no reasonable potential to cause or contribute to an exceedance of the NAWQC criterion for dichlorobromomethane. This is consistent with the federal antibacksliding regulations, because the new data represent new information that was not available at the time the previous Order was adopted.
- ix. **Iron.** Effluent monitoring data collected between December 2011 and November 2014 indicates that iron in the discharge does not exhibit reasonable potential to cause or contribute to an exceedance of the Secondary MCL. This is consistent with the federal antibacksliding regulations, because the new data represent new information that was not available at the time the previous Order was adopted.
- x. **Lead.** Effluent and receiving water monitoring data collected between December 2011 and November 2014 indicates that lead in the discharge does not exhibit reasonable potential to cause or contribute to an exceedance of the CTR criteria for the protection of freshwater aquatic life. This is consistent with the federal antibacksliding regulations, because the new data represent new information that was not available at the time the previous Order was adopted.
- xi. **Manganese.** Effluent monitoring data collected between December 2011 and November 2014 indicates that manganese in the discharge does not exhibit reasonable potential to cause or contribute to an exceedance of the Secondary MCL. This is consistent with the federal antibacksliding regulations, because the new data represent new information that was not available at the time the previous Order was adopted.

#### 4. Antidegradation Policies

As discussed in section II.E of this II.E of this Fact Sheet, the Discharger is planning an upgrade and expansion project that would increase the design capacity of the Facility from 3.0 MGD to 4.5 MGD. Order R5-2010-0099 provided antidegradation findings and authorized an increase in the permitted average discharge flow to 4.5 MGD for the expanded Facility. This Order does not provide for an expansion from the previously authorized discharge rate of 4.5 MGD. Therefore, a complete antidegradation analysis is not necessary. The Order requires compliance with applicable federal technology-based standards and with WQBEL's where the discharge could have the reasonable potential to cause or contribute to an exceedance of water quality standards. The permitted discharge is consistent with the antidegradation provisions of 40 C.F.R. section 131.12 and State Water Board Resolution No. 68-16. Compliance with these requirements will result in the use of best practicable treatment or control of the discharge. The impact on existing water quality will be insignificant.

This Order removes effluent limitations for aluminum, bis(2-ethylhexyl)phthalate, carbon tetrachloride, chlorodibromomethane, cyanide, dichlorobromomethane, and lead based on updated monitoring data demonstrating that the effluent does not cause or contribute to an exceedance of the applicable water quality criteria or objectives in the receiving water. The removal of WQBEL's for these parameters will not result in an increase in pollutant concentration or loading, a decrease in the level of treatment or control, or a reduction of water quality. Therefore, the Central Valley Water Board finds that the removal of the effluent limitations does not result in an allowed increase in pollutants or any additional degradation of the receiving water. Thus, the removal of effluent limitations is consistent with the antidegradation provisions of 40 C.F.R. section 131.12 and State Water Board Resolution No. 68-16.

- a. **Copper.** This Order removes effluent limitations for copper based on the Central Valley Water Board's approval of the Discharger's site specific WER study for copper. The removal of the WQBEL's for copper may result in an increase in pollutant concentration or loading, or a reduction in water quality. However, in accordance with State Water Board Resolution No. 68-16, the Discharger is employing the best practicable treatment or control (BPTC) of the discharge necessary to assure that (a) a pollution or nuisance will not occur and (b) the highest water quality consistent with maximum benefit to the people of the State will be maintained. Therefore, the Central Valley Water Board finds that the removal of the effluent limitations for copper are consistent with State Water Board Resolution No. 68-16.

## 5. Stringency of Requirements for Individual Pollutants

This Order contains both technology-based effluent limitations and WQBEL's for individual pollutants. The technology-based effluent limitations consist of restrictions on BOD<sub>5</sub> and TSS. Restrictions on these parameters are discussed in section IV.B.2 of this Fact Sheet. This Order's technology-based pollutant restrictions implement the minimum, applicable federal technology-based requirements. In addition, this Order contains effluent limitations more stringent than the minimum, federal technology-based requirements that are necessary to meet water quality standards.

### Summary of Final Effluent Limitations Discharge Point EFF-001

Table F-12. Summary of Final Effluent Limitations

| Parameter                                | Units                | Effluent Limitations               |                |               |                       |                       | Basis <sup>1</sup> |
|------------------------------------------|----------------------|------------------------------------|----------------|---------------|-----------------------|-----------------------|--------------------|
|                                          |                      | Average Monthly                    | Average Weekly | Maximum Daily | Instantaneous Minimum | Instantaneous Maximum |                    |
| Average Dry Weather Flow                 | MGD                  | 3.0 <sup>2</sup> /4.5 <sup>3</sup> |                |               |                       |                       | DC                 |
| Biochemical Oxygen Demand (5-day @ 20°C) | mg/L                 | 10                                 | 15             | 20            | --                    | --                    | TTC                |
|                                          | lbs/day <sup>4</sup> | 375                                | 560            | 750           | --                    | --                    |                    |
|                                          | % Removal            | 85                                 | --             | --            | --                    | --                    | CFR                |
| pH                                       | Standard units       | --                                 | --             | --            | 6.5                   | 8.2                   | BP                 |
| Total Suspended Solids                   | mg/L                 | 10                                 | 15             | 20            | --                    | --                    | TTC                |
|                                          | lbs/day <sup>4</sup> | 375                                | 560            | 750           | --                    | --                    |                    |

| Parameter                      | Units       | Effluent Limitations             |                  |                 |                       |                       | Basis <sup>1</sup> |
|--------------------------------|-------------|----------------------------------|------------------|-----------------|-----------------------|-----------------------|--------------------|
|                                |             | Average Monthly                  | Average Weekly   | Maximum Daily   | Instantaneous Minimum | Instantaneous Maximum |                    |
|                                | % Removal   | 85                               | --               | --              | --                    | --                    | CFR                |
| Ammonia Nitrogen, Total (as N) | mg/L        | 1.3                              | 2.9              | --              | --                    | --                    | NAWQC              |
|                                | lbs/day     | 49                               | 109              | --              | --                    | --                    | NAWQC              |
| Arsenic                        | µg/L        | 10                               | --               | 19              | --                    | --                    | MCL                |
| Nitrate Plus Nitrite           | mg/L        | 10                               | 19               | --              | --                    | --                    | MCL                |
| Zinc, Total Recoverable        | µg/L        | 35                               | --               | 51              | --                    | --                    | CTR                |
| Total Coliform Organisms       | MPN/ 100 mL | --                               | 2.2 <sup>5</sup> | 23 <sup>6</sup> | --                    | 240                   | Title 22           |
| Acute Toxicity                 | % Survival  | 70 <sup>7</sup> /90 <sup>8</sup> | --               | --              | --                    | --                    | BP                 |

- <sup>1</sup> DC – Based on the design capacity of the Facility.  
TTC – Based on tertiary treatment capability. These effluent limitations reflect the capability of a properly operated tertiary treatment plant.  
CFR – Based on secondary treatment standards contained in 40 CFR part 133.  
BP – Based on water quality objectives contained in the Basin Plan.  
CTR – Based on water quality criteria contained in the California Toxics Rule and applied as specified in the SIP.  
NAWQC – Based on USEPA's National Ambient Water Quality Criteria for the protection of freshwater aquatic life.  
MCL – Based on the Primary Maximum Contaminant Level.  
Title 22 – Based on CA Department of Public Health Reclamation Criteria, CCR, Division 4, Chapter 3 (Title 22).
- <sup>2</sup> Effective until the Discharger demonstrates compliance with Special Provision VI.C.6.b of this Order, the average dry weather flow shall not exceed 3.0 MGD.
- <sup>3</sup> Effective upon compliance with Special Provision VI.C.6.b of this Order, the average dry weather flow shall not exceed 4.5 MGD.
- <sup>4</sup> Based on an average dry weather flow of 4.5 MGD.
- <sup>5</sup> Applied as a 7-day median effluent limitation.
- <sup>6</sup> Not to be exceeded more than once in any 30-day period.
- <sup>7</sup> 70% minimum of any one bioassay.
- <sup>8</sup> 90% median for any three consecutive bioassays.

**E. Interim Effluent Limitations – Not Applicable**

**F. Land Discharge Specifications – Not Applicable.**

**G. Recycling Specifications – Not Applicable.**

**V. RATIONALE FOR RECEIVING WATER LIMITATIONS**

**A. Surface Water**

1. CWA section 303(a-c), requires states to adopt water quality standards, including criteria where they are necessary to protect beneficial uses. The Central Valley Water Board adopted water quality criteria as water quality objectives in the Basin Plan. The Basin Plan states that “[t]he numerical and narrative water quality objectives define the least stringent standards that the Regional Water Board will apply to regional waters in order to protect the beneficial uses.” The Basin Plan includes numeric and narrative water quality objectives for various beneficial uses and water bodies. This Order contains receiving surface water limitations based on the Basin Plan numerical and narrative



water quality objectives for ammonia, bacteria, biostimulatory substances, color, chemical constituents, dissolved oxygen, floating material, oil and grease, pH, pesticides, radioactivity, suspended sediment, settleable substances, suspended material, tastes and odors, temperature, toxicity, and turbidity.

**B. Groundwater – Not Applicable.**

**VI. RATIONALE FOR PROVISIONS**

**A. Standard Provisions**

Standard Provisions, which apply to all NPDES permits in accordance with 40 C.F.R. section 122.41, and additional conditions applicable to specified categories of permits in accordance with 40 C.F.R. section 122.42, are provided in Attachment D. The discharger must comply with all standard provisions and with those additional conditions that are applicable under section 122.42.

Sections 122.41(a)(1) and (b) through (n) of 40 C.F.R. establish conditions that apply to all state-issued NPDES permits. These conditions must be incorporated into the permits either expressly or by reference. If incorporated by reference, a specific citation to the regulations must be included in the Order. Section 123.25(a)(12) of 40 C.F.R. allows the state to omit or modify conditions to impose more stringent requirements. In accordance with 40 C.F.R. section 123.25, this Order omits federal conditions that address enforcement authority specified in 40 C.F.R. sections 122.41(j)(5) and (k)(2) because the enforcement authority under the Water Code is more stringent. In lieu of these conditions, this Order incorporates by reference Water Code section 13387(e).

**B. Special Provisions**

**1. Reopener Provisions**

- a. **Mercury.** This provision allows the Central Valley Water Board to reopen this Order in the event mercury is found to be causing toxicity based on acute or chronic toxicity test results, or if a TMDL program is adopted. In addition, this Order may be reopened if the Central Valley Water Board determines that a mercury offset program is feasible for dischargers subject to NPDES permits.
- b. **Whole Effluent Toxicity.** This Order requires the Discharger to investigate the causes of, and identify corrective actions to reduce or eliminate effluent toxicity through a Toxicity Reduction Evaluation (TRE). This Order may be reopened to include a numeric chronic toxicity limitation, a new acute toxicity limitation, and/or a limitation for a specific toxicant identified in the TRE. Additionally, if a numeric chronic toxicity water quality objective is adopted by the State Water Board, this Order may be reopened to include a numeric chronic toxicity limitation based on that objective.
- c. **Water Effects Ratio (WER) and Metal Translators.** A default WER of 1.0 has been used in this Order for calculating criteria for applicable inorganic constituents. In addition, default dissolved-to-total metal translators have been used to convert water quality objectives from dissolved to total recoverable when developing effluent limitations for zinc. If the Discharger performs studies to determine site-specific WERs and/or site-specific dissolved-to-total metal translators, this Order may be reopened to modify the effluent limitations for the applicable inorganic constituents.
- d. **Drinking Water Policy.** On 26 July 2013 the Central Valley Water Board adopted Resolution No. R5-2013-0098 amending the Basin Plan and establishing a Drinking Water Policy. The State Water Board approved the Drinking Water Policy on

3 December 2013. This Order may be reopened to incorporate monitoring of drinking water constituents to implement the Drinking Water Policy.

- e. **Ultraviolet Light (UV) Disinfection Operating Specifications** . UV system operating specifications are required to ensure that the UV system is operated to achieve the required pathogen removal. UV disinfection system specifications and monitoring and reporting requirements are required to ensure that adequate UV dosage is applied to the wastewater to inactivate pathogens (e.g., viruses) in the wastewater. UV dosage is dependent on several factors such as UV transmittance, UV power setting, wastewater turbidity, and wastewater flow through the UV disinfection system. The UV specifications in this Order are based on the National Water Research Institute (NWRI) and American Water Works Association Research Foundation (AWWRF) “*Ultraviolet Disinfection Guidelines for Drinking Water and Water Reuse*” first published in December 2000 and revised as a Third Edition dated August 2012 (NWRI guidelines). If the Discharger conducts a site-specific UV engineering study that identifies site-specific UV operating specifications that will achieve the virus inactivation required by Title 22 for disinfected tertiary recycled water, this Order may be reopened to modify the UV specifications, in accordance with Reopener Provision VI.C.1.e.

## 2. Special Studies and Additional Monitoring Requirements

- a. **Chronic Whole Effluent Toxicity Requirements.** The Basin Plan contains a narrative toxicity objective that states, “*All waters shall be maintained free of toxic substances in concentrations that produce detrimental physiological responses in human, plant, animal, or aquatic life.*” (Basin Plan at page III-8.00.) Based on whole effluent chronic toxicity testing performed by the Discharger from December 2010 through October 2014, the discharge does not have reasonable potential to cause or contribute to an in-stream excursion above of the Basin Plan's narrative toxicity objective.

The Monitoring and Reporting Program of this Order requires chronic WET monitoring for demonstration of compliance with the narrative toxicity objective. The provision also includes a numeric toxicity monitoring trigger, requirements for accelerated monitoring, and requirements for TRE initiation, in accordance with the TRE Workplan previously submitted by the Discharger, if toxicity is demonstrated.

**Monitoring Trigger.** A numeric toxicity monitoring trigger of  $> 1$  TUC (where TUC = 100/NOEC) is applied in the provision, because this Order does not allow any dilution for the chronic condition. Therefore, a TRE is triggered when the effluent exhibits toxicity at 100% effluent.

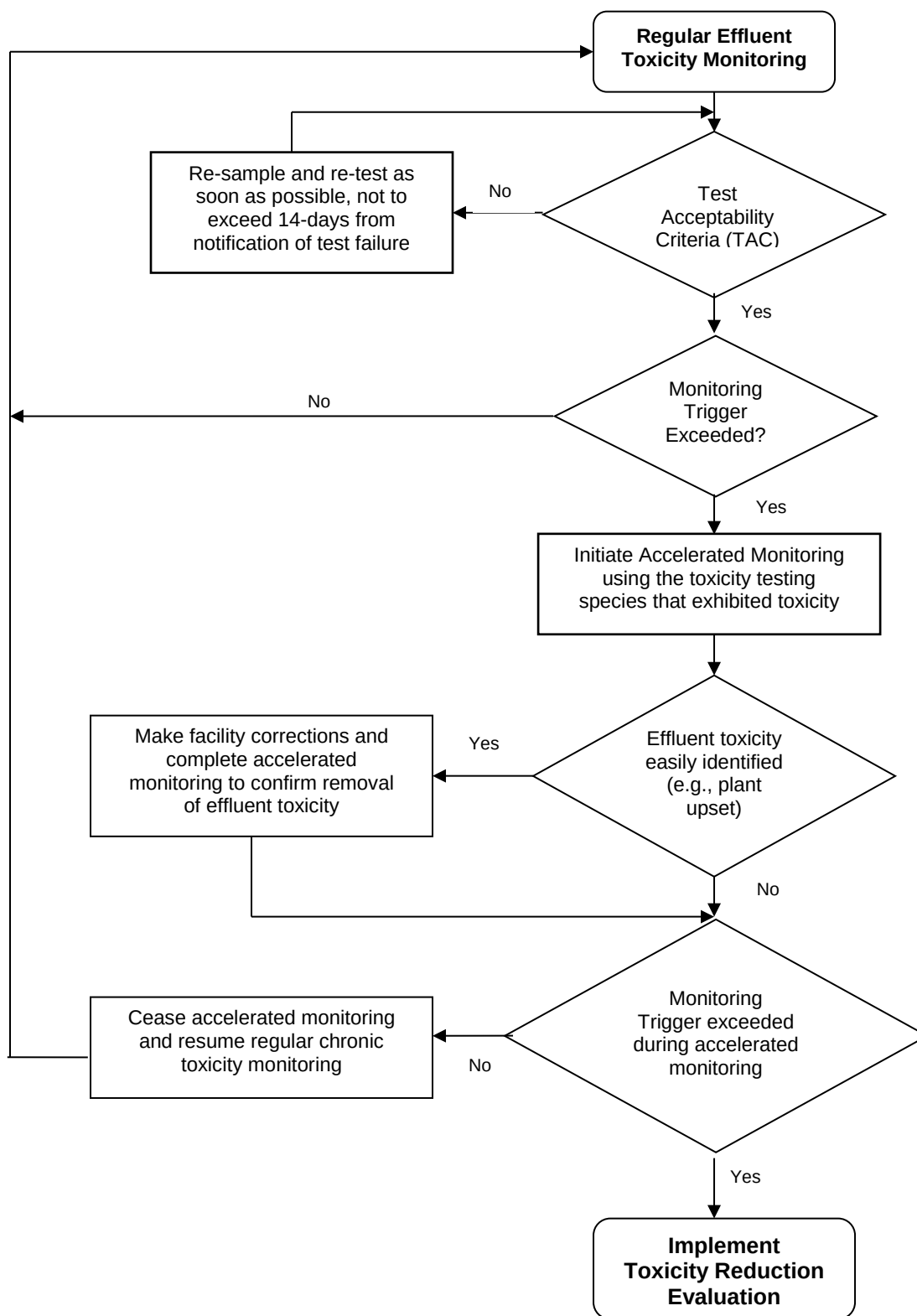
**Accelerated Monitoring.** The provision requires accelerated WET testing when a regular WET test result exceeds the monitoring trigger. The purpose of accelerated monitoring is to determine, in an expedient manner, whether there is toxicity before requiring the implementation of a TRE. Due to possible seasonality of the toxicity, the accelerated monitoring should be performed in a timely manner, preferably taking no more than 2 to 3 months to complete.

The provision requires accelerated monitoring consisting of four chronic toxicity tests in a six-week period (i.e., one test every two weeks) using the species that exhibited toxicity. Guidance regarding accelerated monitoring and TRE initiation is provided in the *Technical Support Document for Water Quality-based Toxics Control*, EPA/505/2-90-001, March 1991 (TSD). The TSD at page 118 states, “EPA

*recommends if toxicity is repeatedly or periodically present at levels above effluent limits more than 20 percent of the time, a TRE should be required.”* Therefore, four accelerated monitoring tests are required in this provision. If no toxicity is demonstrated in the four accelerated tests, then it demonstrates that toxicity is not present at levels above the monitoring trigger more than 20 percent of the time (only 1 of 5 tests are toxic, including the initial test). However, notwithstanding the accelerated monitoring results, if there is adequate evidence of effluent toxicity (i.e. toxicity present exceeding the monitoring trigger more than 20 percent of the time), the Executive Officer may require that the Discharger initiate a TRE.

See the WET Accelerated Monitoring Flow Chart (Figure F-1), below, for further clarification of the accelerated monitoring requirements and for the decision points for determining the need for TRE initiation.

**Figure F-1**  
**WET Accelerated Monitoring Flow Chart**



### 3. Best Management Practices and Pollution Prevention

- a. **Salinity Evaluation and Minimization Plan.** The Discharger submitted a Salinity Evaluation and minimization Plan in June 2011. This Order requires that the existing Salinity Evaluation and Minimization Plan be maintained in order to ensure that adequate measures are developed and implemented by the Discharger to reduce the discharge of salinity to Laguna Creek. The Discharger shall evaluate the effectiveness of the salinity evaluation and minimization plan and provide a summary with the Report of Waste Discharge, due 180 days prior to the permit expiration date.

### 4. Construction, Operation, and Maintenance Specifications

- a. **Filtration System Operating Specifications.** Turbidity is included as an operational specification as an indicator of the effectiveness of the filtration system for providing adequate disinfection. The tertiary treatment process utilized at this Facility is capable of reliably meeting a turbidity limitation of 2 nephelometric turbidity units (NTU) as a daily average. Failure of the treatment system such that virus removal is impaired would normally result in increased particles in the effluent, which result in higher effluent turbidity and could impact UV dosage. Turbidity has a major advantage for monitoring filter performance, allowing immediate detection of filter failure and rapid corrective action. The operational specification requires that turbidity prior to disinfection shall not exceed 2 NTU as a daily average; 5 NTU, more than 5 percent of the time within a 24-hour period, and an instantaneous maximum of 10 NTU.
- b. **Ultraviolet (UV) Disinfection System Operating Specifications.** This Order requires that when discharging to surface water, wastewater shall be oxidized, coagulated, filtered, and adequately disinfected pursuant to the DDW reclamation criteria, CCR, Title 22, division 4, chapter 3, (Title 22), or equivalent. To ensure that the UV disinfection system is operated to achieve the required pathogen removal, this Order includes effluent limits for total coliform organisms, filtration system operating specifications, and UV disinfection system operating specifications. Compliance with total coliform effluent limits alone does not ensure that pathogens in the municipal wastewater have been deactivated by the UV disinfection system. Compliance with the effluent limits and the filtration system and UV disinfection operating specifications demonstrates compliance with the equivalency to Title 22 disinfection requirement.

The NWRI guidelines include UV operating specifications for compliance with Title 22. For water recycling in accordance with Title 22, the UV system shall be an approved system included in the *Treatment Technology Report for Recycled Water*, December 2009 (or a later version, as applicable) published by the DPH. The UV system shall also conform to all requirements and operating specifications of the NWRI guidelines. A memorandum dated 1 November 2004 issued by DPH to Regional Water Board executive offices recommended that provisions be included in permits for water recycling treatment plants employing UV disinfection requiring dischargers to establish fixed cleaning frequency of lamp sleeves, as well as, include provisions that specify minimum delivered UV dose that must be maintained (per the NWRI Guidelines).

For granular media filtration, the NWRI Guidelines recommend a minimum hourly average UV dose of 100 mJ/cm<sup>2</sup>. Therefore, this Order includes UV operating specifications requiring a minimum hourly average UV dose of 100 mJ/cm<sup>2</sup> and a minimum hourly average UV transmittance of 55%, per the NWRI Guidelines. If the

Discharger conducts a site-specific UV engineering study that demonstrates a lower UV dose meets a Title 22 equivalent virus removal, this Order may be reopened to revise the UV operating specifications accordingly.

**5. Special Provisions for Municipal Facilities (POTW's Only)**

**a. Pretreatment Requirements.**

The federal CWA section 307(b), and federal regulations, 40 C.F.R. part 403, require publicly owned treatment works to develop an acceptable industrial pretreatment program. For POTW's with a total design flow greater than 5 MGD, and that are receiving pollutants from industrial users, a pretreatment program is required to prevent the introduction of pollutants, which will interfere with treatment plant operations or sludge disposal, and prevent pass through of pollutants that exceed water quality objectives, standards or permit limitations. The discharger does receive pollutants from industrial users, however, the total design flow of the Facility is less than 5 MGD, and the industrial influent has not been found to cause or contribute to any of the above mentioned conditions. The Discharger also enforces the following pretreatment programs relevant to industrial discharges:

- i. City of Galt Municipal Code (City of Galt, 2002)
- ii. Title 40 Code of Federal Regulations

Therefore, this Order does not impose further pretreatment requirements pursuant to 40 C.F.R. part 403.

- b. The State Water Board issued General Waste Discharge Requirements for Sanitary Sewer Systems, Water Quality Order 2006-0003-DWQ (General Order) on 2 May, 2006. The Monitoring and Reporting Requirements for the General Order were amended by Water Quality Order WQ 2008-0002-EXEC on 20 February , 2008. The General Order requires public agencies that own or operate sanitary sewer systems with greater than one mile of pipes or sewer lines to enroll for coverage under the General Order. The General Order requires agencies to develop sanitary sewer management plans (SSMPs) and report all sanitary sewer overflows (SSOs), among other requirements and prohibitions.

Furthermore, the General Order contains requirements for operation and maintenance of collection systems and for reporting and mitigating sanitary sewer overflows. Inasmuch that the Discharger's collection system is part of the system that is subject to this Order, certain standard provisions are applicable as specified in Provisions, section VI.C.5. For instance, the 24-hour reporting requirements in this Order are not included in the General Order. The Discharger must comply with both the General Order and this Order. The Discharger and public agencies that are discharging wastewater into the facility were required to obtain enrollment for regulation under the General Order by 1 December, 2006.

## VII. RATIONALE FOR MONITORING AND REPORTING REQUIREMENTS

CWA section 308 and 40 C.F.R. sections 122.41(h), (j)-(l), 122.44(i), and 122.48 require that all NPDES permits specify monitoring and reporting requirements. Water Code sections 13267 and 13383 also authorize the Central Valley Water Board to establish monitoring, inspection, entry, reporting, and recordkeeping requirements. The Monitoring and Reporting Program (MRP), Attachment E of this Order establishes monitoring, reporting, and recordkeeping requirements that implement federal and state requirements. The following provides the rationale for the monitoring and reporting requirements contained in the MRP for this facility.

### A. Influent Monitoring

1. Influent monitoring is required to collect data on the characteristics of the wastewater and to assess compliance with effluent limitations (e.g., BOD<sub>5</sub> and TSS reduction requirements). The monitoring frequencies for flow (continuous), BOD<sub>5</sub> (weekly), and TSS (weekly) have been retained from Order R5-2010-0099.

### B. Effluent Monitoring

1. Pursuant to the requirements of 40 C.F.R. section 122.44(i)(2) effluent monitoring is required for all constituents with effluent limitations. Effluent monitoring is necessary to assess compliance with effluent limitations, assess the effectiveness of the treatment process, and to assess the impacts of the discharge on the receiving stream and groundwater.
2. Effluent monitoring frequencies and sample types for flow (continuous), pH (weekly), BOD<sub>5</sub> (weekly), TSS (weekly), have been retained from Order R5-2010-0099 to determine compliance with effluent limitations for these parameters.
3. Monitoring data collected over the previous permit term for aluminum, bis(2-ethylhexyl)phthalate, Carbon tetrachloride, chlorine, chlorodibromomethane, copper, cyanide, dichlorobromomethane, lead, iron, and manganese did not demonstrate reasonable potential to exceed water quality objectives/criteria. Thus, specific monitoring requirements for these parameters have not been retained from Order R5-2010-0099.
4. Water Code section 13176, subdivision (a), states: "*The analysis of any material required by [Water Code sections 13000-16104] shall be performed by a laboratory that has accreditation or certification pursuant to Article 3 (commencing with Section 100825) of Chapter 4 of Part 1 of Division 101 of the Health and Safety Code.*" The DDW certifies laboratories through its Environmental Laboratory Accreditation Program (ELAP).

Section 13176 cannot be interpreted in a manner that would violate federal holding time requirements that apply to NPDES permits pursuant to the CWA. (Wat. Code §§ 13370, subd. (c), 13372, 13377.) Section 13176 is inapplicable to NPDES permits to the extent it is inconsistent with CWA requirements. (Wat. Code § 13372, subd. (a).) The holding time requirements are 15 minutes for chlorine residual, dissolved oxygen, and pH and immediate analysis is required for temperature. (40 C.F.R. § 136.3(e), Table II)

### C. Whole Effluent Toxicity Testing Requirements

1. **Acute Toxicity.** Annual 96-hour bioassay testing is required to demonstrate compliance with the effluent limitation for acute toxicity.
2. **Chronic Toxicity.** Quarterly chronic whole effluent toxicity testing is required in order to demonstrate compliance with the Basin Plan's narrative toxicity objective.

#### **D. Receiving Water Monitoring**

##### **1. Surface Water**

- a. Receiving water monitoring is necessary to assess compliance with receiving water limitations and to assess the impacts of the discharge on the receiving stream.

#### **E. Other Monitoring Requirements**

##### **1. Water Supply Monitoring**

Water supply monitoring is required to evaluate the source of constituents in the wastewater.

##### **2. UV Disinfection System Monitoring**

UV system monitoring and reporting are required to ensure that the UV system is operated to adequately inactivate pathogens in the wastewater. UV disinfection system monitoring is imposed to achieve equivalency to requirements established by the DDW), and the NWRI, Guidelines

##### **3. Discharge Monitoring Report-Quality Assurance (DMR-QA) Study Program**

Under the authority of section 308 of the CWA (33 U.S.C. § 1318), U.S. EPA requires major permittees under the NPDES Program to participate in the annual DMR-QA Study Program. The DMR-QA Study evaluates the analytical ability of laboratories that routinely perform or support self-monitoring analyses required by NPDES permits. There are two options to satisfy the requirements of the DMR-QA Study Program: (1) The Discharger can obtain and analyze a DMR-QA sample as part of the DMR-QA Study; or (2) Per the waiver issued by U.S.EPA to the State Water Board, the Discharger can submit the results of the most recent Water Pollution Performance Evaluation Study from their own laboratories or their contract laboratories. A Water Pollution Performance Evaluation Study is similar to the DMR-QA Study. Thus, it also evaluates a laboratory's ability to analyze wastewater samples to produce quality data that ensure the integrity of the NPDES Program. The Discharger shall submit annually the results of the DMR-QA Study or the results of the most recent Water Pollution Performance Evaluation Study to the State Water Board. The State Water Board's Quality Assurance Program Officer will send the DMR-QA Study results or the results of the most recent Water Pollution Performance Evaluation Study to U.S. EPA's DMR-QA Coordinator and Quality Assurance Manager.

### **VIII. PUBLIC PARTICIPATION**

The Central Valley Water Board has considered the issuance of WDR's that will serve as an NPDES permit for the City of Galt Wastewater Treatment Plant and Recamation Facility. As a step in the WDR adoption process, the Central Valley Water Board staff has developed tentative WDR's and has encouraged public participation in the WDR adoption process.

#### **A. Notification of Interested Parties**

The Central Valley Water Board notified the Discharger and interested agencies and persons of its intent to prescribe WDR's for the discharge and provided an opportunity to submit written comments and recommendations. Notification was provided through publication in the Galt Herald on 30 September 2015.

The public had access to the agenda and any changes in dates and locations through the Central Valley Water Board's website at:  
[http://www.waterboards.ca.gov/centralvalley/board\\_info/meetings/](http://www.waterboards.ca.gov/centralvalley/board_info/meetings/)



**B. Written Comments**

Interested persons were invited to submit written comments concerning tentative WDR's as provided through the notification process. Comments were due either in person or by mail to the Executive Office at the Central Valley Water Board at the address on the cover page of this Order.

To be fully responded to by staff and considered by the Central Valley Water Board, the written comments were due at the Central Valley Water Board office by 5:00 p.m.

**19 October 2015**

**C. Public Hearing**

The Central Valley Water Board held a public hearing on the tentative WDR's during its regular Board meeting on the following date and time and at the following location:

Date: 10/11 December 2015  
Time: 8:30 a.m.  
Location: Regional Water Quality Control Board, Central Valley Region  
11020 Sun Center Dr., Suite #200  
Rancho Cordova, CA 95670

Interested persons were invited to attend. At the public hearing, the Central Valley Water Board heard testimony pertinent to the discharge, WDR's, and permit. For accuracy of the record, important testimony was requested in writing.

**D. Reconsideration of Waste Discharge Requirements**

Any aggrieved person may petition the State Water Board to review the decision of the Central Valley Water Board regarding the final WDR's. The petition must be received by the State Water Board at the following address within 30 calendar days of the Central Valley Water Board's action:

State Water Resources Control Board  
Office of Chief Counsel  
P.O. Box 100, 1001 I Street  
Sacramento, CA 95812-0100

For instructions on how to file a petition for review, see  
[http://www.waterboards.ca.gov/public\\_notices/petitions/water\\_quality/wgpetition\\_instr.shtml](http://www.waterboards.ca.gov/public_notices/petitions/water_quality/wgpetition_instr.shtml)

**E. Information and Copying**

The Report of Waste Discharge, other supporting documents, and comments received are on file and may be inspected at the address above at any time between 8:30 a.m. and 4:45 p.m., Monday through Friday. Copying of documents may be arranged through the Central Valley Water Board by calling (916) 464-3291.

**F. Register of Interested Persons**

Any person interested in being placed on the mailing list for information regarding the WDR's and NPDES permit should contact the Central Valley Water Board, reference this facility, and provide a name, address, and phone number.

**G. Additional Information**

Requests for additional information or questions regarding this order should be directed to Tyson Pelkofer at 916-464-4853 or [Tyson.Pelkofer@waterboards.ca.gov](mailto:Tyson.Pelkofer@waterboards.ca.gov).

**ATTACHMENT G – SUMMARY OF REASONABLE POTENTIAL ANALYSIS**

| Constituent                        | Units    | MEC                 | B                   | C               | CMC              | CCC              | Water & Org | Org. Only | Basin Plan | MCL             | Reasonable Potential |
|------------------------------------|----------|---------------------|---------------------|-----------------|------------------|------------------|-------------|-----------|------------|-----------------|----------------------|
| Aluminum, Total Recoverable        | µg/L     | 54 <sup>1</sup>     | 1100                | 200             | 750 <sup>2</sup> | --               | --          | --        | --         | 200             | No                   |
| Ammonia Nitrogen, Total (as N)     | mg/L     | 19                  | 4.7                 | 3.8             | 3.8 <sup>2</sup> | 2.4 <sup>3</sup> | --          | --        | --         | --              | Yes                  |
| Arsenic, Total Recoverable         | µg/L     | 24                  | 18                  | 10              | 340              | 150              | --          | --        | 10         | 10              | Yes                  |
| Bis (2-ethylhexyl) Phthalate       | µg/L     | 1.4                 | 2.0                 | 1.8             | --               | --               | 1.8         | 5.9       | --         | 4               | No                   |
| Carbon Tetrachloride               | µg/L     | <0.077 <sup>5</sup> | <0.077 <sup>5</sup> | 0.25            | --               | --               | 0.25        | 4.4       | --         | 0.5             | No <sup>4</sup>      |
| Chlorine, Total Residual           | mg/L     |                     |                     |                 |                  |                  |             |           |            |                 | No                   |
| Chlorodibromomethane               | µg/L     | <0.049              | <0.049              | 0.41            | --               | --               | 0.41        | 34        | --         | 80 <sup>6</sup> | No <sup>4</sup>      |
| Copper, Total Recoverable          | µg/L     | 17                  | 5.7                 | 60 <sup>7</sup> | 5.5              | 4                | 1,300       | --        | --         | 1,000           | No                   |
| Cyanide, Total (as CN)             | µg/L     | 5.0                 | <4.0                | 5.2             | 22               | 5.2              | --          | 220,000   | --         | 150             | No                   |
| Dichlorobromomethane               | µg/L     | <0.031              | <0.031              | 0.56            | --               | --               | 0.56        | 46        | --         | --              | No <sup>4</sup>      |
| Electrical Conductivity @ 25°C     | µmhos/cm | 590                 | 540                 | <sup>4</sup>    | --               | --               | --          | --        | --         | --              | No                   |
| Iron, Total Recoverable            | µg/L     | 56 <sup>1</sup>     | 1300                | 300             | --               | --               | --          | --        | --         | 300             | No                   |
| Lead, Total Recoverable            | µg/L     | 0.34                | 0.78                | 0.87            | 22.2             | 0.87             | --          | --        | --         | 15              | No                   |
| Manganese, Total Recoverable       | µg/L     | 7 <sup>1</sup>      | 760                 | 50              | --               | --               | --          | --        | --         | 50              | No                   |
| Nitrate plus Nitrite, Total (as N) | mg/L     | 28                  | --                  | 10              | --               | --               | --          | --        | --         | 10              | Yes                  |
| Zinc, Total Recoverable            | µg/L     | 53                  | 11                  | 52              | 52               | 52               | 7,400       | 26,000    | --         | 5000            | Yes                  |

| Constituent | Units | MEC | B | C | CMC | CCC | Water & Org | Org. Only | Basin Plan | MCL | Reasonable Potential |
|-------------|-------|-----|---|---|-----|-----|-------------|-----------|------------|-----|----------------------|
|-------------|-------|-----|---|---|-----|-----|-------------|-----------|------------|-----|----------------------|

General Note: All inorganic concentrations are given as a total recoverable.

MEC = Maximum Effluent Concentration

B = Maximum Receiving Water Concentration or lowest detection level, if non-detect

C = Criterion used for Reasonable Potential Analysis

CMC = Criterion Maximum Concentration (CTR or NTR)

CCC = Criterion Continuous Concentration (CTR or NTR)

Water & Org = Human Health Criterion for Consumption of Water & Organisms (CTR or NTR)

Org. Only = Human Health Criterion for Consumption of Organisms Only (CTR or NTR)

Basin Plan = Numeric Site-specific Basin Plan Water Quality Objective

MCL = Drinking Water Standards Maximum Contaminant Level

NA = Not Available

ND = Non-detect

Footnotes:

- (1) Represents the maximum observed average annual concentration for comparison with the Secondary MCL or site-specific objective.
- (2) U.S. EPA National Recommended Ambient Water Quality Criteria, Freshwater Aquatic Life Protection, 1-hour average.
- (3) U.S. EPA National Recommended Ambient Water Quality Criteria, Freshwater Aquatic Life Protection, 30-day average.
- (4) See section IV.C.3 of the Fact Sheet (Attachment F) for a discussion of the RPA results.
- (5) The Discharger no longer uses chlorine in the treatment process.
- (6) Represents the Primary MCL for total trihalomethanes, which include bromoform, chlorodibromomethane, chloroform, and dichlorobromomethane.
- (7) Water Effects Ratio for copper of 15.0 is applied to C. See Section IV.C.2.e.ii of the Fact Sheet (Attachment F) for more information.

### ATTACHMENT H – CALCULATION OF WQBEL'S

| Human Health WQBEL's Calculations |       |          |                               |                 |                      |                      |                 |                 |                 |      |
|-----------------------------------|-------|----------|-------------------------------|-----------------|----------------------|----------------------|-----------------|-----------------|-----------------|------|
| Parameter                         | Units | Criteria | Mean Background Concentration | Dilution Factor | MDEL/AMEL Multiplier | AWEL/AMEL Multiplier | AMEL Multiplier | AMEL            | MDEL            | AWEL |
| Arsenic                           | µg/L  | 10       | 8.56 <sup>1</sup>             | --              | 1.91                 | --                   | --              | 10 <sup>2</sup> | 19 <sup>3</sup> | --   |
| Nitrate Nitrogen, Total (as N)    | mg/L  | 10       | 1.0 <sup>1,4</sup>            | --              | --                   | 1.94                 | --              | 10 <sup>2</sup> | --              | 19   |

<sup>1</sup> Maximum background concentration.

<sup>2</sup> AMEL is based on the Primary MCL.

<sup>3</sup> Maximum Daily Effluent Limitations are calculated according to Section 1.4 of the SIP using a 99<sup>th</sup> percentile occurrence probability.

<sup>4</sup> Annual Average.

| Aquatic Life WQBEL's Calculations |       |          |      |                  |     |                                 |                      |                                   |                        |                               |                 |                               |                            |                   |                   |
|-----------------------------------|-------|----------|------|------------------|-----|---------------------------------|----------------------|-----------------------------------|------------------------|-------------------------------|-----------------|-------------------------------|----------------------------|-------------------|-------------------|
| Parameter                         | Units | Criteria |      | Dilution Factors |     | Aquatic Life Calculations       |                      |                                   |                        |                               |                 |                               | Final Effluent Limitations |                   |                   |
|                                   |       | CMC      | CCC  | CMC              | CCC | ECA Multiplier <sup>acute</sup> | LTA <sup>acute</sup> | ECA Multiplier <sup>chronic</sup> | LTA <sup>chronic</sup> | AMEL Multiplier <sup>95</sup> | AWEL Multiplier | MDEL Multiplier <sup>99</sup> | AMEL <sup>1</sup>          | AWEL <sup>2</sup> | MDEL <sup>3</sup> |
| Ammonia Nitrogen, Total (as N)    | mg/L  | 3.83     | 2.43 | --               | --  | 0.14                            | 0.6                  | 0.56                              | 1.35                   | 2.40                          | 5.16            | --                            | 1.3                        | 2.9               | --                |
| Zinc, Total Recoverable           | µg/L  | 52       | 52   | --               | --  | 0.54                            | 27.7                 | 0.72                              | 37.28                  | 1.25                          | --              | 1.86                          | 35                         | --                | 52                |

<sup>1</sup> Average Monthly Effluent Limitations are calculated according to Section 1.4 of the SIP using a 95<sup>th</sup> percentile occurrence probability.

<sup>2</sup> Average Weekly Effluent Limitations are calculated according to Section 1.4 of the SIP using a 98<sup>th</sup> percentile occurrence probability.

<sup>3</sup> Maximum Daily Effluent Limitations are calculated according to Section 1.4 of the SIP using a 99<sup>th</sup> percentile occurrence probability.